Counting $\mathbb{F}_q$-points of orbital varieties in ad-nilpotent ideals of type A_n

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Abstract. Let $\mathfrak{b}_n(\mathbb{F}_q)$ denote the Lie algebra of upper triangular $n \times n$ matrices with entries in the finite field $\mathbb{F}_q$. For every $\mathfrak{b}_n(\mathbb{F}_q)$-stable ideal $\mathfrak{a}$ of the Lie algebra $\mathfrak{u}_n(\mathbb{F}_q)$ of strictly upper triangular matrices, and every partition μ of n, we obtain an explicit formula for the number of elements of $\mathfrak{a}$ of Jordan type μ. Up to an explicit power of q, our formula is given by the Hall scalar product of a modified Hall-Littlewood function indexed by μ and a chromatic quasisymmetric function associated to $\mathfrak{a}$. In the special case that $\mathfrak{a} = \mathfrak{u}_\Lambda(\mathbb{F}_q)$, where $\mathfrak{u}_\Lambda(\mathbb{F}_q)$ is the nilradical of the standard parabolic subalgebra of $\mathfrak{gl}_n(\mathbb{F}_q)$ associated to a composition Λ of n, our formula reduces to the statement (recently proved in [KT23, Theorem 5.13]) that up to an explicit polynomial factor in q, the number of elements in $\mathfrak{u}_\Lambda(\mathbb{F}_q)$ of Jordan type μ is equal to the coefficient of the monomial $\mathbf{x}^\Lambda$ in the specialization of the dual Macdonald symmetric function $\mathrm{Q}_{\mu'}(\mathbf{x}; q^{-1}, t)$ at $t = 0$. We obtain a new and shorter proof of the latter assertion using a parabolic variation of Borodin's division algorithm [Bor97, Kir95b].

We give four applications of our main result stated above. First, we obtain an analogous formula for the number of points of a nilpotent Hessenberg variety. Second, we demonstrate that a recurrence relation of Kirillov in [Kir95a] for the number of strictly upper triangular matrices of Jordan type μ is a straightforward consequence of our main result combined with the Cauchy identity for Macdonald polynomials. Third, we give an explicit formula for the number of $X \in \mathfrak{u}_\Lambda(\mathbb{F}_q)$ that satisfy $X^2 = 0$. In the special case $\Lambda = (1^n)$, our formula is different from the one conjectured by Kirillov and Melnikov [KM97] (and proved by Ekhad and Zeilberger [EZ96]). We obtain a new proof of the Kirillov-Melnikov-Ekhad-Zeilberger formula using our latter result and a formula for two-rowed Macdonald polynomials by Jing and Józefiak [JJ92]. Fourth, we give an explicit formula for the number of double cosets $\mathsf{U}_1 \backslash \mathrm{GL}_n(\mathbb{F}_q) / \mathsf{U}_2$ where U_1 and U_2 denote unipotent subgroups of $\mathrm{GL}_n(\mathbb{F}_q)$ that correspond to two $\mathfrak{b}_n(\mathbb{F}_q)$-stable ideals of $\mathfrak{u}_n(\mathrm{F}_q)$.

§ 1. Introduction

Let $\mathsf{U}_n(\mathbb{F}_q)$ be the group of $n \times n$ upper-triangular matrices over the finite field $\mathbb{F}_q$ with all the diagonal entries equal to 1, and let $\mathfrak{u}_n(\mathbb{F}_q)$ denote the Lie algebra of $n \times n$ upper-triangular nilpotent matrices over $\mathbb{F}_q$. In an influential and elegant paper [Kir95b], Kirillov initiated a comprehensive study of the orbit method for $\mathsf{U}_n(\mathbb{F}_q)$, establishing various connections between the adjoint and coadjoint orbit structures of $\mathsf{U}_n(\mathbb{F}_q)$ and $\mathfrak{u}_n(\mathbb{F}_q)$ and representation theory. In connection to these ideas, Kirillov posed the following question:

Key words and phrases. Finite fields, Jordan form, Macdonald polynomials, q-Hermite polynomials, Hessenberg functions, chromatic quasisymmetric polynomials.

Given a partition μ of n, count the number of $X \in \mathfrak{u}_n(\mathbb{F}_q)$ of Jordan type μ.

This problem was investigated by Kirillov himself and later on revisited by Borodin [Bor97], who used a technique called "the division algorithm" to provide a recursive formula for the number of these matrices. Borodin applied this formula to study the asymptotic behavior of the Jordan type of strictly upper triangular $n \times n$ matrices as $n \to \infty$.

One can think of $\mathfrak{u}_n(\mathbb{F}_q)$ as the nilradical of the standard Borel subalgebra of $\mathfrak{gl}_n(\mathbb{F}_q)$. Then Kirillov's question is about the number of points of the variety obtained as the intersection of $\mathfrak{u}_n(\mathbb{F}_q)$ and a nilpotent orbit of $\mathfrak{gl}_n(\mathbb{F}_q)$. At least over $\mathbb{C}$, such varieties are sometimes called *orbital varieties*; see for example [Jos98]. From this viewpoint it is natural to ask Kirillov's question for nilradicals of other parabolic subalgebras, or even more generally, for any ideal of $\mathfrak{u}_n(\mathbb{F}_q)$ that is stable under the action of the standard Borel subalgebra $\mathfrak{b}_n(\mathbb{F}_q)$ of $\mathfrak{gl}_n(\mathbb{F}_q)$. The latter ideals are sometimes simply called *ad-nilpotent ideals* of $\mathfrak{u}_n(\mathbb{F}_n)$. Their interaction with nilpotent orbits has attracted attention (see [FS21] and the references therein). Our primary goal in this paper is to answer this more general form of Kirillov's question (see Theorems 1.7, 1.10 and 1.13). Subsequently, we give different applications of the latter theorems.

Recall that a composition Λ of n is a finite tuple of positive integers whose sum equals n. When the parts of the composition form a weakly decreasing sequence, we call it a partition. The parabolic subgroup of $\mathrm{G} = \mathrm{GL}_n(\mathbb{F}_q)$ associated to Λ is denoted by $\mathrm{P}_\Lambda = \mathrm{P}_\Lambda(\mathbb{F}_q)$, and the unipotent radical of P_Λ is denoted by $\mathrm{U}_\Lambda = \mathrm{U}_\Lambda(\mathbb{F}_q)$. The nilradical of the parabolic subalgebra of $\mathfrak{gl}_n(\mathbb{F}_q)$ associated to Λ will be denoted by $\mathfrak{u}_\Lambda = \mathfrak{u}_\Lambda(\mathbb{F}_q)$. For instance when $\Lambda = (2,3,1)$ we have:

$$\mathrm{P}_\Lambda = \begin{pmatrix} * & * & * & * & * & * \\ * & * & * & * & * & * \\ 0 & 0 & * & * & * & * \\ 0 & 0 & * & * & * & * \\ 0 & 0 & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & * \end{pmatrix}, \quad \mathrm{U}_\Lambda = \begin{pmatrix} 1 & 0 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 1 & 0 & 0 & * \\ 0 & 0 & 0 & 1 & 0 & * \\ 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathfrak{u}_\Lambda = \begin{pmatrix} 0 & 0 & * & * & * & * \\ 0 & 0 & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & * \\ 0 & 0 & 0 & 0 & 0 & * \\ 0 & 0 & 0 & 0 & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

where the $*$'s represent arbitrary elements of $\mathbb{F}_q$. For $\Lambda = (1^n)$, we obtain $\mathfrak{u}_\Lambda = \mathfrak{u}_n(\mathbb{F}_q)$.

There is a simple yet quite useful characterization of the ad-nilpotent ideals of $\mathfrak{u}_n(\mathbb{F}_q)$, connecting them to Dyck paths. To give this characterization, we need the notion of a Hessenberg function.

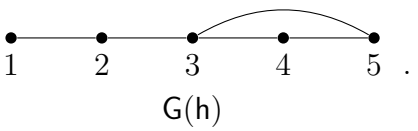
Definition 1.1 (Hessenberg function)**.** A Hessenberg function is a non-decreasing function $\mathsf{h}\colon [n] \to [n]$ such that $\mathsf{h}(i) \geq i$ for every $i \in [n]$, where $[n] := \{1, \ldots, n\}$. The set of all Hessenberg functions on $[n]$ will be denoted by $\mathcal{H}_n$.

One can naturally associate a graph to h which is the graph with vertex set $[n]$ and set of edges $E = \{\{i,j\} \mid i < j \leq \mathsf{h}(i)\}$. This graph, known as the *indifference graph* of h, will be denoted by $\mathsf{G}(\mathsf{h})$. A Hessenberg function $\mathsf{h} \in \mathcal{H}_n$ also produces a partial order on $[n]$ by declaring $i \prec_{\mathsf{h}} j$ if and only if $\mathsf{h}(i) < j$. We denote the poset $([n], \preceq_{\mathsf{h}})$ by $\mathcal{P}_{\mathsf{h}}$. Associated to h, we also define the following Lie subalgebra of $\mathfrak{u}_n(\mathbb{F}_q)$:

$$\mathfrak{u}_{\mathsf{h}} = \mathfrak{u}_{\mathsf{h}}(\mathbb{F}_q) \stackrel{\text{def}}{=} \{(a_{ij}) \in \mathfrak{u}_n(\mathbb{F}_q) : a_{ij} = 0, \quad j \leq \mathsf{h}(i)\}. \tag{1}$$

Example 1.2. For instance, when $\mathsf{h} = (2,3,5,5,5)$ we have

$$\mathfrak{u}_{\mathsf{h}} = \left\{ \begin{pmatrix} 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} : * \in \mathbb{F}_q \right\}, \qquad$$



The following assertion is well-known and straightforward to prove.

Proposition 1.3. *The assignment* $\mathsf{h} \mapsto \mathfrak{u}_{\mathsf{h}}$ *is a bijective correspondence between* $\mathcal{H}_n$ *and ad-nilpotent ideals of* $\mathfrak{u}_n(\mathbb{F}_q)$.

Throughout the paper, we denote $\sum_{i=1}^{n} \mathsf{h}(i)$ by $|\mathsf{h}|$. The (non-commutative) product of $\mathsf{h}_1 \in \mathcal{H}_n$ and $\mathsf{h}_2 \in \mathcal{H}_m$ is defined by

$$\mathsf{h} = \mathsf{h}_1 \sqcup \mathsf{h}_2 \in \mathcal{H}_{n+m} \qquad \mathsf{h}(i) = \begin{cases} \mathsf{h}_1(i) & 1 \le i \le n; \\ n + \mathsf{h}_2(i-n) & n+1 \le i \le n+m. \end{cases}$$

The Hessenberg function $k_n = (n, \dots, n) \in \mathcal{H}_n$ is called the complete Hessenberg function of size n. To each composition $\Lambda = (\Lambda_1, \dots, \Lambda_s)$ of n we assign the Hessenberg function

$$k_\Lambda \stackrel{\text{def}}{=} k_{\Lambda_1} \sqcup k_{\Lambda_2} \sqcup \cdots \sqcup k_{\Lambda_s}, \tag{2}$$

and we note that $\mathfrak{u}_{k_\Lambda} = \mathfrak{u}_\Lambda$.

For any $k \ge 1$ let J_k be the $k \times k$ nilpotent (upper triangular) Jordan block. Given a partition $\mu = (\mu_1, \dots, \mu_k)$ of n, we use J_μ to denote the square block-diagonal matrix of size $|\mu| = \mu_1 + \cdots + \mu_k$ with diagonal blocks $J_{\mu_1}, \dots, J_{\mu_k}$. We say that the Jordan type of a matrix is μ when its Jordan form is J_μ.

Definition 1.4 (Jordan type counting function)**.** Let μ be a partition of n. For a given composition Λ of n we set

$$\mathrm{F}_{\mu\Lambda}(q) \stackrel{\text{def}}{=} \#\left(\mathcal{C}_\mu \cap \mathfrak{u}_\Lambda(\mathbb{F}_q)\right), \tag{3}$$

where $\mathcal{C}_\mu = \mathcal{C}_\mu(\mathbb{F}_q)$ is the $\mathrm{GL}_n(\mathbb{F}_q)$-conjugacy class of J_μ. Similarly, for a Hessenberg function $h \in \mathcal{H}_n$ we set

$$\mathrm{F}_{\mu\mathsf{h}}(q) \stackrel{\text{def}}{=} \#\left(\mathcal{C}_\mu \cap \mathfrak{u}_\mathsf{h}(\mathbb{F}_q)\right). \tag{4}$$

Recall that the length of a partition μ, denoted by $\ell(\mu)$, is the number of its parts. It is convenient to define $\mu_i = 0$ for all $i > \ell(\mu)$. The conjugate of μ, denoted by $\mu' = (\mu'_1, \dots, \mu'_l)$, is defined by

$$\mu'_j = \#\{1 \le i \le k : \mu_i \ge j\}.$$

In particular we have $\mu'_1 = \ell(\mu)$ and $l = \ell(\mu') = \mu_1$. For instance the conjugate of $\mu = (3,2,2,1)$ is $\mu' = (4,3,1)$. Given partitions $\mu = (\mu_1, \dots, \mu_k)$ and $\nu = (\nu_1, \dots, \nu_s)$ of n, we say that ν "dominates" μ, and we write $\mu \trianglelefteq \nu$, if and only if

$$\mu_1 + \mu_2 + \cdots + \mu_i \le \nu_1 + \nu_2 + \cdots + \nu_i, \quad \text{for } i \ge 1.$$

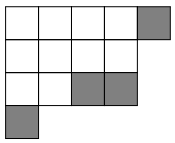
Note that conjugation reverses the dominance order, i.e., $\nu' \trianglelefteq \mu'$. We also remark that $\ell(\nu) \le \ell(\mu)$ when $\mu \trianglelefteq \nu$.

Assigned to each partition μ we define

$$\mathsf{n}(\mu) \stackrel{\text{def}}{=} \sum_{i \ge 1} (i-1)\mu_i = \sum_{i \ge 1} \binom{\mu'_i}{2}. \tag{5}$$

As usual, to any $\mu = (\mu_1, \dots, \mu_k)$ we associate a Young diagram of shape μ: a left-justified shape of k rows of boxes, where the i-th row from the top has μ_i boxes for $1 \le i \le k$. A **standard** Young tableau is a filling of the Young diagram of μ by $\{1, \dots, |\mu|\}$ with strictly increasing rows and columns. A **semi-standard** Young tableau of shape μ is a filling of this Young diagram by positive integers with weakly increasing rows and strictly increasing columns. A semi-standard Young tableau with shape μ has **content** Λ if there are Λ_i occurrences of i for every $i \ge 1$. The number of semi-standard Young tableau with shape μ and content Λ is called a **Kostka number** and it is denoted by $\mathrm{K}_{\mu\Lambda}$.

For partitions ν and μ, we write $\nu \subseteq \mu$ to mean that the Young diagram of μ contains the Young diagram of ν, i.e. $\nu_i \le \mu_i$ for all $i \ge 1$. The set-theoretic difference of the Young diagrams of μ and ν, denoted by μ/ν, is called a **skew diagram**. A **horizontal k-strip** is a skew diagram with k boxes with no two squares in the same column. For instance the skew diagram $(5,4,4,1)/(4,4,2)$ is a 4-horizontal strip:



For a composition Λ of n we use $\mathrm{sort}(\Lambda)$ to denote the partition obtained by sorting the parts of Λ in decreasing order. For a non-negative integer n we adopt the following conventions for q-integers, q-factorials, and q-binomial coefficients:

$$[0]_q \stackrel{\text{def}}{=} 0, \quad [n]_q \stackrel{\text{def}}{=} \frac{q^n-1}{q-1}, \quad [0]_q! \stackrel{\text{def}}{=} 1, \quad [n]_q! \stackrel{\text{def}}{=} [1]_q \cdots [n]_q, \quad \begin{bmatrix} n \\ k \end{bmatrix}_q \stackrel{\text{def}}{=} \frac{[n]_q!}{[k]_q![n-k]_q!}.$$

Definition 1.5. Let $\lambda = (\lambda_1, \ldots, \lambda_s)$ be a partition of n. For any partition $\mu \vdash n$, we define

$$b_{\mu\lambda}(q) \stackrel{\text{def}}{=} \sum_{\mathcal{F}} \prod_{j=1}^{s} \theta_{\mu^j/\mu^{j-1}}(q), \quad \text{with} \quad \theta_{\eta/\rho}(q) \stackrel{\text{def}}{=} \frac{[|\eta|-|\rho|]_q!}{[\eta_1-\rho_1]_q!} \prod_{i\geq 1} \begin{bmatrix} \rho_i - \rho_{i+1} \\ \eta_{i+1} - \rho_{i+1} \end{bmatrix}_q, \tag{6}$$

where η/ρ is a horizontal strip and $\mathcal{F} := \mathcal{F}(\mu, \lambda)$ is the set of all chains of partitions

$$\emptyset = \mu^0 \subseteq \mu^1 \subseteq \cdots \subseteq \mu^s = \mu,$$

such that μ^j/μ^{j-1} is a horizontal λ_j-strip. By convention, $b_{\mu\lambda}(q) = 0$ when $\mathcal{F} = \emptyset$. We note that in (6) we have $\rho_i - \rho_{i+1} \geq \eta_{i+1} - \rho_{i+1}$ because η/ρ is a horizontal strip.

Remark 1.6. Let λ and μ be partitions of n with $\lambda \trianglelefteq \mu$. Then $\#\mathcal{F}(\mu, \lambda) = \mathrm{K}_{\mu\lambda} > 0$.

We now state the first main result of the paper.

Theorem 1.7. *Let Λ be a composition of n and set $\lambda = \mathrm{sort}(\Lambda)$. Let μ be a partition of n. Then $\mathrm{F}_{\mu\Lambda}(q) \neq 0$ if and only if $\lambda \trianglelefteq \mu'$. When $\lambda \trianglelefteq \mu'$, the following hold:*

1. $\mathrm{F}_{\mu\Lambda}(q) = \mathrm{F}_{\mu\lambda}(q) = (q-1)^{n-\ell(\mu)} q^{\binom{n}{2}+\ell(\mu)-\mathsf{n}(\mu)-n}\, b_{\mu'\lambda}(1/q)$.
2. $\mathrm{F}_{\mu\lambda}(q) \in \mathbb{Z}[q]$, *the degree of* $\mathrm{F}_{\mu\lambda}(q)$ *is* $\binom{n}{2} - \mathsf{n}(\mu)$, *and its leading coefficient is* $\mathrm{K}_{\mu'\lambda}$, *the Kostka number of shape* μ' *and content* λ.
3. *There exists a polynomial* $\mathrm{R}_{\mu\lambda}(q) \in \mathbb{Z}[q]$ *satisfying* $\mathrm{R}_{\mu\lambda}(0) = 1$ *and* $\mathrm{R}_{\mu\lambda}(1) > 0$ *such that*

$$\mathrm{F}_{\mu\lambda}(q) = (q-1)^{n-\ell(\mu)} q^{\binom{n}{2} - \binom{\ell(\mu)}{2} - \sum_{j\geq 1} \mu'_j \mu'_{j+1}}\, \mathrm{R}_{\mu\lambda}(q). \tag{7}$$

Remark 1.8. When the composition Λ has only one part i.e., $\Lambda = (n)$, we have $\mathfrak{u}_\Lambda(\mathbb{F}_q) = \{0\}$. Thus $\mathcal{C}_\mu \cap \mathfrak{u}_\Lambda(\mathbb{F}_q) \neq \emptyset$ if and only if $\mu = (1^n)$. In this case the zero matrix is the only element in $\mathcal{C}_\mu \cap \mathfrak{u}_\Lambda(\mathbb{F}_q)$ and so $\mathrm{F}_{\mu\Lambda}(q) = 1$.

Parts (2) and (3) of Theorem 1.7 extend some of the results of [Bor97] and [FK22, Theorem 4.1]. The main tool in the proof of Theorem 1.7 is a parabolic variant of Borodin's division algorithm to reduce the counting problem from $\mathfrak{u}_\Lambda(\mathbb{F}_q)$ to a smaller nilradical.

For $\lambda = (1^n)$, Fulman [Ful00], showed that $\mathrm{F}_{\mu\lambda}(q)$ can be expressed in terms of the Hall-Littlewood polynomials. In the special case $\lambda = (1^n)$, the formula given for $\mathrm{F}_{\mu\lambda}$ in Theorem 1.7 is different from Fulman's result. The $b_{\mu\lambda}(q)$ are related to the coefficients of Macdonald polynomials up to an explicit normalization factor, as we explain below.

Definition 1.9. Let $\lambda = (\lambda_1, \ldots, \lambda_s)$ be a partition of n. For any partition $\mu \vdash n$, we define

$$a_{\mu\lambda}(q) \stackrel{\text{def}}{=} \sum_{\mathcal{F}} \prod_{j=1}^{s} \psi_{\mu^j/\mu^{j-1}}(q) \quad \text{with} \quad \psi_{\eta/\rho}(q) \stackrel{\text{def}}{=} \prod_{i\geq 1} \begin{bmatrix} \eta_i - \eta_{i+1} \\ \eta_i - \rho_i \end{bmatrix}_q, \tag{8}$$

where η/ρ is a horizontal strip and $\mathcal{F}$ represents $\mathcal{F}(\mu, \lambda)$ as in Definition 1.5. By convention, $a_{\mu\lambda}(q) = 0$ when $\mathcal{F} = \emptyset$.

Let $\mathrm{P}_\mu(\mathbf{x};q,t)$ denote the Macdonald symmetric function in $\mathbf{x}=(x_1,x_2,\ldots)$ indexed by μ. Also, let the $\mathrm{Q}_\mu(\mathbf{x};q,t)$ denote the duals of the $\mathrm{P}_\mu(\mathbf{x};q,t)$ with respect to the q,t-Hall scalar product [Mac15, p. 323, Eq. (4.12)]. From [Mac15, p. 346, Eq. (7.13')], one obtains

$$\mathrm{P}_\mu(\mathbf{x};q,0)=\sum_\nu a_{\mu\nu}(q)m_\nu(\mathbf{x}), \tag{9}$$

where $m_\nu(\mathbf{x})$ is the monomial symmetric polynomial associated to ν. Then we have

$$b_{\mu\lambda}(q)=\left(\prod_{i\geq 1}\frac{[\lambda_i]_q!}{[\mu_i-\mu_{i+1}]_q!}\right)a_{\mu\lambda}(q). \tag{10}$$

The calculations to verify (10) are given in Proposition 5.2. We denote the coefficient of $\mathbf{x}^\lambda=x_1^{\lambda_1}x_2^{\lambda_2}\cdots$ in $\Phi(\mathbf{x})$ by $[\mathbf{x}^\lambda]\Phi(\mathbf{x})$. Then using (10), we can state the formula for $\mathrm{F}_{\mu\lambda}(q)$ in Theorem 1.7 as follows.

Theorem 1.10. *Let μ and $\lambda=(\lambda_1,\ldots,\lambda_s)$ be partitions of n and $\lambda\trianglelefteq\mu'$. Then*

$$\mathrm{F}_{\mu\lambda}(q)=\frac{(q-1)^{n-\ell(\mu)}q^{\sum_{i\geq 1}\binom{\mu'_i-\mu'_{i+1}}{2}+\binom{n}{2}+\ell(\mu)-\mathsf{n}(\mu)-\mathsf{n}(\lambda')-n}[\lambda]_q!}{\prod_{i\geq 1}[\mu'_i-\mu'_{i+1}]_q!}[\mathbf{x}^\lambda]\mathrm{P}_{\mu'}(\mathbf{x};1/q,0) \tag{11}$$

$$=(q-1)^n q^{\binom{n}{2}-\mathsf{n}(\mu)-\mathsf{n}(\lambda')-n}[\lambda]_q![\mathbf{x}^\lambda]\mathrm{Q}_{\mu'}(\mathbf{x};1/q,0), \tag{12}$$

where $[\lambda]_q!\stackrel{\text{def}}{=}[\lambda_1]_q!\ldots[\lambda_s]_q!$.

Equation (11) has also been proved recently by Karp and Thomas [KT23]. We would like to highlight the key differences between [KT23] and the present paper. In [KT23], the authors obtain (11) as a byproduct of a combination of results, including a probabilistic bijection between non-negative integer matrices and pairs of semistandard Young tableaux of the same shape, which they call the q-Burge correspondence. Our approach is more direct: first we obtain a recursive formula for $\mathrm{F}_{\mu\lambda}(q)$ with explicit coefficients (see Theorem 3.1). Then we solve this recursion by a simple iterative substitution. This leads in a natural way to a solution for the recursion in terms of the well-known tableau product formulas that occur in the coefficients of Macdonald polynomials!

Example 1.11. Let $\lambda=(2^k)$ and $\mu=(k,k)$ be two partitions of $2k$. From (11) we have:

$$\mathrm{F}_{\mu\lambda}(q)=(q-1)^{2k-2}q^{\binom{2k-2}{2}}(q+1)^{k-1}. \tag{13}$$

To see this, we first observe that $\mu'=\lambda$. One can show [Mac15, p. 325, Eq. (4.17)] that $\mathrm{P}_{\mu'}(x_1,\ldots,x_k;q,t)=(x_1\cdots x_k)^2$. Thus from (11) we obtain:

$$\mathrm{F}_{\mu\lambda}(q)=(q-1)^{2k-\ell(\mu)}q^{\sum_{i\geq 1}\binom{\mu'_i-\mu'_{i+1}}{2}+\binom{2k}{2}+\ell(\mu)-\mathsf{n}(\mu)-\mathsf{n}(\lambda')-n}\left(\prod_{i\geq 1}\frac{[\lambda_i]_q!}{[\mu'_i-\mu'_{i+1}]_q!}\right)$$

$$=(q-1)^{2k-2}q^{\binom{2k-2}{2}}\frac{([2]_q!)^k}{[2]_q!}=(q-1)^{2k-2}q^{\binom{2k-2}{2}}(q+1)^{k-1}.$$

We now proceed towards stating our main result about $\mathrm{F}_{\mu\mathsf{h}}(q)$, which is Theorem 1.13. To motivate Theorem 1.13, first we give a different interpretation of the formula of Theorem 1.10 in Theorem 1.12. To this end, we start with the *transformed Hall–Littlewood functions* $\mathrm{H}_\mu(\mathbf{x};q)$, which are defined by

$$\mathrm{H}_\mu(\mathbf{x};q)\stackrel{\text{def}}{=}\sum_\eta \mathrm{K}_{\eta\mu}(q)s_\eta(\mathbf{x}),$$

where the $s_\eta(\mathbf{x})$ are the Schur functions and the $\mathrm{K}_{\eta\mu}(q)$ are the Kostka polynomials [Mac15, Chapter III, Section 6]. Let $J_\mu(\mathbf{x};q,t)$ be the integral form of $\mathrm{P}_\mu(\mathbf{x};q,t)$ as in [Mac15, p. 352,

Eq. (8.3)]. By [Mac15, p. 354, Eq. (8.11)] we have

$$J_{\mu'}(\mathbf{x};q,t)=\sum_{\eta}\mathrm{K}_{\eta\mu'}(q,t)S_{\eta}(\mathbf{x};t),$$

where the $\mathrm{K}_{\eta\mu'}(q,t)$ are the q,t-Kostka polynomials, and the $S_\eta(\mathbf{x},t)$ satisfy $S_\eta(\mathbf{x},0)=s_\eta(\mathbf{x})$. When $t=0$ we have $J_{\mu'}(\mathbf{x};q,0)=\mathrm{P}_{\mu'}(\mathbf{x},q,0)$ and [Mac15, p.354, Eq. (8.15)] implies that $\mathrm{K}_{\eta\mu'}(q,0)=\mathrm{K}_{\eta'\mu}(q)$. Therefore

$$\mathrm{P}_{\mu'}(\mathbf{x};q,0)=\sum_{\eta}\mathrm{K}_{\eta'\mu}(q)s_{\eta}(\mathbf{x}). \tag{14}$$

From this we deduce

$$\omega\mathrm{P}_{\mu'}(\mathbf{x};q,0)=\mathrm{H}_{\mu}(\mathbf{x};q),$$

where ω denotes the standard involution of symmetric functions that satisfies $\omega s_\mu=s_{\mu'}$ [Mac15, p. 42, Eq. (3.8)]. The *modified Hall-Littlewood functions* are defined by

$$\tilde{\mathrm{H}}_{\mu}(\mathbf{x};t)\overset{\text{def}}{=}q^{\mathsf{n}(\mu)}\mathrm{H}_{\mu}(\mathbf{x};1/q)=\omega q^{\mathsf{n}(\mu)}\mathrm{P}_{\mu'}(\mathbf{x};1/q,0).$$

For other interpretations and the significance of the *q-Whittaker symmetric functions* $\mathrm{P}_\mu(\mathbf{x};q,0)$ we refer the reader to [Ber20].

Recall that $\mathrm{G}=\mathrm{GL}_n(\mathbb{F}_q)$ and $\mathsf{U}_\lambda=\{I+X:X\in\mathfrak{u}_\lambda(\mathbb{F}_q)\}$. The size of the centralizer of $I+J_\mu$ in G can be computed using the following formula [Mac15, p. 181, Eq. (1.6)]:

$$|\mathrm{Z}_{\mathrm{G}}(I+J_\mu)|=q^{n+2\mathsf{n}(\mu)}\prod_{i\geq1}\varphi_{m_i(\mu)}\left(1/q\right), \tag{15}$$

where $\varphi_m(t)=(1-t)(1-t^2)\cdots(1-t^m)$ and $m_i(\mu)=\mu'_i-\mu'_{i+1}$. One can easily verify that

$$\varphi_m\left(1/q\right)=[m]_q!(q-1)^m q^{-\binom{m}{2}-m}. \tag{16}$$

By combining Eqs. (15) and (16) we obtain

$$|\mathrm{Z}_{\mathrm{G}}(I+J_\mu)|=(q-1)^{\ell(\mu)}q^{n+2\mathsf{n}(\mu)-\ell(\mu)-\sum_{i\geq1}\binom{\mu'_i-\mu'_{i+1}}{2}}\prod_{i\geq1}[\mu'_i-\mu'_{i+1}]_q!. \tag{17}$$

Now $|\mathsf{U}_\lambda|=q^{\binom{n}{2}-\mathsf{n}(\lambda')}$. Furthermore, if $\langle\,,\,\rangle$ denotes the Hall scalar product [Mac15, Sec. I.4], then $\langle h_\mu(\mathbf{x}),m_\lambda(\mathbf{x})\rangle=\delta_{\mu\lambda}$, where the h_μ are the complete symmetric functions. It follows that

$$\begin{aligned}[\mathbf{x}^\lambda]\,q^{\mathsf{n}(\mu)}\mathrm{P}_{\mu'}(\mathbf{x};1/q,0)&=\left\langle h_\lambda(\mathbf{x}),q^{\mathsf{n}(\mu)}\mathrm{P}_{\mu'}(\mathbf{x};1/q,0)\right\rangle\\&=\left\langle h_\lambda(\mathbf{x}),\omega\tilde{\mathrm{H}}_\mu(\mathbf{x};q)\right\rangle=\left\langle\omega h_\lambda(\mathbf{x}),\tilde{\mathrm{H}}_\mu(\mathbf{x};q)\right\rangle=\left\langle e_\lambda(\mathbf{x}),\tilde{\mathrm{H}}_\mu(\mathbf{x};q)\right\rangle,\end{aligned}$$

where the $e_\lambda(\mathbf{x})$ are the elementary symmetric functions. Thus, Theorem 1.10 is equivalent to the following assertion.

Theorem 1.12. *Let μ and λ be partitions of n with $\lambda\trianglelefteq\mu'$. Then*

$$\mathrm{F}_{\mu\lambda}(q)=\frac{(q-1)^n|\mathsf{U}_\lambda|}{|\mathrm{Z}_G(I+J_\mu)|}[\lambda]_q!\,[\mathbf{x}^\lambda]q^{\mathsf{n}(\mu)}\mathrm{P}_{\mu'}(\mathbf{x};1/q,0)=\frac{(q-1)^n|\mathsf{U}_\lambda|}{|\mathrm{Z}_G(I+J_\mu)|}\left\langle[\lambda]_q!e_\lambda(\mathbf{x}),\tilde{\mathrm{H}}_\mu(\mathbf{x};q)\right\rangle. \tag{18}$$

Now the key observation is that $[\lambda]_q!e_\lambda(\mathbf{x})$ is the chromatic quasisymmetric function of a disjoint union of complete graphs. Theorem 1.13 establishes that this interpretation of the formula for $\mathrm{F}_{\mu\lambda}(q)$ extends to $\mathrm{F}_{\mu\mathsf{h}}(q)$.

In order to state Theorem 1.13, we start by reviewing the definition of the chromatic quasisymmetric functions introduced in an influential paper by Shareshian and Wachs [SW16]. Let $\mathsf{G}=(V,E)$ be a graph with V a set of positive integers. Given a subset S of $\mathbb{Z}_{\geq1}$, a proper S-coloring of G is a function $\kappa\colon V\to S$ such that $\kappa(i)\neq\kappa(j)$ whenever $\{i,j\}\in E$. Let $\mathcal{C}(\mathsf{G})$ be

the set of proper $\mathbb{Z}_{\geq 1}$-colorings of G. Stanley [Sta95] defined the chromatic symmetric function of G as

$$X_{\mathsf{G}}(\mathbf{x}) := \sum_{\kappa \in \mathcal{C}(\mathsf{G})} \prod_{v \in V} x_{\kappa(v)},$$

The *chromatic quasisymmetric function* of $\mathsf{G} = (V, E)$, is a q-deformation of Stanley's chromatic symmetric function defined by:

$$X_{\mathsf{G}}(\mathbf{x}; q) := \sum_{\kappa \in \mathcal{C}(\mathsf{G})} q^{\mathrm{asc}(\kappa)} \prod_{v \in V} x_{\kappa(v)},$$

where $\mathrm{asc}(\kappa) := |\{\{i, j\} \in E(\mathsf{G}) : i < j \text{ and } \kappa(i) < \kappa(j)\}|$, is the number of ascents of the coloring κ. The chromatic quasisymmetric function is not symmetric in general. However, for the graphs $\mathsf{G}(\mathsf{h})$, where $\mathsf{h} \in \mathcal{H}_n$, Shareshian and Wachs' theorem [SW16, Theorem 4.5] shows that $X_{\mathsf{G}(\mathsf{h})}(\mathbf{x}, q)$ is a symmetric function with coefficients in $\mathbb{Z}[q]$. Therefore, $X_{\mathsf{G}(\mathsf{h})}(\mathbf{x}, q)$ can be expressed in terms of the elementary symmetric functions:

$$X_{\mathsf{G}(\mathsf{h})}(\mathbf{x}; q) = \sum_{\lambda \vdash n} c_{\lambda \mathsf{h}}(q) e_{\lambda}(\mathbf{x}).$$

An explicit formula for the coefficients $c_{\lambda \mathsf{h}}(q)$ was recently given by Hikita [Hik24, Theorem 3]. It is conjectured by Shareshian and Wachs [SW16, Conjecture 1.3] that the coefficients of $c_{\lambda \mathsf{h}}(q)$ are nonnegative. However, Shareshian and Wachs [SW16, Theorem 6.3] proved that $X_{\mathsf{G}(\mathsf{h})}(\mathbf{x}; q)$ is Schur-positive, meaning that

$$X_{\mathsf{G}(\mathsf{h})}(\mathbf{x}; q) = \sum_{\lambda \vdash n} g_{\lambda \mathsf{h}}(q) s_{\lambda}(\mathbf{x}), \qquad g_{\lambda \mathsf{h}} \in \mathbb{Z}_{\geq 0}[q]. \tag{19}$$

Theorem 1.13. *Let $\mathsf{h} \in \mathcal{H}_n$ be a Hessenberg function and let μ be a partition of n. Then $\mathrm{F}_{\mu \mathsf{h}}(q) \in \mathbb{Z}[q]$ is a polynomial in q and*

$$\mathrm{F}_{\mu \mathsf{h}}(q) = |\mathfrak{u}_{\mathsf{h}}| \sum_{\lambda \vdash n} \frac{c_{\lambda \mathsf{h}}(q)}{[\lambda]_q!} \frac{\mathrm{F}_{\mu \lambda}(q)}{|\mathfrak{u}_{\lambda}|} = \frac{q^{n^2 - |\mathsf{h}|}(q-1)^n}{|\mathrm{Z}_{\mathrm{G}}(I + J_{\mu})|} \langle \tilde{\mathrm{H}}_{\mu}(\mathbf{x}; q), X_{\mathsf{G}(\mathsf{h})}(\mathbf{x}; q) \rangle. \tag{20}$$

The strategy of the proof of Theorem 1.13 is a reduction process from $\mathrm{F}_{\mu \mathsf{h}}(q)$ to the $\mathrm{F}_{\mu \lambda}(q)$ using the *modular law* of Abreu and Nigro [AN21]. For the definition of the modular law see Section 7.1.

As in Theorem 1.7, it is interesting to determine for which μ and h we have $\mathrm{F}_{\mu \mathsf{h}}(q) \neq 0$. In Theorem 1.17, we address this problem and indeed prove a stronger result over an arbitrary field. Let $\mathcal{P}$ be any poset of cardinality n and let $c_k(\mathcal{P})$ denote the maximum possible cardinality of a union $\bigcup_{i=1}^{k} C_i$ where $C_i \, (1 \leq i \leq k)$ is a chain in $\mathcal{P}$. For each $i \geq 1$, define integers λ_i by $c_k(\mathcal{P}) = \lambda_1 + \cdots + \lambda_k$ for $k \geq 1$. Then it can be shown that the sequence $\lambda_i \, (i \geq 1)$ is weakly decreasing and forms a partition of n (see Britz and Fomin [BF01, Theorem 2.1]).

Definition 1.14 (Greene-Kleitman shape of a poset)**.** The partition λ defined above is called the Greene-Kleitman shape of $\mathcal{P}$.

Denote by $\lambda_{\mathcal{P}}$ the Greene-Kleitman shape of $\mathcal{P}$ and write λ_{h} for $\lambda_{\mathcal{P}_{\mathsf{h}}}$ when $\mathsf{h} \in \mathcal{H}_n$ is a Hessenberg function. A simple algorithm for determining λ_{h} essentially goes back to an early influential paper of Gerstenhaber [Ger61, p. 535] (also see Fenn and Sommers [FS21, Proposition 6.8]). We now describe Gerstenhaber's algorithm.

Given $\mathsf{h} : [n] \to [n]$ define a sequence $I_1, I_2, \ldots$ of disjoint subsets of $[n]$, called *characteristic sequences* as follows. The first element of I_1 is 1. For each $i \in I_1$, the next element in I_1 is the least positive integer j such that $i \prec_{\mathsf{h}} j$ if such a j exists, otherwise i is the last element of I_1. Once $I_1, \ldots, I_{r-1}$ have been constructed, we declare the smallest element of I_r to be the least positive integer in $[n] \setminus \bigcup_{i=1}^{r-1} I_i$. Successive elements in I_r are defined as before: for each $i \in I_r$, the next element in I_r is the least positive integer $j \in [n] \setminus \bigcup_{i=1}^{r-1} I_i$ such that $i \prec_{\mathsf{h}} j$ if such a j exists, otherwise i is the last element of I_r. It is clear that we must have $[n] = \bigcup_{i=1}^{s} I_i$ for some integer s. If $\lambda_i = |I_i|$, then the Greene-Kleitman shape of $\mathcal{P}_{\mathsf{h}}$ is given by $\lambda_{\mathsf{h}} := (\lambda_1, \ldots, \lambda_s)$.

Example 1.15. For the Hessenberg function $\mathsf{h} = (3, 4, 6, 7, 8, 8, 8, 8)$, Gerstenhaber's algorithm yields the characteristic sequences $I_1 = \{1, 4, 8\}$, $I_2 = \{2, 5\}$, $I_3 = \{3, 7\}$, $I_4 = \{6\}$. Therefore $\lambda_{\mathsf{h}} = (3, 2, 2, 1)$.

Remark 1.16. Let λ be a partition of n. One can easily verify that the Greene-Kleitman shape of the poset generated by the Hessenberg function k_λ, defined in (2), is λ'.

Theorem 1.17. *Let $\mathbb{F}$ be an arbitrary field, and let $\mathsf{h} \in \mathcal{H}_n$ be a Hessenberg function with Greene–Kleitman shape λ_{h}. Let $\mathcal{C}_\mu = \mathcal{C}_\mu(\mathbb{F})$ denote the $\mathrm{GL}_n(\mathbb{F})$-conjugacy class of J_μ. Then, for any partition μ of n, we have $\mathcal{C}_\mu \cap \mathfrak{u}_{\mathsf{h}}(\mathbb{F}) \neq \emptyset$ if and only if $\mu \trianglelefteq \lambda_{\mathsf{h}}$.*

We remark that for $\mathbb{F} = \mathbb{C}$, Theorem 1.17 can be deduced from [FS21]. Our proof is completely different and works over a general field. The following assertion is an immediate consequence of Theorems 1.13 and 1.17.

Corollary 1.18. *Let $\mathsf{h} \in \mathcal{H}_n$ be a Hessenberg function and let let μ be a partition of n with $\mu \trianglelefteq \lambda_{\mathsf{h}}$. Then the exact exponent of $q-1$ in $\mathrm{F}_{\mu\mathsf{h}}(q)$ is $n - \ell(\mu)$.*

Remark 1.19. Table 1 contains the explicit computation of $\mathrm{F}_{\mu\mathsf{h}}(q)$ when $\mathsf{h} = (1, 3, 5, 6, 7, 7, 7)$.

μ	$\mathrm{F}_{\mu\mathsf{h}}(q)$
(7)	0
(6, 1)	0
(5, 2)	0
(5, 1, 1)	0
(4, 3)	0
(4, 2, 1)	$(q-1)^4q^9$
(4, 1, 1, 1)	$(q-1)^3q^9$
(3, 3, 1)	$(q-1)^4q^8$
(3, 2, 2)	$(q+1)^2(q-1)^4q^6$
(3, 2, 1, 1)	$(q^3+5q^2+4q+1)(q+1)(q-1)^3q^5$
(3, 1, 1, 1, 1)	$(q+1)^3(q-1)^2q^5$
(2, 2, 2, 1)	$(q^4+6q^3+7q^2+4q+1)(q-1)^3q^3$
(2, 2, 1, 1, 1)	$(q^6+5q^5+10q^4+10q^3+7q^2+3q+1)(q-1)^2q$
(2, 1, 1, 1, 1, 1)	$(q^5+2q^4+4q^3+3q^2+2q+1)(q-1)$
(1, 1, 1, 1, 1, 1, 1)	1

TABLE 1. Calculation of $\mathrm{F}_{\mu\mathsf{h}}$ when $\mathsf{h} = (1, 3, 5, 6, 7, 7, 7)$

Another perspective on $\mathrm{F}_{\mu\mathsf{h}}(q)$ comes from representation theory, specifically via induced representations. Let G be a finite group. For $u \in \mathrm{G}$ the conjugacy class of u and the centralizer of u are denoted by $\mathrm{Cl}_{\mathrm{G}}(u)$ and $\mathrm{Z}_{\mathrm{G}}(u)$, respectively. Then for any subgroup H of G we have

$$\#\{g \in \mathrm{G} : g^{-1}ug \in \mathrm{H}\} = |\mathrm{Cl}_{\mathrm{G}}(u) \cap \mathrm{H}| \cdot |\mathrm{Z}_{\mathrm{G}}(u)|. \tag{21}$$

Now suppose $\mathrm{ch}(\mathrm{Ind}_{\mathrm{H}}^{\mathrm{G}}1)(u)$ denotes the value of the character of the induced representation $\mathrm{Ind}_{\mathrm{H}}^{\mathrm{G}}1$ at $u \in \mathrm{G}$. Then

$$\mathrm{ch}(\mathrm{Ind}_{\mathrm{H}}^{\mathrm{G}}1)(u) = \#\left\{g\mathrm{H} : g^{-1}ug \in \mathrm{H}\right\} = \frac{\#\left\{g \in \mathrm{G} : g^{-1}ug \in \mathrm{H}\right\}}{|\mathrm{H}|} = \frac{|\mathrm{Z}_{\mathrm{G}}(u)|}{|\mathrm{H}|}|\mathrm{Cl}_{\mathrm{G}}(u) \cap \mathrm{H}|. \tag{22}$$

Let

$$\mathsf{U}_{\mathsf{h}} = \mathsf{U}_{\mathsf{h}}(\mathbb{F}_q) \stackrel{\text{def}}{=} \{I + A : A \in \mathfrak{u}_{\mathsf{h}}(\mathbb{F}_q)\} \tag{23}$$

denote the unipotent subgroup of $\mathrm{G} = \mathrm{GL}_n(\mathbb{F}_q)$ associated with h. Let χ_{h} be the character of the induced representation $\mathrm{Ind}_{\mathsf{U}_{\mathsf{h}}}^{\mathrm{G}} \mathbf{1}$. Then from Eqs. (21) and (22) we have:

$$\chi_{\mathsf{h}}(I + J_\mu) = \frac{|\mathrm{Z}_{\mathrm{G}}(I + J_\mu)|}{|\mathsf{U}_{\mathsf{h}}|} \mathrm{F}_{\mu\mathsf{h}}(q). \tag{24}$$

Therefore, understanding $\mathrm{F}_{\mu\mathsf{h}}(q)$ reduces to analyzing the induced representation of the trivial representation from U_{h} to G. This approach was explored by Gagnon, among other things, in his recent and inspiring paper [Gag24]. To describe the connection with Gagnon's work, we need to recall the work of Zelevinsky [Zel81] which introduces a family of Hopf algebras, known as PSH-algebras, that unify the representation theories of S_n and $\mathrm{GL}_n(\mathbb{F}_q)$. In Gagnon's notation, Zelevinsky's construction produces a graded Hopf algebra isomorphism

$$\mathbf{p}_{\{1\}} \colon \mathsf{cf}^{\mathrm{uni}}_{\mathrm{supp}}(\mathrm{GL}_\bullet) \to \mathrm{Sym}, \quad \delta_\lambda \mapsto \tilde{\mathrm{P}}_\lambda(\mathbf{x}; q) := q^{-\mathsf{n}(\lambda)} \mathrm{P}_\lambda(\mathbf{x}; q^{-1}),$$

where Sym is the ring of symmetric functions with coefficients in $\mathbb{Q}(q)$ and $\mathrm{P}_\lambda(\mathbf{x}; t) \stackrel{\mathrm{def}}{=} \mathrm{P}_\lambda(\mathbf{x}; 0, t)$ denotes a Hall-Littlewood function. Using this, Gagnon proves [Gag24, Theorem 3.1] that

$$\chi_{\mathsf{h}} = (q-1)^n \, \mathbf{p}^{-1}_{\{1\}}(X_{\mathsf{G}(\mathsf{h})}(\mathbf{x}; q)).$$

Gagnon then applies this identity to derive a formula for χ_{h} which can be reproved using Theorem 1.13. We remark that $\tilde{\mathrm{P}}_\lambda(\mathbf{x}; q)$ is dual to the modified Hall-Littlewood basis $\tilde{\mathrm{H}}_\lambda(\mathbf{x}; q)$ with respect to the Hall scalar product.

In the rest of this section, we explain the applications of our results on $\mathrm{F}_{\mu\lambda}(q)$ and $\mathrm{F}_{\mu\mathsf{h}}(q)$. Our first application is in counting the number of points on nilpoent Hessenberg varieties. Varieties of this kind have been studied in [Tym06, JP22, PS25]. We recall the definition of a nilpotent Hessenberg variety from [PS25]. In what follows, the conjugate of a Hessenberg function $\mathsf{h} \in \mathcal{H}_n$ is defined by

$$\mathsf{h}' : [n] \to [n], \quad i \mapsto \#\{j \in [n] : \mathsf{h}(j) \geq n + 1 - i\}.$$

Definition 1.20 (Nilpotent Hessenberg variety)**.** Let $\mathsf{h} \in \mathcal{H}_n$ be a Hessenberg function, and let X be a nilpotent $n \times n$ matrix with entries in $\mathbb{F}_q$. Let $\mathscr{F}_n$ denote the set of complete flags in $\mathbb{F}_q^n$. The nilpotent Hessenberg variety associated to h and X is defined by

$$\mathcal{H}\mathrm{ess}_{\mathrm{nil}}(\mathsf{h}, X) \stackrel{\mathrm{def}}{=} \left\{ V_\bullet = (V_i)_{1 \leq i \leq n} \in \mathscr{F}_n : XV_i \subseteq V_{\mathsf{e}(i)} \right\},$$

where $\mathsf{e}(i) := n - \mathsf{h}'(n + 1 - i)$.

Example 1.21. For the Hessenberg function $\mathsf{h} = (2, 3, 5, 5, 5)$ (see Example 1.2), we have $\mathsf{h}' = (3, 3, 4, 5, 5)$ and $\mathsf{e} = (0, 0, 1, 2, 2)$.

Remark 1.22. We have $\#\mathcal{H}\mathrm{ess}_{\mathrm{nil}}(\mathsf{h}, X) = \#\mathcal{H}\mathrm{ess}_{\mathrm{nil}}(\mathsf{h}, g^{-1}Xg)$ for any $g \in \mathrm{GL}_n(\mathbb{F}_q)$. In Definition 1.20, the function e is nondecreasing and satisfies $\mathsf{e}(i) < i$. We refer the reader to [ANR24] for counting the number of $\mathbb{F}_q$-points on a variant of the above Hessenberg varieties, defined by

$$\mathcal{H}\mathrm{ess}(\mathsf{h}, X) \stackrel{\mathrm{def}}{=} \left\{ V_\bullet = (V_i)_{1 \leq i \leq n} \in \mathscr{F}_n : XV_i \subseteq V_{\mathsf{h}(i)} \right\}.$$

Theorem 1.23. *Let $\mathsf{h} \in \mathcal{H}_n$ be a Hessenberg function and let X be a nilpotent $n \times n$ matrix with entries in $\mathbb{F}_q$, with Jordan form J_μ. Then*

$$\#\mathcal{H}\mathrm{ess}_{\mathrm{nil}}(\mathsf{h}, X) = \frac{|\mathrm{Z}_{\mathrm{G}}(I + J_\mu)|}{(q-1)^n q^{\binom{n}{2}}} \mathrm{F}_{\mu\mathsf{h}}(q) = q^{-E_{\mathsf{h}}} \langle \tilde{\mathrm{H}}_\mu(\mathbf{x}; q), X_{\mathsf{G}(\mathsf{h})}(\mathbf{x}; q) \rangle,$$

where E_{h} denotes the number of edges of the graph $\mathsf{G}(\mathsf{h})$.

Our second application concerns another elegant idea introduced by Kirillov. Define

$$a_{nk} \stackrel{\mathrm{def}}{=} \#\left\{ X \in \mathfrak{u}_n(\mathbb{F}_q) \, : \, X^2 = 0 \text{ and } \mathrm{rank}(X) = k \right\}.$$

In [Kir95a], Kirillov discovered the mysterious relation

$$\sum_{k\geq 0} a_{nk}\, q^{k(k-n)}\, H_{n-2k}(w \mid 1/q) = (2w)^n, \tag{25}$$

where the $H_{n-2k}(w \mid 1/q)$ are the Hermite polynomials in the Askey–Wilson scheme. The proof of (25) in [Kir95a] is based on an algebraic structure which Kirillov calls "fusion rules" on the set of coadjoint orbits of $\mathrm{GL}_n(\mathbb{F}_q)$ on $\mathfrak{u}_n(\mathbb{F}_q)$. In [Kir95b], Kirillov writes "This algebra is in fact one of the realizations of the Hall algebra, but we shall not discuss this phenomenon here." In Theorem 1.24 below, we obtain a generalization of Kirillov's mysterious formula (25), and show that it follows easily from the Cauchy identity for Macdonald polynomials combined with (12). Recall that the continuous q-Hermite polynomials $H_n(w \,|\, q)$ are defined recursively by

$$H_{n+1}(w \,|\, q) = 2wH_n(w \,|\, q) + (q^n - 1)H_{n-1}(w \,|\, q), \qquad n \geq 0,$$

with the initial conditions $H_0(w \,|\, q) := 1$ and $H_n(w \,|\, q) := 0$ when $n \leq -1$. One can also show [Ism09, Chapter 13] that

$$H_n(\cos(\theta) \,|\, q) = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q e^{i(n-2k)\theta}. \tag{26}$$

It is a classical result that the $H_n(w \,|\, 1/q)$ form a sequence of orthogonal polynomials, i.e.,

$$\frac{1}{2\pi}\int_0^\pi H_n(\cos(\theta) \,|\, 1/q) H_m(\cos(\theta) \,|\, 1/q)(e^{2i\theta}, e^{-2i\theta}; 1/q)_\infty = \frac{(1/q; 1/q)_n}{(1/q; 1/q)_\infty}\delta_{nm}, \tag{27}$$

where

$$(e^{2i\theta}, e^{-2i\theta}\,; 1/q)_\infty \stackrel{\text{def}}{=} \left(e^{2i\theta}\,; 1/q\right)_\infty \left(e^{-2i\theta}; 1/q\right)_\infty = \prod_{j\geq 0}\left(1 - q^{-j}e^{2i\theta}\right)\prod_{j\geq 0}\left(1 - q^{-j}e^{-2i\theta}\right).$$

From this, one can conclude the following remarkable formula (due to Kirillov) [1]

$$a_{nk} = q^{k(n-k)}\prod_{j=1}^{n-2k}\left(1 - \frac{1}{q^j}\right)^{-1}\frac{1}{2\pi}\int_0^\pi (2\cos(\theta))^n H_{n-2k}(\cos(\theta) \,|\, 1/q)(e^{2i\theta}, e^{-2i\theta}\,; 1/q)_\infty\, d\theta \tag{28}$$

Below we discuss a generalization of (25). To this end, first we introduce some notation. Recall that the Chebyshev polynomials of the first kind $T_n(w)$ are defined by $T_n(\cos(\theta)) = \cos(n\theta)$, which are also derived from the recurrence relation for $n \geq 2$:

$$T_{n+1}(w) = 2wT_n(w) - T_{n-1}(w), \qquad T_0(w) = 1, \quad T_1(w) = w.$$

For a partition $\rho = (\rho_1, \rho_2, \dots)$, define

$$T_\rho(w) \stackrel{\text{def}}{=} T_{\rho_1}(w)T_{\rho_2}(w)\cdots. \tag{29}$$

For any partition $\rho \vdash n$, we set

$$z_\rho \stackrel{\text{def}}{=} \prod_{i\geq 1} i^{m_i}\cdot m_i!\,, \tag{30}$$

where $m_i = m_i(\rho)$ is the number of parts of ρ equal to i. Note that z_ρ is the size of the centralizer of a permutation of type ρ in the symmetric group S_n.

Now let Λ be a composition of n and define

$$a_{nk}(\Lambda) \stackrel{\text{def}}{=} \#\left\{X \in \mathfrak{u}_\Lambda(\mathbb{F}_q) : X^2 = 0, \mathrm{rank}(X) = k\right\}. \tag{31}$$

From Theorem 1.7 it follows that

$$a_{nk}(\Lambda) = a_{nk}(\lambda), \qquad \text{for } \lambda = \mathrm{sort}(\Lambda).$$

[1] From (26) it is probably more natural to express Eqs. (27) and (28) in terms of Roger-Szegő polynomials $h_n(x; q) = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q x^k$. However, we have kept our exposition loyal to Kirillov's articles [Kir95a, Kir95b]. We thank Ole Warnaar for bringing this point to our attention.

Our analysis of $a_{nk}(\lambda)$ starts with the observation that a matrix $X \in \mathfrak{u}_\lambda(\mathbb{F}_q)$, with rank k and $X^2 = 0$, must have a Jordan form of type $\mu = (2^k, 1^{n-2k})$ and so $\mu' = (n-k, k)$, where $0 \le k \le n/2$. Therefore, thanks to (12), we have

$$a_{nk}(\lambda) = (q-1)^n q^{nk-k^2-\mathsf{n}(\lambda')-n} \left(\prod_{1\le i\le s} [\lambda_i]_q! \right) [\mathbf{x}^\lambda] \mathrm{Q}_{\mu'}(\mathbf{x}; 1/q, 0). \tag{32}$$

Evidently we have $a_{nk}(\lambda) = 0$ when $k > n/2$. We prove the following assertion.

Theorem 1.24. *Let $\lambda = (\lambda_1, \dots, \lambda_s)$ be a partition of n. Then*

$$\sum_{k=0}^{n} a_{nk}(\lambda) q^{k(k-n)} H_{n-2k}(w \,|\, 1/q) = [\lambda]_q!(q-1)^n q^{-\mathsf{n}(\lambda')} \prod_{i=1}^{s} \sum_{\rho\vdash\lambda_i} \frac{2^{\ell(\rho)} T_\rho(w)}{z_\rho \prod_{i=1}^{\ell(\rho)} (q^{\rho_i}-1)}. \tag{33}$$

This theorem is indeed a special case of Theorem 8.4. To illustrate this formula we provide two examples.

Example 1.25. Consider $\lambda = (1^n)$. In this case we have $[\lambda]_q! = 1$, $\mathsf{n}(\lambda') = 0$. Moreover for any $1 \le i \le n$, the only partition ρ that contributes in the right hand side of (33) is $\rho = (1)$ which gives $z_\rho = 1$. Then the right hand side of (33) is

$$(q-1)^n \prod_{i=1}^{n} \frac{2T_1(w)}{q-1} = (2T_1(w))^n = (2w)^n.$$

Thus, (25) is a special case of Theorem 1.24.

Example 1.26. Now for $\lambda = (2^l) \vdash n$ we obtain $[\lambda]_q! = (q+1)^l$, $\mathsf{n}(\lambda') = l$. Moreover for any $1 \le i \le l$, the partitions ρ that contributes in the right hand side of (33) are $\rho_1 = (1,1)$ and $\rho_2 = (2)$ which gives $z_{\rho_1} = z_{\rho_2} = 2$. Thus the right hand side of (33) is

$$(q+1)^l(q-1)^{2l}q^{-l} \prod_{i=1}^{l} \left(\frac{4T_1^2(w)}{2(q-1)^2} + \frac{2T_2(w)}{2(q^2-1)} \right) = (q+1)^l(q-1)^{2l}q^{-l} \left(\frac{4qw^2 - q + 1}{(q-1)^2(q+1)} \right)^l.$$

Thus for $\lambda = (2^l) \vdash n$ we obtain

$$\sum_{k\ge 0} a_{nk}(\lambda) q^{k(k-n)} H_{n-2k}(w \,|\, 1/q) = \left(4w^2 - 1 + \frac{1}{q} \right)^l.$$

We can use orthogonality to represent $a_{nk}(\lambda)$ as an integral, similar to (27). This integral can be evaluated using Jacobi's triple product, but we do not discuss this calculation.

Let the $f^\lambda_{\nu\rho}(t)$ be the polynomials defined in [Mac15, Sec. III.3, p. 215] which describe the structural coefficients of products of Hall-Littlewood polynomials. Indeed the $f^\lambda_{\nu\rho}(t)$ are related to the Hall polynomials $g^\lambda_{\nu\rho}(t)$ as given in [Mac15, Eq. (3.6), p. 217]. Using (12), we prove the following combinatorial formula for $a_{nk}(\lambda)$:

Theorem 1.27. *Let λ be a partition of n. Then $a_{nk}(\lambda)$ for $k \le n/2$ is equal to*

$$q^{nk-k^2-\mathsf{n}(\lambda')} \sum_{j=0}^{k} (-1)^j q^{-\binom{j+1}{2}-(n-2k)j} \begin{bmatrix} n-2k+j \\ j \end{bmatrix}_q \frac{q^{n-2k+2j}-1}{q^{n-2k+j}-1} \sum_{\substack{\nu\vdash n-k+j \\ \rho\vdash k-j}} \frac{q^{\mathsf{n}(\nu')+\mathsf{n}(\rho')} f^\lambda_{\nu\rho}(1)}{[\nu]_q!\,[\rho]_q!},$$

where by convention $(q^0-1)/(q^0-1) = 1$.

Note that $f^\lambda_{\nu\rho}(t) = 0$ unless $\nu, \rho \subseteq \lambda$ and $|\nu| + |\rho| = |\lambda|$. When $\lambda = (1^n)$, from the definition of $f^\lambda_{\nu\rho}(t)$ it follows easily that

$$\sum_{\substack{\nu\vdash n-k+j \\ \rho\vdash k-j}} \frac{q^{\mathsf{n}(\nu')+\mathsf{n}(\rho')} f^\lambda_{\nu\rho}(1)}{[\nu]_q!\,[\rho]_q!} = \binom{n}{k-j}. \tag{34}$$

Thus for $\lambda = (1^n)$ the formula in Theorem 1.27 for $a_{nk}(\lambda)$ specializes to

$$a_{nk} = \sum_{j=0}^{k} (-1)^j q^{nk-k^2-\binom{j+1}{2}-(n-2k)j} \begin{bmatrix} n-2k+j \\ j \end{bmatrix}_q \binom{n}{k-j} \frac{q^{n-2k+2j}-1}{q^{n-2k+j}-1}. \tag{35}$$

Now define

$$\mathrm{C}_n = \mathrm{C}_n(q) \stackrel{\text{def}}{=} \#\left\{X \in \mathfrak{u}_n(\mathbb{F}_q) \,:\, X^2 = 0\right\}.$$

Formula (35) has the following immediate consequence:

Corollary 1.28. *For $n \geq 1$ we have*

$$\mathrm{C}_n = \sum_{0 \leq k_1+k_2 \leq \lfloor \frac{n}{2} \rfloor} q^{nk_1-k_1^2} \binom{n}{k_1} (-1)^{k_2} q^{\binom{k_2}{2}} \begin{bmatrix} n-2k_1-k_2 \\ k_2 \end{bmatrix}_q \frac{[n-2k_1]_q}{[n-2k_1-k_2]_q}.$$

The problem of computing C_n was investigated by Kirillov and Melnikov in [KM97], where they conjectured a different formula for C_n that was later proved by Ekhad and Zeilberger [EZ96] (see Theorem 1.29 below). The Ekhad-Zeilberger proof of Theorem 1.29 is very short, but R. Stanley notes in [Sta12, p. 194] that "no conceptual reason is known for such a simple formula". It turns out that Corollary 1.28 sheds some light on the formulas for C_n in Theorem 1.29. Indeed we give a short proof that Corollary 1.28 implies Theorem 1.29, using a formula for Macdonald polynomials $\mathrm{Q}_{(\mu_1,\mu_2)}(\mathbf{x}\,;q,t)$ due to Jing and Józefiak [JJ92, Eq. (1)].

We now give the Kirillov-Melnikov-Ekhad-Zeilberger formula. Let χ_3 be the Dirichlet character defined by

$$\chi_3(n) := \begin{cases} 0 & n \equiv 0 \pmod 3; \\ 1 & n \equiv 1 \pmod 3; \\ -1 & n \equiv 2 \pmod 3. \end{cases} \tag{36}$$

Theorem 1.29. *(Ekhad–Zeilberger) For $n \geq 1$ we have*

$$\begin{aligned} \mathrm{C}_{2n+1} &= \sum_{j=0}^{n} \binom{2n+1}{n-j} \left[\chi_3(j+1) q^{\frac{1}{3}\binom{2j+1}{2}} - \chi_3(j) q^{\frac{1}{3}\binom{2j+2}{2}}\right] q^{n^2+n-j^2-j} \\ &= \sum_{j \in \mathbb{Z}} \left[\binom{2n+1}{n-3j} - \binom{2n+1}{n-3j-1}\right] q^{n^2+n-3j^2-2j}, \end{aligned}$$

and

$$\begin{aligned} \mathrm{C}_{2n} &= \binom{2n}{n} q^{n^2} + \sum_{j=1}^{n} \binom{2n}{n-j} \left[\chi_3(2j+1) q^{\frac{1}{3}\binom{2j}{2}} - \chi_3(2j-1) q^{\frac{1}{3}\binom{2j+1}{2}}\right] q^{n^2-j^2} \\ &= \sum_{j \in \mathbb{Z}} \left[\binom{2n}{n-3j} - \binom{2n}{n-3j-1}\right] q^{n^2-3j^2-2j}. \end{aligned}$$

Finding the polynomials $\mathrm{F}_{\mu\lambda}(q)$ based on Theorem 1.10 can be a highly time consuming computation. Here we explain an alternative and significantly more efficient method for this computation that is based on a result of Ram and Schlosser [RS23]. To this end, we first introduce two functions $r(T)$ and $\mathrm{s}(T)$ of a semistandard Young tableau T ($r(T)$ was initially introduced by Prasad and Ram [PR23] for standard Young tableaux). Henceforth, the set of semistandard Young tableaux of shape μ and content λ will be denoted by $\mathrm{SSYT}(\mu, \lambda)$.

Definition 1.30. Suppose $T \in \mathrm{SSYT}(\mu, \lambda)$. For each $i \geq 1$, let $\beta_j^i = \beta_j^i(T)$ denote the number of occurrences of the integer i in the jth row of T for $j \geq 1$. Define $\beta^i = \beta^i(T) := (\beta_1^i, \beta_2^i, \ldots)$ for $i \geq 1$.

Note that β^i is not necessarily a partition, but it is a composition. Also, we have $|\beta^i| = \lambda_i$.

Definition 1.31. Given $T \in \mathrm{SSYT}(\mu, \lambda)$, let $\beta^i = \beta^i(T)$ and define

$$s_q(T) \stackrel{\text{def}}{=} \prod_{i\geq 1} \begin{bmatrix} \lambda_i \\ \beta^i \end{bmatrix}_q \qquad \text{where} \qquad \begin{bmatrix} \lambda_i \\ \beta^i \end{bmatrix}_q \stackrel{\text{def}}{=} \frac{[\lambda_i]_q!}{\prod_{j\geq 1}[\beta^i_j]_q!}\,.$$

Definition 1.32. Suppose $T \in \mathrm{SSYT}(\mu, \lambda)$. Let b_{ij} denote the entry in row i and column j of T. For each box of T in the (i, j) position where $i \geq 2$, we define $r_{ij}(T)$ to be

$$r_{ij}(T) \stackrel{\text{def}}{=} \#\{j' : j' \geq j \text{ and } b_{i-1,j'} < b_{ij}\}. \tag{37}$$

Now we set

$$r_q(T) \stackrel{\text{def}}{=} \prod_{\substack{(i,j)\in T \\ i\geq 2}} [r_{ij}(T)]_q.$$

We illustrate these definitions by the following example.

Example 1.33. Consider the semistandard tableau T below. The table on the right describes $r_{ij}(T)$ as defined in (37):

$$T = \begin{array}{|c|c|c|}\hline 1&1&2\\\hline 2&3&3\\\hline 4\\\cline{1-1}\end{array} \qquad r_{ij}(T): \begin{array}{|c|c|c|}\hline &&\\\hline 2&2&1\\\hline 3\\\cline{1-1}\end{array} \tag{38}$$

Thus $r_q(T) = [2]_q^2[3]_q = (1+q)^2(1+q+q^2)$. Moreover, for this tableau T, we have $\beta^1 = (2, 0, 0)$, $\beta^2 = (1, 1, 0)$, $\beta^3 = (0, 2, 0)$ and $\beta^4 = (0, 0, 1)$, which yield $s_q(T) = [2]_q = q + 1$.

Ram and Schlosser observed the following relation [RS23, Definition 3.7 and Proposition 5.1]. For $T \in \mathrm{SSYT}(\mu', \lambda)$ we have

$$s_q(T)r_q(T) = \prod_{j=1}^{s} \theta_{\mu^j/\mu^{j-1}}(q),$$

where $\emptyset = \mu^0 \subseteq \mu^1 \subseteq \cdots \subseteq \mu^s = \mu'$ is a chain of partitions such that every μ^j/μ^{j-1} is a horizontal λ_j-strip. As elements of $\mathcal{F}(\mu', \lambda)$ are in one-to-one correspondence with elements of $\mathrm{SSYT}(\mu', \lambda)$, from the definition of $b_{\mu'\lambda}(q)$ we infer that

$$b_{\mu'\lambda}(q) = \sum_{T\in \mathrm{SSYT}(\mu',\lambda)} s_q(T)r_q(T). \tag{39}$$

The next example explains how to compute $\mathrm{F}_{\mu\lambda}(q)$ using Theorem 1.7 combined with (39).

Example 1.34. Let $\mu = (3, 2, 2)$ and $\lambda = (2, 2, 2, 1)$. Then $\binom{n}{2} + \ell(\mu) - \mathsf{n}(\mu) - n = 11$ and $n - \ell(\mu) = 4$. There are only three tableaux of shape μ' and content λ:

$$T_1 = \begin{array}{|c|c|c|}\hline 1&1&2\\\hline 2&3&3\\\hline 4\\\cline{1-1}\end{array} \qquad T_2 = \begin{array}{|c|c|c|}\hline 1&1&2\\\hline 2&3&4\\\hline 3\\\cline{1-1}\end{array} \qquad T_3 = \begin{array}{|c|c|c|}\hline 1&1&3\\\hline 2&2&4\\\hline 3\\\cline{1-1}\end{array}$$

Hence $s_q(T_1)r_q(T_1) = [2]_q^3[3]_q$, $s_q(T_2)r_q(T_2) = [2]_q^4$ and $s_q(T_3)r_q(T_3) = [2]_q^3$. From (39) we obtain:

$$b_{\mu'\lambda}(q) = (q+1)^3(q^2+2q+3).$$

This, in combination with the first part of Theorem 1.7, gives:

$$\mathrm{F}_{\mu\lambda}(q) = (q-1)^4 q^{11}\,(1+1/q)^3\,\left(3+2/q+1/q^2\right) = (q-1)^4q^6(q+1)^3(3q^2+2q+1).$$

Remark 1.35. When $\lambda = (2^k)$ and $\mu = (k, k)$, we obtain $\mathrm{K}_{\mu'\lambda} = 1$. The same argument as in Example 1.34 can also be used to show (13). Moreover, using the same approach as in Example 1.34, one can show that for $\mu = \lambda = (k^k)$, we have:

$$\mathrm{F}_{\mu\lambda}(q) = (q-1)^{k^2-k} q^{(k^2-k-1)\binom{k}{2}}\,\left([k]_q!\right)^{k-1}.$$

For arbitrary partitions λ and μ, from (7), we know the exact exponent of q and $q-1$ in $\mathrm{F}_{\mu\lambda}(q)$. Table 2 shows that $\mathrm{F}_{\mu\lambda}(q)$ for $\lambda=(2,2,2,2)$ is a product of powers of q, $q\pm1$, q-integers and another polynomial that is irreducible over $\mathbb{Q}$. It is tempting to conjecture that this holds in general, and in particular to compute the exponent of $q+1$.

μ	$\mathrm{F}_{\mu\lambda}(q)$
(4^2)	$(q-1)^6q^{15}(q+1)^3$
$(4,3,1)$	$(q-1)^5q^{13}(q+1)^4(3q+1)$
$(4,2^2)$	$(q-1)^5q^{12}(q+1)^4(2q+1)$
$(4,2,1^2)$	$(q-1)^4q^{11}(q+1)^4(3q^2+2q+1)$
$(4,1^4)$	$(q-1)^3q^{11}(q+1)^4$
$(3^2,2)$	$(q-1)^5q^{10}(q+1)^4(3q^2+2q+1)$
$(3^2,1^2)$	$(q-1)^4q^{10}(q+1)^2(6q^4+9q^3+9q^2+3q+1)$
$(3,2^2,1)$	$(q-1)^4q^7(q+1)^4(7q^4+7q^3+7q^2+2q+1)$
$(3,2,1^3)$	$(q-1)^3q^6(q+1)^3(8q^5+9q^4+10q^3+6q^2+2q+1)$
$(3,1^5)$	$(q-1)^2q^6(q+1)^3(3q^2+1)$
(2^4)	$(q-1)^4q^6(q+1)^2(3q^2+q+1)(q^2+q+1)$
$(2^3,1^2)$	$(q-1)^3q^3(q+1)^3(6q^6+6q^5+10q^4+5q^3+5q^2+q+1)$
$(2^2,1^4)$	$(q-1)^2q(q+1)(6q^8+11q^7+16q^6+15q^5+14q^4+8q^3+5q^2+2q+1)$
$(2,1^6)$	$(q-1)(q+1)^2(3q^4+2q^2+1)$
(1^8)	1

TABLE 2. Calculation of $\mathrm{F}_{\mu\lambda}$ when $\lambda=(2,2,2,2)$

When $\lambda=(1^n)$, Yip [Yip18] obtained several interesting closed formulas of $\mathrm{F}_{\mu\lambda}(q)$, including for the case $\mu=(k+1,1^{n-k-1})$. Theorem 1.36 below generalizes Yip's results. Consider an arbitrary partition $\lambda=(\lambda_1,\dots,\lambda_s)$ of n and a hook partition $\mu=(k+1,1^{n-k-1})$ where $k\le s-1$ (the latter condition is equivalent to $\lambda\trianglelefteq\mu'$). Recall that $[n]_{1/q}=q^{1-n}[n]_q$.

Theorem 1.36. *Let $\lambda=(\lambda_1,\dots,\lambda_s)$ be a partition of n and let $\mu=(k+1,1^{n-k-1})$ with $k\le s-1$. Then*

$$\mathrm{F}_{\mu\lambda}(q)=(q-1)^kq^{\binom{n}{2}-\binom{n-k}{2}-k}\sum_{j=2}^{s-k+1}[\lambda_1+\cdots+\lambda_{j-1}]_{1/q}\sum_{\substack{A\subseteq\{2,\dots,s\}\\|A|=k\\\min A=j}}\prod_{i\in A}[\lambda_i]_{1/q}\,. \tag{40}$$

Example 1.37. We illustrate some special cases of Theorem 1.36. For $\lambda=(1^n)$, from Theorem 1.36 and the identity $\binom{n}{k}+\binom{n-1}{k}+\cdots+\binom{k}{k}=\binom{n+1}{k+1}$, we obtain

$$\begin{aligned}\mathrm{F}_{\mu\lambda}(q)&=(q-1)^kq^{\binom{n}{2}-\binom{n-k}{2}-k}\sum_{j=2}^{n-k+1}[j-1]_{1/q}\binom{n-j}{k-1}\\&=(q-1)^kq^{\binom{n-1}{2}-\binom{n-k-1}{2}}\sum_{j=1}^{n-k}\binom{n-j}{k}q^{-(j-1)}.\end{aligned} \tag{41}$$

When $\lambda = (2^l) \vdash n = 2l$ with $k \leq l-1$, a similar argument to the one above gives the identity:

$$\begin{aligned} \mathrm{F}_{\mu\lambda}(q) &= (q-1)^k q^{\binom{n}{2}-\binom{n-k}{2}-k} \sum_{j=2}^{l-k+1} [2(j-1)]_{1/q} \binom{l-j}{k-1} ([2]_{1/q})^k \\ &= (q-1)^k (q+1)^k q^{\binom{n-1}{2}-\binom{n-k-1}{2}-k} \sum_{j=1}^{2(l-k)} \binom{l-\lceil j/2 \rceil}{k} q^{-(j-1)}, \end{aligned} \tag{42}$$

where $\lceil x \rceil$ is the ceiling function.

We end this section by stating the last result of this paper. For a partition μ of n, as before we denote the nilpotent Jordan matrix of type μ by J_μ.

Theorem 1.38. *Let h_1 and h_2 be two Hessenberg functions on $[n]$, and let $\mathsf{U}_{\mathsf{h}_1}$ and $\mathsf{U}_{\mathsf{h}_2}$ be the unipotent subgroups of $\mathrm{G} = \mathrm{GL}_n(\mathbb{F}_q)$ associated with h_1 and h_2 as defined in (23). Then*

$$|\mathsf{U}_{\mathsf{h}_1} \backslash \mathrm{G} / \mathsf{U}_{\mathsf{h}_2}| = q^{|\mathsf{h}_1|+|\mathsf{h}_2|-2n^2} \sum_{\mu \vdash n} |\mathrm{Z}_{\mathrm{G}}(I + J_\mu)| \, \mathrm{F}_{\mu \mathsf{h}_1}(q) \, \mathrm{F}_{\mu \mathsf{h}_2}(q), \tag{43}$$

where $\mathrm{Z}_{\mathrm{G}}(I + J_\mu)$ is the centralizer of $I + J_\mu$ in G.

From (17) we infer that $|\mathrm{Z}_{\mathrm{G}}(I + J_\mu)|$ is a polynomial in q of degree $n + 2\mathsf{n}(\mu)$. Recall that $\mathsf{U}_{k_\Lambda} = \mathsf{U}_\Lambda$ where k_Λ is the Hessenberg function defined in (2). Moreover, for a composition Λ of n, one can easily verify that

$$n^2 - |k_\Lambda| = \binom{n}{2} - \mathsf{n}(\lambda'), \tag{44}$$

where $\lambda = \mathrm{sort}(\Lambda)$. Thus, for $h_1 = k_\Lambda$ and $h_2 = k_\Gamma$ where Λ and Γ are compositions of n, from Theorem 1.38, the equality (44), the second part of Theorem 1.7, and the well-known fact that $\mathrm{K}_{\mu\Lambda} = \mathrm{K}_{\mu\lambda}$ [Loe18, Theorem 9.25], we obtain

Corollary 1.39. *Let Λ and Γ be two compositions of n. Set $\lambda = \mathrm{sort}(\Lambda)$ and $\gamma = \mathrm{sort}(\Gamma)$. Then $|\mathsf{U}_\Lambda \backslash \mathrm{G} / \mathsf{U}_\Gamma|$ is a polynomial in q of degree $n + \mathsf{n}(\lambda') + \mathsf{n}(\gamma')$, with leading coefficient $\sum_\mu \mathrm{K}_{\mu'\lambda} \mathrm{K}_{\mu'\gamma}$.*

Remark 1.40. This corollary has an interesting combinatorial interpretation. First note that when $\lambda = (1^n)$ we have $\mathrm{K}_{\mu'\lambda} = \mathrm{K}_{\mu\lambda}$ and in this case $\mathrm{K}_{\mu\lambda}$ is the number of standard Young tableaux of shape μ, denoted by f^μ. Moreover, when $\lambda = \gamma = (1^n)$, from Bruhat decomposition we have $\mathrm{GL}_n(\mathbb{F}_q) = \sqcup_{t \in T} \sqcup_{w \in \mathrm{S}_n} UtwU$ where $\mathsf{U} = \mathsf{U}_\lambda(\mathbb{F}_q)$ and T is the subgroup of diagonal matrices of G. From this we get $|\mathsf{U} \backslash G / \mathsf{U}| = n!(q-1)^n$. Thus from Corollary 1.39 we obtain the well-known formula $n! = \sum_{\mu \vdash n} (f^\mu)^2$.

§ 2. The division algorithm

The goal of this section is to describe a parabolic extension of the *division algorithm*, originally introduced in [Bor97, Kir95b] in the special case of $\lambda = (1^n)$, which will be used in Section 3 to find a recursive formula for $\mathrm{F}_{\mu\lambda}$.

Throughout this section, we consider matrices over an arbitrary field $\mathbb{F}$, and we use $\mathcal{C}_\mu = \mathcal{C}_\mu(\mathbb{F})$ to denote the $\mathrm{GL}_n(\mathbb{F})$-conjugacy class of J_μ. We abbreviate $\mathfrak{u}_\Lambda(\mathbb{F})$ to $\mathfrak{u}_\Lambda$.

2.1. Properties of the rank sequence.

We fix an integer $n \geq 2$.

Definition 2.1 (Eligible partition and admissible matrix)**.** Let $1 \leq m \leq n$, and let ν be a partition of $n - m$. A partition μ of n is called (ν, m)-eligible, if there exists an $(n-m) \times m$ matrix Z, such that

$$\left(\begin{array}{c|c} J_\nu & \mathrm{Z} \\ \hline 0_{m \times (n-m)} & 0_{m \times m} \end{array} \right) \in \mathcal{C}_\mu. \tag{45}$$

The set of all (ν, m)-eligible partitions will be denoted by $\mathrm{H}_m(\nu)$. Moreover a matrix Z satisfying (45) will be called a (μ, ν, m)-admissible matrix.

Definition 2.2 (Rank sequence)**.** Let Z be a (μ,ν,m)-admissible matrix with

$$A := \left(\begin{array}{c|c} J_\nu & \mathrm{Z} \\ \hline 0_{m\times(n-m)} & 0_{m\times m}\end{array}\right) \in \mathcal{C}_\mu. \tag{46}$$

The rank sequence of Z is defined by $\mathfrak{r}(\mathrm{Z}) = (\omega_j)_{j\geq 0}$ where

$$\omega_j = \omega_j(\mu,\nu,\mathrm{Z}) \stackrel{\text{def}}{=} \begin{cases} m & j=0, \\ \mathrm{rank}(A^j) - \mathrm{rank}(J_\nu^j) & j\geq 1. \end{cases} \tag{47}$$

For any $j \geq 1$ we have

$$A^j = \left(\begin{array}{c|c} J_\nu^j & J_\nu^{j-1}\mathrm{Z} \\ \hline 0_{m\times(n-m)} & 0_{m\times m}\end{array}\right), \tag{48}$$

and, following [Yip18], we can describe $J_\nu^{j-1}\mathrm{Z}$ combinatorially in terms of Z and ν as follows. Let $z_1,\dots,z_{n-m}$ be the rows of Z from top to bottom. We put these vectors, from top right to bottom left, in the boxes of the Young diagram of ν. To obtain the rows of $J_\nu^{j-1}\mathrm{Z}$ we shift the entries in each row of ν to the right by $j-1$ boxes (the vectors that fall out are eliminated and we write the zero vector in the boxes that become vacant). Henceforth the matrix whose rows are the ones that appear in the j'th column of ν after $j-1$ shifts will be denoted by Z_j.

Example 2.3. Let $\nu = (4,3,1) \vdash 8$ and consider the following diagrams filled with vectors:

z_4	z_3	z_2	z_1
z_7	z_6	z_5	
z_8			

,

0	z_4	z_3	z_2
0	z_7	z_6	
0			

,

0	0	z_4	z_3
0	0	z_7	
0			

0	0	0	z_4
0	0	0	
0			

The first diagram says that the rows of Z are $z_1,\dots,z_8$, while the second diagram indicates that the rows of $J_\nu\mathrm{Z}$ are $z_2, z_3, z_4, 0, z_6, z_7, 0, 0$ and so on. We also have

$$\mathrm{Z}_1 = \begin{pmatrix} z_4 \\ z_7 \\ z_8 \end{pmatrix}, \qquad \mathrm{Z}_2 = \begin{pmatrix} z_4 \\ z_7 \end{pmatrix}, \qquad \mathrm{Z}_3 = \begin{pmatrix} z_4 \\ z_7 \end{pmatrix}, \qquad \mathrm{Z}_4 = \begin{pmatrix} z_4 \end{pmatrix}.$$

From nilpotence of J_ν and (48) it follows that

$$\omega_j = \mathrm{rank}(A^j) - \mathrm{rank}(J_\nu^j) = \mathrm{rank}(\mathrm{Z}_j), \qquad j \geq 1. \tag{49}$$

The following description of Z_j also emerges from Example 2.3.

Lemma 2.4. *Let $\nu = (\nu_1,\dots,\nu_k)$ be a partition of $n-m$ with the conjugate $\nu' = (\nu'_1,\dots,\nu'_\ell)$ and let* Z *be a (μ,ν,m)-admissible matrix. Then for $1 \leq j \leq \ell$ we have*

$$\mathrm{Z}_j = \begin{pmatrix} z_{\varepsilon(1)} \\ \vdots \\ z_{\varepsilon(\nu'_j)} \end{pmatrix}, \tag{50}$$

where $\varepsilon(l) := \sum_{i\leq l}\nu_i$.

Definition 2.5 (Essential and free vectors)**.** Let Z be a (μ,ν,m)-admissible matrix. We call the rows $\{z_{\varepsilon(1)},\dots,z_{\varepsilon(k)}\}$ of Z the *essential vectors*, and its other rows the *free vectors*.

Remark 2.6. Lemma 2.4 in particular implies that Z_1 has k rows and Z_ℓ has ν'_ℓ rows. Moreover the number of new rows in Z_j that were not in Z_{j+1} is $\nu'_j - \nu'_{j+1}$.

The proof of the following lemma from [Yip18, Lemma 2] is straightforward.

Lemma 2.7. *Let $\nu = (\nu_1,\dots,\nu_k)$ be a partition of n with conjugate $\nu' = (\nu'_1,\dots,\nu'_\ell)$. Then*

$$\mathrm{rank}\left(J_\nu^j\right) = \begin{cases} \sum_{i=j+1}^{\ell}\nu'_i & 1\leq j\leq \ell-1, \\ 0 & j\geq \ell. \end{cases} \tag{51}$$

From this we deduce the following lemma.

Lemma 2.8. *Let ν be a partition of $n-m$ with $\ell = \ell(\nu') = \nu_1$, and let $(\omega_j)_{j\geq 0}$ be the sequence defined in (47). The following statements hold:*

(1) $\omega_{\ell+1} = 0$.

(2) $0 \leq \cdots \leq \omega_{j+1} \leq \omega_j \leq \cdots \leq \omega_1 \leq \omega_0 = m$.

(3) $0 \leq \omega_j - \omega_{j+1} \leq \nu'_j - \nu'_{j+1}$ *for* $j \geq 1$.

(4) $\mu'_j = \nu'_j + (\omega_{j-1} - \omega_j)$ *for* $j \geq 1$.

(5) $\sum_{j\geq 1}(\omega_{j-1} - \omega_j) = m$.

Proof. Note that $\ell = \nu_1$ and since $J_\nu^\ell = J_\nu^{\nu_1} = 0$, we obtain $A^{\ell+1} = 0$ which proves the first statement. For the second statement, note that $\omega_j = \operatorname{rank}(\mathrm{Z}_j)$ and the set of rows of Z_{j+1} is a subset of the set of rows of Z_j. Moreover, since Z_1 has m columns, $\omega_1 \leq m$. For the third statement, observe that the matrix Z_j^T has ν'_j columns and thus its nullity is $\mathrm{Nul}_j := \nu'_j - \omega_j$. Since the columns of Z_{j+1}^T form a subset of the columns of Z_j^T, we have $\mathrm{Nul}_j \geq \mathrm{Nul}_{j+1}$. For the fourth one, note that by Lemma 2.7 we have

$$\operatorname{rank}(A^j) = \sum_{i\geq j+1} \mu'_j = n - \sum_{i\leq j} \mu'_i, \qquad \operatorname{rank}(J_\nu^j) = \sum_{j\geq j+1} \nu'_j = n - m - \sum_{i\leq j} \nu'_i.$$

Thus for $j \geq 1$ we have $\omega_j = m - \sum_{i\leq j}(\mu'_i - \nu'_i)$, or equivalently $\sum_{i\leq j}(\mu'_i - \nu'_i) = \omega_0 - \omega_j$. Taking the difference of two such relations for successive values of j yields the fourth statement. Finally, the last assertion follows from the first and the second ones. □

Lemma 2.9. *Let μ and ν be partitions of n and $n-m$, respectively. Set $\ell = \ell(\nu') = \nu_1$, and let $(\omega_j)_{j\geq 0}$ satisfy conditions (1)–(4) of Lemma 2.8. Then, there exist a (μ, ν, m)-admissible matrix* Z *whose rank sequence is $(\omega_j)_{j\geq 0}$.*

Proof. We construct an $(n-m)\times m$ matrix Z with row vectors $\{z_1, \ldots, z_{n-m}\}$ in $\mathbb{F}^m$ such that $\operatorname{rank}(\mathrm{Z}_j) = \omega_j$ for $1 \leq j \leq \ell$, where Z_j is as in (50). The value of $\operatorname{rank}(\mathrm{Z}_j)$ is determined by the essential vectors that occur in Z_j.

We construct the essential vectors inductively, starting with those that occur in Z_ℓ. By condition (3) of Lemma 2.9, we have $\omega_\ell \leq \nu'_\ell$, and by Remark 2.6, the number of rows in Z_ℓ is ν'_ℓ. Thus, we can select ν'_ℓ rows in $\mathbb{F}^m$ for Z_ℓ such that $\operatorname{rank}(\mathrm{Z}_\ell) = \omega_\ell$. To construct Z_j from Z_{j+1} with $\operatorname{rank}(\mathrm{Z}_j) = \omega_j$, observe from Remark 2.6 that the number of new essential vectors in Z_j not appearing in Z_{j+1} is $\nu'_j - \nu'_{j+1}$, which by condition (4) of Lemma 2.9 is at least $\omega_j - \omega_{j+1}$. Hence, we can select new essential vectors in Z_j, linearly independent of those in Z_{j+1}, such that their span has dimension $\omega_j - \omega_{j+1}$ and so we obtain Z_j with $\operatorname{rank}(\mathrm{Z}_j) = \omega_j$. Now construct Z by choosing the free vectors arbitrarily. This completes the construction of Z. From Lemma 2.7 and the assumption $\mu'_j = \nu'_j + (\omega_{j-1} - \omega_j)$ for $j \geq 1$, we deduce that $A \in \mathcal{C}_\mu$. □

Example 2.10. Let $\mu = (4,4,2,1,1,1)$, $\nu = (4,3,1)$, and $\omega_0 = 5$, $\omega_1 = 2$, $\omega_2 = \omega_3 = 1$, $\omega_4 = 0$, with $n = 13$ and $m = 5$. The essential vectors are $\{z_4, z_7, z_8\}$ as in Example 2.3. In this example we can pick $z_4 = 0$ and two linearly independent vectors z_7, z_8. The rest of vectors are arbitrary.

We are now ready to state the main theorem of this section.

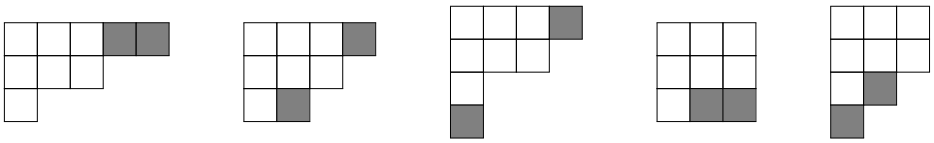
Theorem 2.11 (Division algorithm)**.** *Let $1 \leq m \leq n$ and let ν be a partition of $n-m$. Then $\mu \in \mathrm{H}_m(\nu)$ if and only if μ'/ν' is a horizontal strip of size m.*

Proof. Assume $\mu \in \mathrm{H}_m(\nu)$. From parts (4) and (5) of Lemma 2.8 it follows that μ' is obtained by adding m boxes into the Young diagram of ν'. From parts (1) and (3) of Lemma 2.8 it follows that no two boxes are in the same row of μ. Thus, μ'/ν' is a horizontal strip of size m.

Conversely, assume that μ'/ν' is a horizontal strip of size m and let $\ell = \ell(\nu')$. Then $\ell(\mu') \leq \ell + 1$ since otherwise μ'/ν' cannot be a horizontal strip. Choose non-negative integers α_j for

$1 \le j \le \ell+1$ with $\mu'_j = \nu'_j + \alpha_j$ and $\alpha_1 + \cdots + \alpha_{\ell+1} = m$. Since μ'/ν' is a horizontal strip, we have $\alpha_{j+1} \le \nu'_j - \nu'_{j+1}$. Define $\omega_0 = m$ and $\omega_j = \omega_{j-1} - \alpha_j$ for $j \ge 1$. It is straightforward to verify that the $(\omega_j)_{j\ge 0}$ satisfy conditions (1)–(5) of Lemma 2.8. Hence, by Lemma 2.9 there exists a corresponding (μ, ν, m)-admissible matrix Z, which proves that $\mu \in \mathrm{H}_m(\nu)$. □

Example 2.12. Let $m = 2$ and let $\nu = (3, 2, 2)$ with the conjugate $\nu' = (3, 3, 1)$. Then $\mathrm{H}_m(\nu)$ has five elements and the following Young diagrams are the conjugates of $\mu \in \mathrm{H}_m(\nu)$



2.2. Non-emptiness of $\mathcal{C}_\mu \cap \mathfrak{u}_\Lambda$. Given a composition $\Lambda = (\Lambda_1, \ldots, \Lambda_s)$ on n, the partition of n obtained by sorting Λ will be denoted by $\lambda = (\lambda_1, \ldots, \lambda_s)$. We define $\Lambda^* = (\Lambda_1, \ldots, \Lambda_{s-1})$ and set $\lambda^* = \mathrm{sort}(\Lambda^*)$. We can write any $X \in \mathfrak{u}_\Lambda$ as

$$X = \left(\begin{array}{c|c} X_1 & Y \\ \hline 0_{\Lambda_s \times (n-\Lambda_s)} & 0_{\Lambda_s \times \Lambda_s} \end{array}\right), \tag{52}$$

where X_1 is a square matrix of size $(n - \Lambda_s)$. The main goal of this section is to prove the following assertion.

Proposition 2.13. *Let $\mu = (\mu_1, \ldots, \mu_k)$ be a partition of n and let $\Lambda = (\Lambda_1, \ldots, \Lambda_s)$ be a composition of n with $\lambda = \mathrm{sort}(\Lambda)$. Then $\mathcal{C}_\mu \cap \mathfrak{u}_\Lambda$ is non-empty if and only if $\lambda \trianglelefteq \mu'$.*

Proof. Suppose that $X \in \mathcal{C}_\mu \cap \mathfrak{u}_\Lambda$. Write X as in (52) and let J_ν be the Jordan form of X_1. Therefore, $\nu \vdash n - \Lambda_s$ and $X_1 \in \mathcal{C}_\nu \cap \mathfrak{u}_{\Lambda^*}$ where $\Lambda^* = (\Lambda_1, \ldots, \Lambda_{s-1})$. Now pick $g \in \mathrm{GL}_{n-\Lambda_s}(\mathbb{F})$ such that $gX_1g^{-1} = J_\nu$ and set $\tilde{g} := \mathrm{diag}(g, \mathrm{I}_{\Lambda_s})$. Then $\tilde{g}X\tilde{g}^{-1} \in \mathcal{C}_\mu$ and thus $\mu \in \mathrm{H}_{\Lambda_s}(\nu)$. Theorem 2.11 implies that μ'/ν' is a horizontal Λ_s-strip. By an iterative application of this process, we obtain a sequence of partitions $\emptyset = \mu^0 \subseteq \cdots \subseteq \mu^{s-2} \subseteq \mu^{s-1} \subseteq \mu^s = \mu$ such that $(\mu^j/\mu^{j-1})'$ is a horizontal Λ_j-strip. Equivalently, we obtain a semistandard Young tableau of shape μ' and content Λ. Thus $\mathrm{K}_{\mu'\Lambda} = \mathrm{K}_{\mu'\lambda} \neq 0$ which implies $\lambda \trianglelefteq \mu'$.

Conversely, assume $\lambda \trianglelefteq \mu'$. Then there exists a semistandard Young tableau of shape μ' and content Λ (see [FH91, Exercise A.11, p. 457]). Therefore there exists a sequence of partitions $\emptyset = \mu^0 \subseteq \cdots \subseteq \mu^{s-2} \subseteq \mu^{s-1} \subseteq \mu^s = \mu$ such that $(\mu^j/\mu^{j-1})'$ is a horizontal Λ_j-strip. Starting from a $\Lambda_1 \times \Lambda_1$ zero matrix and successively using Theorem 2.11 for $(\mu^j/\mu^{j-1})'$, we can enlarge the matrix inductively to construct an element of $\mathcal{C}_\mu \cap \mathfrak{u}_\Lambda$. □

Let $X \in \mathcal{C}_\mu \cap \mathfrak{u}_\Lambda$. We can write X as in (52). As before, let J_ν be the Jordan form of X_1, where $\nu \vdash (n - \Lambda_s)$ and $X_1 \in \mathcal{C}_\nu \cap \mathfrak{u}_{\Lambda^*}$ (in particular, $\lambda^* \trianglelefteq \nu'$). If $g_{X_1} \in \mathrm{GL}_{n-\Lambda_s}(\mathbb{F})$ is chosen such that $gXg^{-1} = J_\nu$, then

$$\tilde{g}X\tilde{g}^{-1} = \left(\begin{array}{c|c} J_\nu & gY \\ \hline 0_{\Lambda_s \times (n-\Lambda_s)} & 0_{\Lambda_s \times \Lambda_s} \end{array}\right) \in \mathcal{C}_\mu, \qquad \text{for } \tilde{g} := \mathrm{diag}(g_{X_1}, \mathrm{I}_{\Lambda_s}). \tag{53}$$

Therefore, $\mu \in \mathrm{H}_{\Lambda_s}(\nu)$ and $g_{X_1}Y$ is a (μ, ν, Λ_s)-admissible matrix. Thus, we obtain a map

$$\pi : \mathcal{C}_\mu \cap \mathfrak{u}_\Lambda \longrightarrow \bigsqcup_{\substack{\nu \vdash n-\Lambda_s \\ \lambda^* \trianglelefteq \nu' \\ \mu \in \mathrm{H}_{\Lambda_s}(\nu)}} (\mathcal{C}_\nu \cap \mathfrak{u}_{\Lambda^*}), \quad \pi(X) = X_1. \tag{54}$$

Lemma 2.14. *The map π defined in (54) is surjective. Moreover for any $X_1 \in \mathcal{C}_\nu \cap \mathfrak{u}_{\Lambda^*}$, there is a bijective correspondence between the fiber $\pi^{-1}(X_1)$ and the set of (μ, ν, Λ_s)-admissible matrices, given by $Y \mapsto g_{X_1}Y$.*

Proof. We first show that π is surjective. Let $X_1 \in \mathcal{C}_\nu \cap \mathfrak{u}_{\Lambda^*}$. Since $\mu \in \mathrm{H}_{\Lambda_s}(\nu)$, there exists also a (μ, ν, Λ_s)-admissible matrix Z satisfying

$$\left(\begin{array}{c|c} J_\nu & \mathrm{Z} \\ \hline 0_{\Lambda_s \times (n-\Lambda_s)} & 0_{\Lambda_s \times \Lambda_s} \end{array}\right) \in \mathcal{C}_\mu. \tag{55}$$

Now define

$$X := \left(\begin{array}{c|c} X_1 & g_{X_1}^{-1}\mathrm{Z} \\ \hline 0_{\Lambda_s \times (n-\Lambda_s)} & 0_{\Lambda_s \times \Lambda_s} \end{array}\right) \in \mathfrak{u}_\Lambda.$$

From (55) we have

$$\tilde{g} X \tilde{g}^{-1} = \left(\begin{array}{c|c} J_\nu & \mathrm{Z} \\ \hline 0_{\Lambda_s \times (n-\Lambda_s)} & 0_{\Lambda_s \times \Lambda_s} \end{array}\right) \in \mathcal{C}_\mu, \qquad \tilde{g} := \mathrm{diag}(g_{X_1}, \mathrm{I}_{\Lambda_s}). \tag{56}$$

This shows $X \in \mathcal{C}_\mu \cap \mathfrak{u}_\Lambda$ and $\pi(X) = X_1$. From this argument we deduce that the map

$$\begin{aligned} \{(\mu, \nu, \Lambda_s)\text{-admissible matrices}\} &\to \pi^{-1}(X_1) \\ \mathrm{Z} &\longmapsto \left(\begin{array}{c|c} X_1 & g_{X_1}^{-1}\mathrm{Z} \\ \hline 0_{\Lambda_s \times (n-\Lambda_s)} & 0_{\Lambda_s \times \Lambda_s} \end{array}\right) \end{aligned}$$

is a bijection, which proves the second part of the lemma. □

§ 3. A recursion for $\mathrm{F}_{\mu\Lambda}(q)$

From now on we assume that $\mathbb{F} = \mathbb{F}_q$. The goal of this section is to provide a recursive formula for $\mathrm{F}_{\mu\Lambda}(q)$. Because of Proposition 2.13 we will assume $\lambda \trianglelefteq \mu'$, where $\lambda = \mathrm{sort}(\Lambda)$. The following theorem is an immediate consequence of Lemma 2.14.

Theorem 3.1. *Let $\Lambda = (\Lambda_1, \dots, \Lambda_s)$ be a composition of n and set $\lambda = \mathrm{sort}(\Lambda)$. Additionally, let $\Lambda^* = (\Lambda_1, \dots, \Lambda_{s-1})$ and set $\lambda^* = \mathrm{sort}(\Lambda^*)$. For any partition μ of n with $\lambda \trianglelefteq \mu'$, we have*

$$\mathrm{F}_{\mu\Lambda}(q) = \sum_{\substack{\nu \vdash n - \Lambda_s \\ \lambda^* \trianglelefteq \nu' \\ \mu \in \mathrm{H}_{\Lambda_s}(\nu)}} \mathrm{G}^*(\mu, \nu, \Lambda_s)\, \mathrm{F}_{\nu\Lambda^*}(q), \tag{57}$$

where $\mathrm{G}^*(\mu, \nu, \Lambda_s)$ is the number of (μ, ν, Λ_s) admissible matrices.

We can also introduce bottom-up notions of eligible partitions and admissible matrices as in Lemma 2.14, but working with matrices of the form

$$\left(\begin{array}{c|c} 0_{\Lambda_1 \times \Lambda_1} & \mathrm{Z} \\ \hline 0_{(n-\Lambda_1) \times \Lambda_1} & J_\nu \end{array}\right).$$

In this fashion, we obtain the following variant of Theorem 3.1: suppose that $\Lambda_* = (\Lambda_2, \dots, \Lambda_s)$ and $\lambda_* = \mathrm{sort}(\Lambda_*)$. Then

$$\mathrm{F}_{\mu\Lambda}(q) = \sum_{\substack{\nu \vdash n - \Lambda_1 \\ \lambda_* \trianglelefteq \nu' \\ \mu \in \mathrm{H}_{\Lambda_1}(\nu)}} \mathrm{G}_*(\mu, \nu, \Lambda_1)\, \mathrm{F}_{\nu\Lambda_*}(q), \tag{58}$$

where $\mathrm{G}_*(\mu, \nu, \Lambda_1)$ the number of bottom-up (μ, ν, Λ_1)-admissible matrices. It is straightforward to verify that $\mathrm{G}_*(\mu, \nu, |\mu| - |\nu|) = \mathrm{G}^*(\mu, \nu, |\mu| - |\nu|)$.

3.1. Symmetry of $\mathrm{F}_{\mu\Lambda}(q)$. As before we assume $\Lambda = (\Lambda_1, \dots, \Lambda_s)$ is a composition of $n \geq 2$ with $\lambda = \mathrm{sort}(\Lambda)$. The goal of this section is to prove the result below.

Proposition 3.2. *Let μ be a partition of n. Then $\mathrm{F}_{\mu\Lambda}(q) = \mathrm{F}_{\mu\lambda}(q)$.*

Proof. We prove this by induction on $s = \ell(\Lambda)$. When $s = 1$, the statement is trivial by Remark 1.8. When $s = 2$, let $\Lambda = (\Lambda_1, \Lambda_2)$ be a composition of n with $\Lambda_1 \leq \Lambda_2$ so that $\lambda = (\Lambda_2, \Lambda_1)$. Let $X \in \mathfrak{u}_\Lambda$ be a matrix of rank k. Since $X^2 = 0$, the Jordan type of such a matrix is $\mu = (2^k, 1^{n-2k})$ with $\mu' = (n-k, k)$. For this partition, $\mathrm{F}_{\mu\Lambda}(q)$ is the number of $\Lambda_1 \times \Lambda_2$ matrices of rank k, which is clearly the same as the number of $\Lambda_2 \times \Lambda_1$ matrices of rank k. Therefore

$$\mathrm{F}_{\mu\Lambda}(q) = \mathrm{F}_{\mu\lambda}(q), \qquad \Lambda = (\Lambda_1, \Lambda_2). \tag{59}$$

Now assume $s \geq 3$. Define $\Lambda^{\#} = (\lambda^*, \Lambda_s)$ and $\Lambda_{\#} = (\Lambda_1, \lambda_*)$ where λ^* and λ_* are defined in Eqs. (57) and (58). From Eqs. (57) and (58) and the induction hypothesis we deduce

$$\mathrm{F}_{\mu\Lambda}(q) = \mathrm{F}_{\mu\Lambda^{\#}}(q) = \mathrm{F}_{\mu\Lambda_{\#}}(q). \tag{60}$$

But we have $\lambda = ((\Lambda_{\#})^{\#})_{\#}$, hence $\mathrm{F}_{\mu\Lambda}(q) = \mathrm{F}_{\mu\lambda}(q)$. □

3.2. Counting admissible matrices. For $0 \leq r \leq l$, set

$$\mathrm{C}_l^r \stackrel{\mathrm{def}}{=} \left(q^l - 1\right)\left(q^l - q\right)\cdots\left(q^l - q^{r-1}\right) = q^{\binom{r}{2}}(q-1)^r\,[r]_q!\begin{bmatrix} l \\ r \end{bmatrix}_q. \tag{61}$$

Proposition 3.3. *Let $1 \leq m \leq n$, and let ν be a partition of $n - m$ with $\ell = \ell(\nu')$. Suppose $\mu \in \mathrm{H}_m(\nu)$. The total number of (μ, ν, m)-admissible matrices is given by*

$$\mathrm{G}^*(\mu, \nu, m) = q^{m(n-m-\nu_1')} \prod_{j=1}^{\ell} q^{\omega_{j+1}(\nu_j' - \nu_{j+1}')} \begin{bmatrix} m - \omega_{j+1} \\ \omega_j - \omega_{j+1} \end{bmatrix}_q \mathrm{C}_{\nu_j' - \nu_{j+1}'}^{\omega_j - \omega_{j+1}}, \tag{62}$$

where $\omega_0 = m$, $\omega_{\ell+1} = 0$ and $\mu_j' = \nu_j' + (\omega_{j-1} - \omega_j)$ for $1 \leq j \leq \ell + 1$.

Proof. By Lemma 2.8 an $(n-m) \times n$ matrix Z is (μ, ν, m)-admissible if and only if (49) holds. There are $n - m - \nu_1'$ free vectors in Z (see Definition 2.5), and there are $q^{m(n-m-\nu_1')}$ ways to fill their entries. Thus, counting the number of admissible matrices boils down to counting the number of (essential) vectors $z_{\varepsilon(1)}, z_{\varepsilon(2)}, \dots, z_{\varepsilon(\nu_1')} \in \mathbb{F}_q^m$ such that for the corresponding Z we have $\mathrm{rank}(\mathrm{Z}_j) = \omega_j$ for $1 \leq j \leq \nu_1'$. The statement follows from setting $n_i = \nu_i'$ and $r_i = \omega_i$ in Lemma 11.2. □

§ 4. Proof of Theorem 1.7: first and second parts

Let λ and μ be partitions of n with $\lambda \trianglelefteq \mu'$. As before we set $s = \ell(\lambda)$ and $\lambda^* = (\lambda_1, \dots, \lambda_{s-1})$. By Proposition 2.13 the intersection $\mathcal{C}_\lambda \cap \mathfrak{u}_\lambda$ is non-empty and from Theorem 3.1 we obtain

$$\mathrm{F}_{\mu\lambda}(q) = \sum_{\substack{\nu \vdash n - \lambda_s \\ \lambda^* \trianglelefteq \nu' \\ \mu \in \mathrm{H}_{\lambda_s}(\nu)}} \mathrm{G}^*(\mu, \nu, \lambda_s)\,\mathrm{F}_{\nu\lambda^*}(q). \tag{63}$$

Setting $m = \lambda_s$ and $n = |\mu|$ in Proposition 3.3 we obtain

$$\mathrm{G}^*(\mu, \nu, \lambda_s) = q^{\lambda_s(|\mu| - \lambda_s - \nu_1')} \prod_{j=1}^{\ell} q^{\omega_{j+1}(\nu_j' - \nu_{j+1}')} \begin{bmatrix} \lambda_s - \omega_{j+1} \\ \omega_j - \omega_{j+1} \end{bmatrix}_q \mathrm{C}_{\nu_j' - \nu_{j+1}'}^{\omega_j - \omega_{j+1}}, \tag{64}$$

where $\ell = \ell(\nu')$ and $\mu_j' = \nu_j' + (\omega_{j-1} - \omega_j)$, for $1 \leq j \leq \ell + 1$, with $\omega_0 = \lambda_s$ and $\nu_{\ell+1}' = \omega_{\ell+1} = 0$. We now simplify $\mathrm{G}^*(\mu, \nu, \lambda_s)$ by relating it to $\theta_{\mu'/\nu'}(q)$, defined in (6).

Lemma 4.1. *Let ν be a partition of $n-\lambda_s$ with $\ell=\ell(\nu')$. Then for any $\mu\in \mathrm{H}_{\lambda_s}(\nu)$ we have*

$$\mathrm{G}^*(\mu,\nu,\lambda_s)=q^{\alpha(\mu,\nu,\lambda_s)}\,(q-1)^{\beta(\mu,\nu,\lambda_s)}\theta_{\mu'/\nu'}(q),$$

where

$$\begin{aligned}\alpha(\mu,\nu,\lambda_s)&=\lambda_s(|\mu|-\lambda_s-\nu_1')+\sum_{j=1}^{\ell}\binom{\mu_{j+1}'-\nu_{j+1}'}{2}+\sum_{j=1}^{\ell}\omega_{j+1}(\nu_j'-\nu_{j+1}'),\\ \beta(\mu,\nu,\lambda_s)&=\lambda_s-(\mu_1'-\nu_1').\end{aligned}\tag{65}$$

Proof. From the definition of C_l^r given in (61) and using the fact that $\omega_j-\omega_{j+1}=\mu_{j+1}'-\nu_{j+1}'$, one obtains

$$\mathrm{C}_{\nu_j'-\nu_{j+1}'}^{\omega_j-\omega_{j+1}}=q^{\binom{\mu_{j+1}'-\nu_{j+1}'}{2}}\,(q-1)^{\mu_{j+1}'-\nu_{j+1}'}\cdot[\omega_j-\omega_{j+1}]_q!\cdot\begin{bmatrix}\nu_j'-\nu_{j+1}'\\ \mu_{j+1}'-\nu_{j+1}'\end{bmatrix}_q.\tag{66}$$

Furthermore, using $\lambda_s-\omega_1=\mu_1'-\nu_1'$, we find

$$\prod_{j=1}^{\ell}\begin{bmatrix}\lambda_s-\omega_{j+1}\\ \omega_j-\omega_{j+1}\end{bmatrix}_q[\omega_j-\omega_{j+1}]_q!\begin{bmatrix}\nu_j'-\nu_{j+1}'\\ \mu_{j+1}'-\nu_{j+1}'\end{bmatrix}_q=\frac{[\lambda_s]_q!}{[\lambda_s-\omega_1]_q!}\cdot\prod_{j=1}^{\ell}\begin{bmatrix}\nu_j'-\nu_{j+1}'\\ \mu_{j+1}'-\nu_{j+1}'\end{bmatrix}_q=\theta_{\mu'/\nu'}(q).$$

Using this and Eqs. (64) and (66), we obtain

$$\begin{aligned}\mathrm{G}^*(\mu,\nu,\lambda_s)&=q^{\lambda_s(|\mu|-\lambda_s-\nu_1')}\prod_{j=1}^{\ell}q^{\omega_{j+1}(\nu_j'-\nu_{j+1}')}\begin{bmatrix}\lambda_s-\omega_{j+1}\\ \omega_j-\omega_{j+1}\end{bmatrix}_q\mathrm{C}_{\nu_j'-\nu_{j+1}'}^{\omega_j-\omega_{j+1}}\\ &=q^{\lambda_s(|\mu|-\lambda_s-\nu_1')+\sum_{j=1}^{\ell}\binom{\mu_{j+1}'-\nu_{j+1}'}{2}+\sum_{j=1}^{\ell}\omega_{j+1}(\nu_j'-\nu_{j+1}')}\,(q-1)^{\sum_{j=1}^{\ell}(\mu_{j+1}'-\nu_{j+1}')}\,\theta_{\mu'/\nu'}(q).\end{aligned}\tag{67}$$

Observe that

$$\sum_{i=1}^{\ell}(\mu_{j+1}'-\nu_{j+1}')=(n-\mu_1')-(n-\lambda_s-\nu_1')=\lambda_s-(\mu_1'-\nu_1').$$

Combining this with (67) proves the statement. □

Recall that a polynomial $f(q)$ of degree N is called palindromic if $q^Nf(1/q)=f(q)$.

Lemma 4.2. *Let ν be a partition of $n-\lambda_s$ with $\ell=\ell(\nu')$. Then for any $\mu\in\mathrm{H}_{\lambda_s}(\nu)$ the polynomial $\theta_{\mu'/\nu'}(q)$ is palindromic of degree*

$$\gamma(\mu,\nu,\lambda_s)\overset{\text{def}}{=}\binom{\lambda_s}{2}-\binom{\mu_1'-\nu_1'}{2}+\sum_{j=1}^{\ell}(\nu_j'-\mu_{j+1}')(\mu_{j+1}'-\nu_{j+1}').$$

Proof. Since $[n]_q$ is palindromic, it follows immediately that $\theta_{\mu'/\nu'}(q)$ is palindromic. Next note that $\deg_q([n]_q!)=\binom{n}{2}$ and also

$$\deg_q\left(\begin{bmatrix}n\\ k\end{bmatrix}_q\right)=\binom{n}{2}-\binom{k}{2}-\binom{n-k}{2}=k(n-k).\tag{68}$$

The definition of $\theta_{\mu'/\nu'}(q)$ and (68) prove the lemma. □

Lemma 4.3. *Let ν be a partition of $n-\lambda_s$. Then for any $\mu\in\mathrm{H}_{\lambda_s}(\nu)$ we have*

$$\alpha(\mu,\nu,\lambda_s)+\gamma(\mu,\nu,\lambda_s)=\lambda_s|\mu|-\lambda_s^2+\mathsf{n}(\nu)-\mathsf{n}(\mu)-\lambda_s+\ell(\mu)-\ell(\nu)+\binom{\lambda_s}{2}.$$

Proof. Recall that $\mu'_{j+1}-\nu'_{j+1}=\omega_j-\omega_{j+1}$ and $\omega_0=\lambda_s$. Thus, $\omega_{j+1}=\lambda_s-\sum_{i=1}^{j+1}(\mu'_i-\nu'_i)$. Set $\ell=\ell(\nu')$. By applying summation by parts, and the fact that $\nu_{\ell+1}=0$, we obtain

$$\begin{aligned}\sum_{j=1}^{\ell}\omega_{j+1}(\nu'_j-\nu'_{j+1}) &= \lambda_s\nu'_1-\sum_{j=1}^{\ell}\sum_{i=1}^{j+1}(\mu'_i-\nu'_i)(\nu'_j-\nu'_{j+1})\\ &= \lambda_s\nu'_1-\nu'_1(\mu'_1-\nu'_1)-\sum_{j=1}^{\ell}\nu'_j(\mu'_{j+1}-\nu'_{j+1}) \qquad (69)\\ &= \lambda_s\nu'_1-\nu'_1(\mu'_1-\nu'_1)-\sum_{j=1}^{\ell}(\nu'_j-\mu'_{j+1})(\mu'_{j+1}-\nu'_{j+1})-\sum_{j=1}^{\ell}\mu'_{j+1}(\mu'_{j+1}-\nu'_{j+1}).\end{aligned}$$

We now simplify $-\sum_{j=1}^{\ell}\mu'_{j+1}(\mu'_{j+1}-\nu'_{j+1})$ using the identity

$$-ab=\binom{b-a}{2}-\binom{b}{2}-\binom{a}{2}-a.$$

Set $b=\mu'_{j+1}$ and $a=\mu'_{j+1}-\nu'_{j+1}$. Then

$$\begin{aligned}-\sum_{j=1}^{\ell}\mu'_{j+1}(\mu'_{j+1}-\nu'_{j+1}) &= \sum_{j=1}^{\ell}\binom{\nu'_{j+1}}{2}-\sum_{j=1}^{\ell}\binom{\mu'_{j+1}}{2}-\sum_{j=1}^{\ell}\binom{\mu'_{j+1}-\nu'_{j+1}}{2}-\sum_{j=1}^{\ell}(\mu'_{j+1}-\nu'_{j+1})\\ &= \mathsf{n}(\nu)-\mathsf{n}(\mu)-\binom{\nu'_1}{2}+\binom{\mu'_1}{2}-\lambda_s+\mu'_1-\nu'_1-\sum_{j=1}^{\ell}\binom{\mu'_{j+1}-\nu'_{j+1}}{2}. \qquad (70)\end{aligned}$$

From now on, the proof is a straightforward simplification. From the definition of $\alpha(\mu,\nu,\lambda_s)$ in (65), Lemma 4.2, and Eqs. (69) and (70) we obtain

$$\begin{aligned}\alpha(\mu,\nu,\lambda_s)+\gamma(\mu,\nu,\lambda_s) &= \lambda_s|\mu|-\lambda_s^2+\mathsf{n}(\nu)-\mathsf{n}(\mu)-\lambda_s+\binom{\lambda_s}{2}\\ &\quad-\nu'_1(\mu'_1-\nu'_1)-\binom{\nu'_1}{2}+\binom{\mu'_1}{2}-\binom{\mu'_1-\nu'_1}{2}+\mu'_1-\nu'_1.\end{aligned}$$

But

$$-\nu'_1(\mu'_1-\nu'_1)-\binom{\nu'_1}{2}+\binom{\mu'_1}{2}-\binom{\mu'_1-\nu'_1}{2}+\mu'_1-\nu'_1=\mu'_1-\nu'_1=\ell(\mu)-\ell(\nu),$$

and this completes the proof of the lemma. □

Proposition 4.4. *Let λ and μ be two partitions of n with $\lambda\trianglelefteq\mu'$. Set $s=\ell(\lambda)$. Furthermore, let $\mathcal{F}':=\mathcal{F}'(\mu,\lambda)$ denote the set of all chains of partitions $\emptyset=\mu^0\subseteq\mu^1\subseteq\cdots\subseteq\mu^s=\mu$ such that $(\mu^j/\mu^{j-1})'$ is a horizontal λ_j-strip. Then*

$$\mathrm{F}_{\mu\lambda}(q)=(q-1)^{n-\ell(\mu)}\sum_{\mathcal{F}'}q^{\sum_{j=1}^{s}\alpha(\mu^j,\mu^{j-1},\lambda_j)+\gamma(\mu^j,\mu^{j-1},\lambda_j)}\prod_{j=1}^{s}\theta_{(\mu^j/\mu^{j-1})'}(1/q). \qquad (71)$$

Proof. This follows from iterative substitution of the $\mathrm{F}_{\nu\lambda^*}$ and the explicit formula for $\mathrm{G}^*(\mu,\nu,\lambda_s)$ given in Lemma 4.1 in the recursive relation (63). The condition $\lambda^*\trianglelefteq\nu'$ can be dropped because it is equivalent to $\mathrm{F}_{\nu\lambda^*}(q)\neq 0$. The condition $\mu\in\mathrm{H}_{\lambda_s}(\nu)$ states that μ'/ν' is a horizontal λ_s-strip. Recall that

$$\beta(\mu^j,\mu^{j-1},\lambda_j)=\lambda_j-\left((\mu^j)'_1-(\mu^{j-1})'_1\right).$$

Therefore

$$\sum_{j=1}^{s}\beta(\mu^j,\mu^{j-1},\lambda_j)=\sum_{j=1}^{s}\left(\lambda_j-\left((\mu^j)'_1-(\mu^{j-1})'_1\right)\right)=n-(\mu^s)'_1=n-\mu'_1=n-\ell(\mu).$$

Finally, Lemma 4.2 allows the substitution of $\theta_{(\mu^j/\mu^{j-1})'}(q)$ by $q^{\gamma(\mu^j,\mu^{j-1},\lambda_j)}\theta_{(\mu^j/\mu^{j-1})'}(1/q)$. □

Proposition 4.5. *Let $\emptyset = \mu^0 \subseteq \mu^1 \subseteq \cdots \subseteq \mu^s = \mu$ be a chain of partitions such that $(\mu^j/\mu^{j-1})'$ is a horizontal λ_j-strip. Then*

$$\sum_{j=1}^{s} \big(\alpha(\mu^j, \mu^{j-1}, \lambda_j) + \gamma(\mu^j, \mu^{j-1}, \lambda_j)\big) = \binom{n}{2} + \ell(\mu) - \mathsf{n}(\mu) - n,$$

where $\mathsf{n}(\mu)$ is defined in (5).

Proof. By Lemma 4.3 we have

$$\begin{aligned}\alpha(\mu^j, \mu^{j-1}, \lambda_j) + \gamma(\mu^j, \mu^{j-1}, \lambda_j) &= \lambda_j(\lambda_1 + \cdots + \lambda_j) \\ &\quad - \binom{\lambda_j}{2} - 2\lambda_j + \mathsf{n}(\mu^{j-1}) - \mathsf{n}(\mu^j) + \ell(\mu^j) - \ell(\mu^{j-1}).\end{aligned}$$

But we have

$$\sum_{j=1}^{s} \lambda_j(\lambda_1 + \cdots + \lambda_j) = \frac{1}{2}\left(\left(\sum_{j=1}^{s} \lambda_j\right)^2 + \sum_{j=1}^{s} \lambda_j^2\right) = \binom{n+1}{2} + \mathsf{n}(\lambda').$$

It follows that

$$\begin{aligned}\sum_{j=1}^{s} \big(\alpha(\mu^j, \mu^{j-1}, \lambda_j) + \gamma(\mu^j, \mu^{j-1}, \lambda_j)\big) &= \sum_{j=1}^{s} \lambda_j(\lambda_1 + \cdots + \lambda_j) - \mathsf{n}(\lambda') - 2n - \mathsf{n}(\mu) + \ell(\mu) \\ &= \binom{n}{2} - n + \ell(\mu) - \mathsf{n}(\mu).\end{aligned}$$

□

Parts (1) and (2) of Theorem 1.7 readily follow from Proposition 4.4 and Proposition 4.5. For Part (2) note that

$$\deg_q\left(\mathrm{F}_{\mu\lambda}(q)\right) = n - \ell(\mu) + \binom{n}{2} + \ell(\mu) - \mathsf{n}(\mu) - n = \binom{n}{2} - \mathsf{n}(\mu).$$

From Remark 1.6 we conclude that the leading coefficient of $\mathrm{F}_{\mu\lambda}(q)$ is $\mathrm{K}_{\mu'\lambda}$.

§ 5. Macdonald polynomials and the proof of Theorem 1.10

The goal of this section is to prove Theorem 1.10. We use the same notation introduced in Definition 1.5 and Definition 1.9. We begin by describing the relation between $\psi_{\eta/\rho}(q)$ and $\theta_{\eta/\rho}(q)$.

Lemma 5.1. *Let η/ρ be a horizontal strip. Then*

$$\theta_{\eta/\rho}(q) = \left(\prod_{i\geq 1} \frac{[\rho_i - \rho_{i+1}]_q!}{[\eta_i - \eta_{i+1}]_q!}\right) [|\eta| - |\rho|]_q!\ \psi_{\eta/\rho}(q).$$

Proof. We have

$$\begin{bmatrix}\eta_i - \eta_{i+1} \\ \eta_i - \rho_i\end{bmatrix}_q = \frac{[\eta_i - \eta_{i+1}]_q!\,[\eta_{i+1} - \rho_{i+1}]_q!}{[\rho_i - \rho_{i+1}]_q!\,[\eta_i - \rho_i]_q!} \begin{bmatrix}\rho_i - \rho_{i+1} \\ \eta_{i+1} - \rho_{i+1}\end{bmatrix}_q.$$

This implies

$$\begin{aligned}[|\eta| - |\rho|]_q!\,\psi_{\eta/\rho}(q) &= \left(\prod_{i\geq 1} \frac{[\eta_i - \eta_{i+1}]_q!}{[\rho_i - \rho_{i+1}]_q!}\right) \left(\prod_{i\geq 1} \frac{[\eta_{i+1} - \rho_{i+1}]_q!}{[\eta_i - \rho_i]_q!}\right) [|\eta| - |\rho|]_q! \prod_{i\geq 1} \begin{bmatrix}\rho_i - \rho_{i+1} \\ \eta_{i+1} - \rho_{i+1}\end{bmatrix}_q \\ &= \left(\prod_{i\geq 1} \frac{[\eta_i - \eta_{i+1}]_q!}{[\rho_i - \rho_{i+1}]_q!}\right) \frac{[|\eta| - |\rho|]_q!}{[\eta_1 - \rho_1]_q!} \prod_{i\geq 1} \begin{bmatrix}\rho_i - \rho_{i+1} \\ \eta_{i+1} - \rho_{i+1}\end{bmatrix}_q \\ &= \left(\prod_{i\geq 1} \frac{[\eta_i - \eta_{i+1}]_q!}{[\rho_i - \rho_{i+1}]_q!}\right) \theta_{\eta/\rho}(q),\end{aligned}$$

which proves the claim. □

An immediate consequence of Lemma 5.1 is the following proposition.

Proposition 5.2. *Let λ and μ be two partitions with $\lambda \trianglelefteq \mu'$. Then*

$$b_{\mu'\lambda}(q) = \left(\prod_{i\geq 1} \frac{[\lambda_i]_q!}{[\mu'_i - \mu'_{i+1}]_q!} \right) a_{\mu'\lambda}(q).$$

Proof. Let $\emptyset = \mu^0 \subseteq \mu^1 \subseteq \cdots \subseteq \mu^s = \mu'$ be a chain of partitions such that each μ^j/μ^{j-1} is a horizontal λ_j-strip. Then from Lemma 5.1 we obtain:

$$\begin{aligned}
\prod_{j=1}^{s} \theta_{\mu^j/\mu^{j-1}}(q) &= \prod_{j=1}^{s} \left(\left(\prod_{i\geq 1} \frac{[\mu_i^{j-1} - \mu_{i+1}^{j-1}]_q!}{[\mu_i^j - \mu_{i+1}^j]_q!} \right) [|\mu^j| - |\mu^{j-1}|]_q! \, \psi_{\mu^j/\mu^{j-1}}(q) \right) \\
&= \prod_{i\geq 1} \prod_{j=1}^{s} \frac{[\mu_i^{j-1} - \mu_{i+1}^{j-1}]_q!}{[\mu_i^j - \mu_{i+1}^j]_q!} \prod_{j=1}^{s} [|\mu^j| - |\mu^{j-1}|]_q! \prod_{j=1}^{s} \psi_{\mu^j/\mu^{j-1}}(q) \\
&= \prod_{i\geq 1} \frac{[\mu_i^0 - \mu_{i+1}^0]_q!}{[\mu_i^s - \mu_{i+1}^s]_q!} \prod_{j=1}^{s} [\lambda_j]_q! \prod_{j=1}^{s} \psi_{\mu^j/\mu^{j-1}}(q) \\
&= \prod_{i\geq 1} \frac{[\lambda_i]_q!}{[\mu'_i - \mu'_{i+1}]_q!} \prod_{j=1}^{s} \psi_{\mu^j/\mu^{j-1}}(q).
\end{aligned} \tag{72}$$

Hence

$$b_{\mu'\lambda}(q) = \sum_{\mathcal{F}} \prod_{j=1}^{s} \theta_{\mu^j/\mu^{j-1}}(q) = \prod_{i\geq 1} \frac{[\lambda_i]_q!}{[\mu'_i - \mu'_{i+1}]_q!} \sum_{\mathcal{F}} \prod_{j=1}^{s} \psi_{\mu^j/\mu^{j-1}}(q) = \left(\prod_{i\geq 1} \frac{[\lambda_i]_q!}{[\mu'_i - \mu'_{i+1}]_q!} \right) a_{\mu'\lambda}(q)\,,$$

where the sum is taken over the set $\mathcal{F} = \mathcal{F}(\mu', \lambda)$ of chains $\emptyset = \mu^0 \subseteq \mu^1 \subseteq \cdots \subseteq \mu^s = \mu'$ of partitions such that μ^j/μ^{j-1} is a horizontal λ_j-strip. □

With these tools at our disposal, we can now prove Theorem 1.10. As before, we work with fixed partitions λ and μ of n with $\lambda \trianglelefteq \mu'$. Note that $\ell(\mu') \leq \ell(\lambda)$ since we assume $\lambda \trianglelefteq \mu'$. Thanks to (9), we have

$$\mathrm{P}_{\mu'}(\mathbf{x}; q, 0) = \sum_{\nu} a_{\mu'\nu}(q) m_\nu(\mathbf{x}), \tag{73}$$

Proof of (11). From $[n]_{1/q}! = q^{-\binom{n}{2}} [n]_q!$, we deduce

$$b_{\mu'\lambda}\,(1/q) = q^{\sum_{i\geq 1} \left(\binom{\mu'_i - \mu'_{i+1}}{2} - \binom{\lambda_i}{2} \right)} \left(\prod_{i\geq 1} \frac{[\lambda_i]_q!}{[\mu'_i - \mu'_{i+1}]_q!} \right) a_{\mu'\lambda}\,(1/q)\,. \tag{74}$$

From (73) we have $a_{\mu'\lambda}\,(1/q) = [\mathbf{x}^\lambda] \mathrm{P}_{\mu'}(x; 1/q, 0)$ and so from (74) and the first part of Theorem 1.7 we obtain

$$\mathrm{F}_{\mu\lambda}(q) = (q-1)^{n-\ell(\mu)} q^{\sum_{i\geq 1} \binom{\mu'_i - \mu'_{i+1}}{2} + \binom{n}{2} + \ell(\mu) - \mathsf{n}(\mu) - \mathsf{n}(\lambda') - n} \prod_{i\geq 1} \frac{[\lambda_i]_q!}{[\mu'_i - \mu'_{i+1}]_q!} [\mathbf{x}^\lambda] \mathrm{P}_{\mu'}(\mathbf{x}; 1/q, 0).$$

This finishes proof of (11). □

In the rest of this section we prove (12). Recall that the q-Pochhammer symbol, also called the q-shifted factorial, is defined for each integer $n \geq 0$ by

$$(a; q)_n \stackrel{\text{def}}{=} (1-a)(1-aq)\cdots(1-aq^{n-1}), \qquad (a; q)_0 \stackrel{\text{def}}{=} 1.$$

Let $\langle \cdot, \cdot \rangle_{q,t}$ denote the q,t-scalar product on symmetric functions [Mac15, Sec. VI.1, p. 306], so that

$$\langle p_\lambda, p_\mu \rangle_{q,t} = \delta_{\lambda\mu} z_\lambda \prod_{i=1}^{\ell(\lambda)} \frac{1-q^{\lambda_i}}{1-t^{\lambda_i}},$$

where the p_λ are the power sum symmetric functions and z_λ is defined in (30). For any partition μ of n, we define

$$b_\mu(q,t) \stackrel{\text{def}}{=} \langle \mathrm{P}_\mu(\mathbf{x};q,t), \mathrm{P}_\mu(\mathbf{x};q,t) \rangle_{q,t}^{-1}.$$

Then $\mathrm{Q}_\mu(\mathbf{x};q,t)$ is defined by

$$\mathrm{Q}_\mu(\mathbf{x};q,t) = b_\mu(q,t)\mathrm{P}_\mu(\mathbf{x};q,t).$$

Obviously we have $\langle \mathrm{P}_\lambda, \mathrm{Q}_\mu \rangle_{q,t} = \delta_{\lambda\mu}$. For each square $s=(i,j)$ in the diagram of a partition μ, we define its arm-length as $a(s) := \mu_i - j$ and its leg-length as $l(s) := \mu'_j - i$. Note that $a(s)$ and $l(s)$ represent the numbers of squares in the diagram of μ to the east and to the south of the square s, respectively. One can show [Mac15, p. 339, Eq. (6.19)] that

$$b_\mu(q,t) = \prod_{s\in\mu} \frac{1-q^{a(s)}t^{l(s)+1}}{1-q^{a(s)+1}t^{l(s)}}. \tag{75}$$

In order to prove (12), we need to express $\mathrm{Q}_\mu(\mathbf{x};1/q,0)$ in terms of $\mathrm{P}_\mu(\mathbf{x};1/q,0)$.

Lemma 5.3. *Let μ be a partition of n. Then*

$$\frac{(q-1)^{\mu_1}}{q^{\mu_1+\sum_{j\geq 1}\binom{\mu_j-\mu_{j+1}}{2}}}\mathrm{Q}_\mu(\mathbf{x};1/q,0) = \prod_{j\geq 1}\frac{1}{[\mu_j-\mu_{j+1}]_q!}\mathrm{P}_\mu(\mathbf{x};1/q,0). \tag{76}$$

Proof. From (75) we infer that $b_\mu(q,0) = \prod_{\substack{s\in\mu \\ l(s)=0}} \frac{1}{1-q^{a(s)+1}} = \prod_{j\geq 1}\frac{1}{(q;q)_{\mu_j-\mu_{j+1}}}$. This combined with the identity $(1/q\,;1/q)_n = \prod_{j=1}^n \left(1-\frac{1}{q^j}\right) = \frac{(q-1)^n}{q^{n(n+1)/2}}[n]_q!$ gives us

$$b_\mu(1/q,0) = \prod_{j\geq 1}\frac{q^{\mu_j-\mu_{j+1}+\binom{\mu_j-\mu_{j+1}}{2}}}{(q-1)^{\mu_j-\mu_{j+1}}[\mu_j-\mu_{j+1}]_q!} = \frac{q^{\mu_1+\sum_{j\geq 1}\binom{\mu_j-\mu_{j+1}}{2}}}{(q-1)^{\mu_1}}\prod_{j\geq 1}\frac{1}{[\mu_j-\mu_{j+1}]_q!}.$$

The lemma follows from this and the definition of $\mathrm{Q}_\mu(\mathbf{x};q,t)$. □

Now we can prove (12).

Proof of (12). From Theorem 1.10 and Lemma 5.3, by replacing μ with μ', we obtain:

$$\begin{aligned}
\mathrm{F}_{\mu\lambda}(q) &= (q-1)^{n-\ell(\mu)}q^{\sum_{i\geq 1}\binom{\mu'_i-\mu'_{i+1}}{2}+\binom{n}{2}+\ell(\mu)-\mathsf{n}(\mu)-\mathsf{n}(\lambda')-n}\prod_{i\geq 1}\frac{[\lambda_i]_q!}{[\mu'_i-\mu'_{i+1}]_q!}[\mathbf{x}^\lambda]\mathrm{P}_{\mu'}(\mathbf{x};1/q,0)\\
&= (q-1)^{n-\ell(\mu)}q^{\sum_{i\geq 1}\binom{\mu'_i-\mu'_{i+1}}{2}+\binom{n}{2}+\ell(\mu)-\mathsf{n}(\mu)-\mathsf{n}(\lambda')-n}\frac{\left(\prod_{j\geq 1}[\lambda_j]_q!\right)(q-1)^{\ell(\mu)}}{q^{\ell(\mu)+\sum_{j\geq 1}\binom{\mu'_j-\mu'_{j+1}}{2}}}[\mathbf{x}^\lambda]\mathrm{Q}_{\mu'}(\mathbf{x};1/q,0)\\
&= (q-1)^n q^{\binom{n}{2}-\mathsf{n}(\mu)-\mathsf{n}(\lambda')-n}\left(\prod_{i\geq 1}[\lambda_i]_q!\right)[\mathbf{x}^\lambda]\mathrm{Q}_{\mu'}(\mathbf{x};1/q,0).
\end{aligned}$$

This completes the proof. □

§ 6. Proof of the third part of Theorem 1.7

We begin by using the first part of Theorem 1.7 to give another proof of the following result due to Fuchs and Kirillov [FK22, Theorem 4.1].

Theorem 6.1. *Take $\lambda = (1^n)$ and let μ be an arbitrary partition. Then there exists an integer polynomial $\mathrm{R}_{\mu\lambda}(q)$ with $\mathrm{R}_{\mu\lambda}(0) = 1$ and $\mathrm{R}_{\mu\lambda}(1) \neq 0$ such that*

$$\mathrm{F}_{\mu\lambda}(q) = (q-1)^{n-\ell(\mu)} q^{\binom{n}{2} - \binom{\ell(\mu)}{2} - \sum_{j\geq 1} \mu'_j \mu'_{j+1}} \mathrm{R}_{\mu\lambda}(q).$$

Proof. From the first part of Theorem 1.7 we have

$$\mathrm{F}_{\mu\lambda}(q) = (q-1)^{n-\ell(\mu)} q^{\binom{n}{2} + \ell(\mu) - \mathsf{n}(\mu) - n}\, b_{\mu'\lambda}(1/q). \tag{77}$$

Let $\mathrm{SYT}(\mu')$ denote the set of standard tableaux of shape μ'. Then $\mathrm{SSYT}(\mu', 1^n) = \mathrm{SYT}(\mu')$. Moreover $s_q(T) = 1$ for any standard tableau T, and so from (39) we have

$$b_{\mu'\lambda}(q) = \sum_{T \in \mathrm{SYT}(\mu')} r_q(T).$$

There is only one standard tableau that maximizes the degree of $r_q(T)$ which is the standard tableau obtained by inserting $\{1, 2, \dots, \mu'_1\}$ into the first row of μ' and so on. For instance when $\mu' = (6, 3, 2)$, the tableau is

$$T = \begin{array}{|c|c|c|c|c|c|} \hline 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 7 & 8 & 9 \\ \cline{1-3} 10 & 11 \\ \cline{1-2} \end{array}\,.$$

Then $r_q(T) = \prod_{j\geq 2} \prod_{i=0}^{\mu'_j - 1} [\mu'_{j-1} - i]_q$. Therefore

$$\begin{aligned} \deg_q(r_q(T)) &= \sum_{j\geq 2} \left(\mu'_j \mu'_{j-1} - \sum_{i=1}^{\mu'_j} i \right) = \sum_{j\geq 1} \mu'_j \mu'_{j+1} - \sum_{j\geq 2} \binom{\mu'_j}{2} - \sum_{j\geq 2} \mu'_j \\ &= \sum_{j\geq 1} \mu'_j \mu'_{j+1} + \binom{\mu'_1}{2} + \mu'_1 - \sum_{j\geq 1} \binom{\mu'_j}{2} - \sum_{j\geq 1} \mu'_j \\ &= \sum_{j\geq 1} \mu'_j \mu'_{j+1} + \binom{\ell(\mu)}{2} + \ell(\mu) - \mathsf{n}(\mu) - n. \end{aligned}$$

Thus we can define

$$\mathrm{R}_{\mu\lambda}(q) \stackrel{\mathrm{def}}{=} q^{\sum_{j\geq 1} \mu'_j \mu'_{j+1} + \binom{\ell(\mu)}{2} + \ell(\mu) - \mathsf{n}(\mu) - n} b_{\mu'\lambda}(1/q) \in \mathbb{Z}[q]$$

From this and (77) we have

$$\mathrm{F}_{\mu\lambda}(q) = (q-1)^{n-\ell(\mu)} q^{\binom{n}{2} - \binom{\ell(\mu)}{2} - \sum_{j\geq 1} \mu'_j \mu'_{j+1}} \mathrm{R}_{\mu\lambda}(q).$$

Note that $b_{\mu'\lambda}(q)$ is monic and so $\mathrm{R}_{\mu\lambda}(0) = 1$. Moreover, $\lim_{q\to 1} [n]_q = n$, and thus $b_{\mu'\lambda}(1) \neq 0$, which confirms $\mathrm{R}_{\mu\lambda}(1) \neq 0$. This proves the theorem. □

For an arbitrary partition λ, unlike in the previous case, $s_q(T)$ contributes and this approach becomes complicated. For that reason, we use Theorem 1.10 to prove (7) for a general partition $\lambda = (\lambda_1, \dots, \lambda_s)$.

6.1. Kostka polynomials.

As before $\mathbf{x} = (x_1, x_2, \dots)$. We recall the following equation obtained in (14):

$$\mathrm{P}_{\mu'}(\mathbf{x}; q, 0) = \sum_{\eta} \mathrm{K}_{\eta'\mu}(q) s_\eta(\mathbf{x}).$$

From $\mathrm{K}_{\eta\lambda} = [\mathbf{x}^\lambda] s_\eta(x)$ we obtain

$$[\mathbf{x}^\lambda] \mathrm{P}_{\mu'}(\mathbf{x}; q, 0) = \sum_{\lambda \trianglelefteq \eta \trianglelefteq \mu'} \mathrm{K}_{\eta\lambda} \mathrm{K}_{\eta'\mu}(q). \tag{78}$$

Note that the exponent of q in (11) is $\sum_{i\geq 1}\binom{\mu'_i-\mu'_{i+1}}{2}+\binom{n}{2}+\ell(\mu)-\mathsf{n}(\mu)-\mathsf{n}(\lambda')-n$, which can easily be shown to be equal to

$$\binom{n}{2}+\mathsf{n}(\mu)-\mathsf{n}(\lambda')-\sum_{j\geq 1}\mu'_j\mu'_{j+1}-\binom{\ell(\mu)}{2}. \tag{79}$$

From Theorem 1.10 and Eqs. (78) and (79) it follows that

$$\mathrm{F}_{\mu\lambda}(q)=(q-1)^{n-\ell(\mu)}q^{\binom{n}{2}+\mathsf{n}(\mu)-\mathsf{n}(\lambda')-\sum_{i\geq 1}\mu'_i\mu'_{i+1}-\binom{\ell(\mu)}{2}}\prod_{i\geq 1}\frac{[\lambda_i]_q!}{[\mu'_i-\mu'_{i+1}]_q!}\sum_{\lambda\trianglelefteq\eta\trianglelefteq\mu'}\mathrm{K}_{\eta\lambda}\mathrm{K}_{\eta'\mu}(1/q). \tag{80}$$

The following lemma is well-known and follows directly from Karamata's inequality.

Lemma 6.2. *Let λ and μ be two partitions of n with $\lambda\trianglelefteq\mu$. Then $\mathsf{n}(\lambda')\leq\mathsf{n}(\mu')$ and the equality holds if and only if $\lambda=\mu$.*

We are now ready to prove (7).

Proof of the third part of Theorem 1.7. It is known [Mac15, p. 242, Eq. (6.5)] that the Kostka polynomial $\mathrm{K}_{\eta'\mu}(q)$ is an integer monic polynomial of degree $\mathsf{n}(\mu)-\mathsf{n}(\eta')$ and $\mathrm{K}_{\eta'\mu}(1)=\mathrm{K}_{\eta'\mu}>0$ when $\eta\trianglelefteq\mu'$. The modified Kostka polynomial is defined as

$$\tilde{\mathrm{K}}_{\eta'\mu}(q):=q^{\mathsf{n}(\mu)}\mathrm{K}_{\eta'\mu}(1/q).$$

Therefore

$$\tilde{\mathrm{K}}_{\eta'\mu}(q)=q^{\mathsf{n}(\eta')}\,\tilde{\mathrm{R}}_{\eta'\mu}(q),$$

where $\tilde{\mathrm{R}}_{\eta'\mu}(q)$ is an integer polynomial of degree $\mathsf{n}(\mu)-\mathsf{n}(\eta')$ with $\tilde{\mathrm{R}}_{\eta'\mu}(0)=1$ and $\tilde{\mathrm{R}}_{\eta'\mu}(1)=\mathrm{K}_{\eta'\mu}>0$. For two distinct partitions λ and η with $\lambda\trianglelefteq\eta$, from Lemma 6.2 we have $\mathsf{n}(\lambda')<\mathsf{n}(\eta')$. Then we can rewrite (80) as

$$\mathrm{F}_{\mu\lambda}(q)=(q-1)^{n-\ell(\mu)}q^{\binom{n}{2}-\sum_{i\geq 1}\mu'_i\mu'_{i+1}-\binom{\ell(\mu)}{2}}\prod_{i\geq 1}\frac{[\lambda_i]_q!}{[\mu'_i-\mu'_{i+1}]_q!}\sum_{\lambda\trianglelefteq\eta\trianglelefteq\mu'}q^{\mathsf{n}(\eta')-\mathsf{n}(\lambda')}\mathrm{K}_{\eta\lambda}\tilde{\mathrm{R}}_{\eta'\mu}(q).$$

Define

$$\mathrm{R}_{\mu\lambda}(q):=\prod_{i\geq 1}\frac{[\lambda_i]_q!}{[\mu'_i-\mu'_{i+1}]_q!}\sum_{\lambda\trianglelefteq\eta\trianglelefteq\mu'}q^{\mathsf{n}(\eta')-\mathsf{n}(\lambda')}\mathrm{K}_{\eta\lambda}\tilde{\mathrm{R}}_{\eta'\mu}(q),$$

so that

$$\mathrm{F}_{\mu\lambda}(q)=(q-1)^{n-\ell(\mu)}q^{\binom{n}{2}-\sum_{j\geq 1}\mu'_j\mu'_{j+1}-\binom{\ell(\mu)}{2}}\,\mathrm{R}_{\mu\lambda}(q).$$

Obviously $\mathrm{R}_{\mu\lambda}(0)=1$ and $\mathrm{R}_{\mu\lambda}(1)>0$ since $\tilde{\mathrm{R}}_{\eta'\mu}(1)>0$ and $\mathrm{K}_{\eta\lambda}>0$ for each $\lambda\trianglelefteq\eta\trianglelefteq\mu'$. Thus $\mathrm{R}_{\mu\lambda}(q)$ is a rational function without a pole at 0 or 1. Since $\mathrm{F}_{\mu\lambda}(q)\in\mathbb{Z}[q]$, we obtain $\mathrm{R}_{\mu\lambda}(q)\in\mathbb{Z}[q]$. □

§ 7. The modular law and the proofs of Theorems 1.13, 1.17 and 1.23

This section is devoted to the proofs of our results on orbital varieties of ad-nilpotent ideals, and their application to nilpotent Hessenberg varieties.

7.1. Proof of Theorem 1.13: modular law and ad-nilpotent ideals.

Throughout this section, for a given Hessenberg function h, we always set $\mathsf{h}(0)=0$ by convention. We begin by recalling the modular law [AN21].

Definition 7.1. Let $\mathsf{h}_0,\mathsf{h}_1,\mathsf{h}_2\in\mathcal{H}_n$ be Hessenberg functions. We say $(\mathsf{h}_0,\mathsf{h}_1,\mathsf{h}_2)$ is a compatible sequence whenever one of the following two conditions holds:

Case I: There exists $i \in [n-1]$ such that

$$\mathsf{h}_1(i-1) < \mathsf{h}_1(i) < \mathsf{h}_1(i+1) \quad \text{and} \quad \mathsf{h}_1(\mathsf{h}_1(i)) = \mathsf{h}_1(\mathsf{h}_1(i)+1). \tag{81}$$

Moreover h_0 and h_2 are obtained from h_1 via:

$$\mathsf{h}_0(l) = \begin{cases} \mathsf{h}_1(l) & l \neq i, \\ \mathsf{h}_1(i)-1 & l = i, \end{cases} \qquad \mathsf{h}_2(l) = \begin{cases} \mathsf{h}_1(l) & l \neq i, \\ \mathsf{h}_1(i)+1 & l = i. \end{cases} \tag{82}$$

Case II: There exists $i \in [n-1]$ such that

$$\mathsf{h}_1(i+1) = \mathsf{h}_1(i)+1 \quad \text{and} \quad \mathsf{h}_1^{-1}(i) = \emptyset. \tag{83}$$

Moreover h_0 and h_2 are obtained from h_1 via:

$$\mathsf{h}_0(l) = \begin{cases} \mathsf{h}_1(l) & l \neq i+1, \\ \mathsf{h}_1(i) & l = i+1, \end{cases} \qquad \mathsf{h}_2(j) = \begin{cases} \mathsf{h}_1(l) & l \neq i, \\ \mathsf{h}_1(i)+1 & l = i. \end{cases} \tag{84}$$

Remark 7.2. In the first case, the condition $\mathsf{h}_1(i-1) < \mathsf{h}_1(i)$ guarantees that $\mathsf{h}_0 \in \mathcal{H}_n$. In particular, when $i = 1$, from (81), we deduce that $\mathsf{h}_1(1) > 1$. Moreover in both cases $\mathsf{h}_1(i) \leq n-1$.

Example 7.3. The function $\mathsf{h}_1 = (2,3,4,6,6,6)$ for $i = 3$ satisfies the first case. Here we have $\mathsf{h}_0 = (2,3,3,6,6,6)$ and $\mathsf{h}_2 = (2,3,5,6,6,6)$. Also the function $\mathsf{h}_1 = (1,2,4,5,6,7,7)$ for $i = 3$ satisfies the second case. For this function we obtain $\mathsf{h}_0 = (1,2,4,4,6,7,7)$ and $\mathsf{h}_2 = (1,2,5,5,6,7,7)$

Definition 7.4 (q-modular law)**.** Let $f : \mathcal{H}_n \to \mathbb{R}$ be a real-valued function and let $q \in \mathbb{R}$. We say that f satisfies the q-modular law if for any compatible sequence $(\mathsf{h}_0, \mathsf{h}_1, \mathsf{h}_2)$ of Hessenberg functions in $\mathcal{H}_n$ we have

$$(q+1)f(\mathsf{h}_1) = qf(\mathsf{h}_0) + f(\mathsf{h}_2). \tag{85}$$

Remark 7.5. The definition of the modular law in [AN21] is slightly different from what we give here. There, the requirement on $f : \mathcal{H}_n \to V$ is that V is a $\mathbb{Q}(q)$-vector space for an indeterminate q. Here, we consider q to be a number. However, the main results of [AN21] (in particular [AN21, Theorem 1.2] and [AN21, Algorithm 2.8]) are still valid for all but finitely many choices of q (more precisely, when q is not a root of finitely many polynomials).

Now let μ be a partition of n. Our next goal is to show that the linear function

$$f_\mu : \mathcal{H}_n \to \mathbb{R}, \qquad \mathsf{h} \mapsto q^{|h|}\mathrm{F}_{\mu\mathsf{h}}(q), \tag{86}$$

where $|\mathsf{h}| = \sum \mathsf{h}(l)$, satisfies the q-modular law. Let $(\mathsf{h}_0, \mathsf{h}_1, \mathsf{h}_2)$ be a compatible sequence. Note that $|\mathsf{h}_0| = |\mathsf{h}_1| - 1$ and $|\mathsf{h}_2| = |\mathsf{h}_1| + 1$. Thus (86) would follow from

$$(q+1)\mathrm{F}_{\mu\mathsf{h}_1}(q) = \mathrm{F}_{\mu\mathsf{h}_0}(q) + q\mathrm{F}_{\mu\mathsf{h}_2}(q). \tag{87}$$

The proof of (87) is given in Lemma 7.6. To begin, we recall basic properties of elementary row and column operations. For $x \in \mathbb{F}_q$ and $1 \leq s,t \leq n$, denote $\mathsf{E}_{st}(x) = I + \mathrm{E}_{st}(x)$, where $\mathrm{E}_{st}(x)$ is the elementary $n \times n$ matrix with x in the (s,t)-entry and zeros in other entries. Given an $n \times n$ matrix, we immediately obtain:

$$\mathrm{Row}_l(\mathsf{E}_{st}(x)A) = \begin{cases} \mathrm{Row}_l(A) & l \neq s; \\ x\,\mathrm{Row}_t(A) + \mathrm{Row}_s(A) & l = s. \end{cases} \qquad \mathrm{Col}_l(A\mathsf{E}_{st}(x)) = \begin{cases} \mathrm{Col}_l(A) & l \neq t; \\ x\,\mathrm{Col}_s(A) + \mathrm{Col}_t(A) & l = t. \end{cases} \tag{88}$$

The permutation matrix associated with the transposition $(s\ t) \in \mathrm{S}_n$ is denoted by $\mathrm{Per}(s,t)$.

Lemma 7.6. *The function $f_\mu : \mathcal{H}_n \to \mathbb{R}$ defined in (86) satisfies the q-modular law.*

Proof. We consider two cases according to Definition 7.1.

Case I: Set $j = \mathsf{h}_1(i)$. Define

$$\mathcal{A}_0 = \{(a_{st}) \in \mathfrak{u}_{\mathsf{h}_0}(\mathbb{F}_q) \cap \mathcal{C}_\mu : a_{ij} \neq 0\} \qquad \mathcal{A}_1 = \{(a_{st}) \in \mathfrak{u}_{\mathsf{h}_1}(\mathbb{F}_q) \cap \mathcal{C}_\mu : a_{i(j+1)} \neq 0\}.$$

One observes that

$$\mathrm{F}_{\mu\mathsf{h}_0}(q) = \mathrm{F}_{\mu\mathsf{h}_1}(q) + |\mathcal{A}_0| \qquad \mathrm{F}_{\mu\mathsf{h}_1}(q) = \mathrm{F}_{\mu\mathsf{h}_2}(q) + |\mathcal{A}_1|.$$

Thus (87) follows from $q|\mathcal{A}_1| = |\mathcal{A}_0|$. To prove this, we consider the following map:

$$\varphi : \mathcal{A}_0 \to \mathcal{A}_1, \qquad A \mapsto \mathrm{Per}(j, j+1)\, \mathsf{E}_{j(j+1)}\left(\frac{a_{i(j+1)}}{a_{ij}}\right) A\, \mathsf{E}_{j(j+1)}\left(-\frac{a_{i(j+1)}}{a_{ij}}\right) \mathrm{Per}(j, j+1).$$

We will show that $\varphi(A) \in \mathcal{A}_1$ with $A_{ij} = \varphi(A)_{i(j+1)}$. It follows from (88) that the $(i, j+1)$ entry of $A\, \mathsf{E}_{j(j+1)}\left(-a_{i(j+1)}/a_{ij}\right)$ is zero. The assumptions $\mathsf{h}_1(i) < \mathsf{h}_1(i+1)$ and $\mathsf{h}_1(\mathsf{h}_1(i)) = \mathsf{h}_1(\mathsf{h}_1(i)+1)$ imply that $i+1 \leq j$. From (88) we deduce $A_1 = \mathsf{E}_{j(j+1)}\left(a_{i(j+1)}/a_{ij}\right) A\, \mathsf{E}_{j(j+1)}\left(-a_{i(j+1)}/a_{ij}\right)$ belongs to $\mathcal{A}_0$ and its $(i, j+1)$ entry is zero. Since $\mathsf{h}_1(i) < \mathsf{h}_1(i+1)$ we have $a_{(i+1)(j+1)} = 0$. Then $A_1\, \mathrm{Per}(j, j+1)$, which swaps the jth and $(j+1)$th column, belongs to $\mathfrak{u}_{\mathsf{h}_1}$ with a non-zero $(i, j+1)$ entry. Thus, the matrix $\mathrm{Per}(j, j+1)\, A_1\, \mathrm{Per}(j, j+1)$ belongs to $\mathfrak{u}_{\mathsf{h}_1}$ with a non-zero $(i, j+1)$ entry, because $i < j$. Since the Jordan form of A_1 is J_μ, we conclude that $\varphi(A) = \mathrm{Per}(j, j+1)\, A_1\, \mathrm{Per}(j, j+1) \in \mathcal{A}_1$.

We now show surjectivity of φ. If $B \in \mathcal{A}_1$ with $b = b_{i(j+1)} \neq 0$, then one can check that for any $x \in \mathbb{F}_q$

$$\mathsf{E}_{j(j+1)}(-x/b)\, \mathrm{Per}(j, j+1)\, B\, \mathrm{Per}(j, j+1)\, \mathsf{E}_{j(j+1)}(x/b) \in \mathcal{A}_0, \tag{89}$$

with (i, j) entry b and $(i, j+1)$ entry x. This matrix maps to B, so φ is surjective. Moreover, the matrices of the form (89) are all distinct. This follows from the identity

$$\mathrm{Per}(j, j+1)\, \mathsf{E}_{j(j+1)}(x)\, \mathrm{Per}(j, j+1) = \mathsf{E}_{(j+1)j}(x)$$

and the fact that $\mathsf{E}_{(j+1)j}(x) B\, \mathsf{E}_{(j+1)j}(-x) = B$, where $B \in \mathcal{A}_1$, implies $x = 0$. Thus the fiber of each element has cardinality q, and so $q|\mathcal{A}_1| = |\mathcal{A}_0|$.

Case II: This case is similar to the previous case. Set $j = \mathsf{h}_1(i)$. Define

$$\mathcal{B}_0 = \{(b_{st}) \in \mathfrak{u}_{h_0} \cap \mathcal{C}_\mu : b_{(i+1)(j+1)} \neq 0\}, \qquad \mathcal{B}_1 = \{(b_{st}) \in \mathfrak{u}_{h_1} \cap \mathcal{C}_\mu : b_{i(j+1)} \neq 0\}.$$

Then we have

$$\mathrm{F}_{\mu\mathsf{h}_0}(q) = \mathrm{F}_{\mu\mathsf{h}_1}(q) + |\mathcal{B}_0| \qquad \mathrm{F}_{\mu\mathsf{h}_1}(q) = \mathrm{F}_{\mu\mathsf{h}_2}(q) + |\mathcal{B}_1|.$$

This implies that (87) follows from $q|\mathcal{B}_1| = |\mathcal{B}_0|$. To prove this we now define

$$\psi : \mathcal{B}_0 \to \mathcal{B}_1, \qquad B \mapsto \mathrm{Per}(i, i+1)\, \mathsf{E}_{i(i+1)}\left(\frac{-b_{i(j+1)}}{b_{(i+1)(j+1)}}\right) B\, \mathsf{E}_{i(i+1)}\left(-\frac{b_{i(j+1)}}{b_{(i+1)(j+1)}}\right) \mathrm{Per}(i, i+1).$$

From (83) we have $j > i$ which implies $\psi(B) \in \mathcal{B}_1$ and a similar argument as above shows the fiber of each element has cardinality q. □

Proposition 7.7. *Let $\mathsf{h} \in \mathcal{H}_n$ for some $n \geq 1$. Then $\mathrm{F}_{\mu\mathsf{h}}(q)$ is a rational function of q.*

Proof. This follows from [AN21, Theorem 1.2, Algorithm 2.8], which also holds for real-valued functions that satisfy the q-modular law for generic q. Indeed [AN21, Algorithm 2.8] expresses $\mathrm{F}_{\mu\mathsf{h}}(q)$ as a linear combination of the $\mathrm{F}_{\mu\lambda}(q)$ with coefficients which are rational functions of q. But by Theorem 1.7 the $\mathrm{F}_{\mu\lambda}(q)$ are polynomials in q. □

We now recall [AN21, Corollary 3.2], which states that for a function $f : \mathcal{H}_n \to \mathbb{Q}(q)$ satisfying the q-modular law, treating q as a variable, we have

$$f(\mathsf{h}) = \sum_{\lambda \vdash n} \frac{c_{\lambda\mathsf{h}}(q)}{[\lambda]_q!} f(k_\lambda), \tag{90}$$

where $k_\lambda = k_{\lambda_1} \sqcup \cdots \sqcup k_{\lambda_l}$ with $l = \ell(\lambda)$. By applying Proposition 7.7 and (90) we deduce

$$\mathrm{F}_{\mu\mathsf{h}}(q) = \frac{1}{q^{|\mathsf{h}|}} \sum_{\lambda \vdash n} \frac{c_{\lambda\mathsf{h}}(q)}{[\lambda]_q!} q^{|k_\lambda|} \mathrm{F}_{\mu\lambda}(q). \tag{91}$$

One can show

$$|k_\lambda| = (n^2+n)/2 + \mathsf{n}(\lambda') = n^2 - \binom{n}{2} + \mathsf{n}(\lambda') = n^2 - \dim \mathfrak{u}_\lambda.$$

This combined with (91) proves the the first equation of Theorem 1.13. The rest of the theorem follows from Theorem 1.12. Finally, to show that $\mathrm{F}_{\mu\mathsf{h}}(q) \in \mathbb{Z}[q]$ note that we have $\mathrm{F}_{\mu\mathsf{h}}(q) = \frac{g(q)}{h(q)}$ where $g(q), h(q) \in \mathbb{Z}[q]$ and $h(q)$ is monic. By long division $f(q) = f_1(q) + (r(q)/h(q))$ for $f_1(q), r(q) \in \mathbb{Z}[q]$ satisfying $\deg(r(q)) < \deg(h(q))$. Since $f(q) \in \mathbb{Z}$ for every prime q, it follows that $r(q) = 0$ when $q \gg 0$, hence $r(q) = 0$.

7.2. Proof of Theorem 1.17: Non-emptiness of $\mathcal{C}_\mu \cap \mathfrak{u}_\mathsf{h}$. In this section $\mathbb{F}$ denotes an arbitrary field, $\mathcal{C}_\mu$ is the $\mathrm{GL}_n(\mathbb{F})$-conjugacy class of J_μ, and $\mathfrak{u}_\mathsf{h} = \mathfrak{u}_\mathsf{h}(\mathbb{F})$. Recall that $[n] := \{1, \dots, n\}$. For positive integers m and n, set $[m : n] = \{m, \dots, n\}$. If $m > n$, we define $[m : n] = \emptyset$. For $i < j$, we write E_{ij} for the elementary $n \times n$ matrix whose only nonzero entry is a 1 in the i-th row and the j-th column. If we denote the standard basis vectors of $\mathbb{F}^n$ by $e_1, \dots, e_n$, then

$$\mathrm{E}_{ij}(e_\ell) = \begin{cases} e_i & \ell = j; \\ 0 & \text{otherwise.} \end{cases} \tag{92}$$

If $J = \{i_1, \dots, i_r\} \subseteq \{1, \cdots, n\}$ with $i_1 < \cdots < i_r$, we write $\mathrm{E}_J := \sum_{l=1}^{r-1} \mathrm{E}_{i_l i_{l+1}}$. We set $\mathrm{E}_J = 0$ when J has only one element. Then from (92) we have

$$\mathrm{E}_J(e_{i_1}) = 0, \quad \mathrm{E}_J(e_{i_2}) = e_{i_1}, \quad \dots \quad \mathrm{E}_J(e_{i_r}) = e_{i_{r-1}}.$$

From this we observe that E_J has the Jordan canonical form of type $(r, 1, \dots, 1)$. Similarly, if $\mathcal{J} = \{J_1, \dots, J_k\}$ is a partition of $[n]$ with $|J_i| = \mu_i$, then the Jordan canonical form of $\mathrm{E}_\mathcal{J} := \sum_{i=1}^k \mathrm{E}_{J_i}$ is $\mu = (\mu_1, \dots, \mu_k)$.

Now let $\mathsf{h} \in \mathcal{H}_n$ be a Hessenberg function. We recall the definition of the partial order $\mathcal{P}_\mathsf{h}$ on $[n]$ given by $i \prec_\mathsf{h} j$ if and only if $\mathsf{h}(i) < j$. Let λ_h be the Greene-Kleitman shape of $\mathcal{P}_\mathsf{h}$.

Proposition 7.8. *Let $\mu \trianglelefteq \lambda = \lambda_\mathsf{h}$ be a partition of n. Then $[n]$ can be partitioned into $\ell(\mu)$ chains in $\mathcal{P}_\mathsf{h}$ whose cardinalities are the parts of μ.*

Before giving the proof of Proposition 7.8, we note that for a given partition $\mu \trianglelefteq \lambda_\mathsf{h}$, by applying Proposition 7.8 we obtain a family of disjoint subsets $\mathcal{J} = \{J_1, \dots, J_{\ell(\mu)}\}$ of $[n]$, where each J_i is a chain with respect to $\preceq_\mathsf{h}$ and $|J_i| = \mu_i$. Thus $\mathrm{E}_\mathcal{J} \in \mathcal{C}_\mu \cap \mathfrak{u}_\mathsf{h}$ which proves one direction of Theorem 1.17. We require the following lemma to prove Proposition 7.8 .

Lemma 7.9. *If C_1, C_2 are disjoint chains in $\mathcal{P}_\mathsf{h}$ with $|C_1| > |C_2| + 1$, then there exist disjoint chains C_1' and C_2' in $\mathcal{P}_\mathsf{h}$ such that $C_1 \cup C_2 = C_1' \cup C_2'$ with $|C_1'| = |C_1| - 1$ and $|C_2'| = |C_2| + 1$.*

Proof. It is known [SW16, Proposition 4.1] that $\mathcal{P}_\mathsf{h}$ is isomorphic to a natural unit interval order (a collection of n closed unit intervals on the real line with intervals $U_1 < U_2$ precisely when U_1 is completely to the left of U_2). Therefore we may assume that $C_1 = \{U_1 < U_2 < \cdots < U_\ell\}$ and $C_2 = \{V_1 < V_2 < \cdots < V_k\}$ for unit intervals U_i and V_j on the real line in the Euclidean plane with $1 \le i \le \ell, 1 \le j \le k$. The proof is by induction on $k = |C_2|$. The cases $k = 0, 1$ are straightforward. Suppose the lemma holds whenever $|C_2| < k$. Now consider the case $|C_2| = k$. If some interval U in C_1 does not intersect any interval in C_2, then $C_1' = C_1 - \{U\}$ and $C_2' = C_2 \cup \{U\}$ have the desired property.

Now suppose no interval can be moved from C_1 to C_2. For $1 \le i \le \ell - 1$ let L_i be a line prependicular to the x-axis which intersects the real line strictly between U_i and U_{i+1}. Since each interval in C_2 intersects at most one such line, it can be seen that some line L_j does not intersect any interval in C_2. For $m = 1, 2$, write $C_m = C_m^L \cup C_m^R$ where C_m^L denotes all intervals in C_m to the left of L_j and C_m^R denotes all intervals in C_m to the right of L_j. By the movability assumption, both $|C_2^L|$ and $|C_2^R|$ are nonempty. Suppose either $|C_1^L| > |C_2^L| + 1$ or $|C_1^R| > |C_2^R| + 1$. Since both $|C_2^L|$ and $|C_2^R|$ are less than k, we can use the inductive hypothesis to construct C_1' and C_2'

as required. If not, then we must have $|C_1^L| = |C_2^L| + 1$ and $|C_1^R| = |C_2^R| + 1$. In this case it is clear that $C_1' = C_2^L \cup C_1^R$ and $C_2' = C_1^L \cup C_2^R$ have the desired property. This completes the inductive step and the proof. □

Remark 7.10. The above lemma fails for arbitrary partial orders. For instance, if P is isomorphic to the direct sum of a 3-element chain and a 1-element chain.

Proof of Proposition 7.8. Write $\lambda = \lambda_{\mathsf{h}}$ and suppose $C = \{C_i\}_{1 \le i \le k}$ is a disjoint chain cover of $\mathcal{P}_{\mathsf{h}}$ with $|C_i| = \lambda_i$ for $1 \le i \le k$ (such a cover exists by Gerstenhaber's algorithm). Now suppose $\mu \trianglelefteq \lambda$ is a partition. Then there is a sequence $\mu = \nu^0 \trianglelefteq \nu^1 \trianglelefteq \cdots \trianglelefteq \nu^m = \lambda_{\mathsf{h}}$ of partitions such that ν^{i-1} is covered by ν^i in the dominance order for $1 \le i \le m$ [Mac15, p. 9]. Therefore, for each i, the partition ν^{i-1} can be obtained from ν^i by moving a single cell in some row of the Young diagram of ν^i to a lower row. If there exists a chain cover for $\mathcal{P}_{\mathsf{h}}$ where the chains have cardinalities given by the parts of ν^i, then by Lemma 7.9 we can construct another chain cover for $\mathcal{P}_{\mathsf{h}}$ where the chains have cardinalities given by the parts of ν^{i-1}. This proves Proposition 7.8. □

We now prove the other direction of Theorem 1.17. Let $\mathcal{P}$ be a finite poset with n elements. Fix a linear extension of $\mathcal{P}$ which identifies its elements with $[n]$ (thus if $i \preceq_P j$ then $i \le j$ as positive integers). The incidence algebra $I(\mathcal{P}, \mathbb{F})$ consists of all $n \times n$ matrices $A = (a_{ij})$ over $\mathbb{F}$ such that $a_{ij} = 0$ unless $i \preceq_{\mathcal{P}} j$. Note that $I(\mathcal{P}, \mathbb{F})$ is an algebra of upper triangular matrices. A matrix $A \in I(\mathcal{P}, \mathbb{F})$ is nilpotent precisely when its diagonal entries are all zero. Write $\mathrm{Nil}(\mathcal{P}, \mathbb{F})$ for the collection of all nilpotent matrices in $I(\mathcal{P}, \mathbb{F})$. Let $\delta_k(A)$ denote the greatest common divisor of all $k \times k$ minors of $xI - A$. Given a matrix $A \in \mathrm{Nil}(\mathcal{P}, \mathbb{F})$, suppose $\delta_k(A) = x^{d_k}$ for $0 \le k \le n$ (by convention $\delta_0(A) = 1$, so $d_0 = 0$) [BF01, Lemma 6.2]. Note that $d_n = n$. Then A is similar to J_λ where $\lambda = (\lambda_1, \lambda_2, \ldots)$ with $\lambda_i = d_{n+1-i} - d_{n-i}$ for $1 \le i \le n$. Therefore $\lambda_1 + \cdots + \lambda_r = n - d_{n-r}$ for $r \ge 1$.

Proposition 7.11. *If* $\mathrm{Nil}(\mathcal{P}, \mathbb{F}) \cap \mathcal{C}_\mu \neq \emptyset$, *then* $\mu \trianglelefteq \lambda_{\mathcal{P}}$. *In particular, if* $\mathcal{C}_\mu \cap \mathfrak{u}_{\mathsf{h}} \neq \emptyset$ *then* $\mu \trianglelefteq \lambda_{\mathsf{h}}$.

Proof. Let x_{ij} $(i, j \ge 1)$ be independent indeterminates over the field $\mathbb{F}$. Consider the generic $n \times n$ matrix $G = (g_{ij})$ where

$$g_{ij} = \begin{cases} x_{ij} & \text{if } i \prec_{\mathcal{P}} j, \\ 0 & \text{otherwise.} \end{cases}$$

By Britz and Fomin [BF01, Theorem 6.1] the matrix G is similar to J_λ where $\lambda = \lambda_{\mathcal{P}}$ (the proof in [BF01] is given for the complex field but carries over verbatim for any field $\mathbb{F}$). Suppose $\delta_i(G) = x^{g_i}$ for $i \ge 1$. Let $A \in \mathrm{Nil}(\mathcal{P}, \mathbb{F}) \cap \mathcal{C}_\mu$ and suppose $\delta_i(A) = x^{a_i}$ for $1 \le i \le n$. Since A can be obtained from G by specializing some entries to elements of F, it follows that $\delta_i(G) = x^{g_i}$ divides $\delta_i(A)$ for each $1 \le i \le n$. In other words, $g_i \le a_i$ for $1 \le i \le n$. From $A \in \mathcal{C}_\mu$, we deduce $\mu_1 + \cdots + \mu_j = n - a_{n-j} \le n - g_{n-i} = \lambda_1 + \cdots + \lambda_j$ which shows $\mu \trianglelefteq \lambda = \lambda_{\mathcal{P}}$. □

7.3. Proof of Corollary 1.18. From Theorem 1.13 and (19) we obtain

$$\mathrm{F}_{\mu\mathsf{h}}(q) = \frac{q^{n^2 - |\mathsf{h}|}(q-1)^n}{|\mathrm{Z}_{\mathrm{G}}(I + J_\mu)|} \sum_{\lambda \vdash n} \tilde{\mathrm{K}}_{\lambda\mu}(q)\, g_{\lambda\mathsf{h}}(q),$$

where $\tilde{\mathrm{K}}_{\lambda\mu}(q) \stackrel{\text{def}}{=} q^{\mathsf{n}(\mu)} \mathrm{K}_{\lambda\mu}(1/q)$ is the modified Kostka polynomial. It is known [Mac15, p. 242, Eq. (6.5)] that the modified Kostka polynomial $\tilde{\mathrm{K}}_{\lambda\mu}(q)$ has nonnegative integer coefficients. Since $\mu \trianglelefteq \lambda_{\mathsf{h}}$, we have $\mathrm{F}_{\mu\mathsf{h}}(q) \neq 0$, and so, using the positivity of $g_{\lambda\mathsf{h}}(q)$, equation (17), and Theorem 1.7, we conclude that the exact power of $q - 1$ in $\mathrm{F}_{\mu\mathsf{h}}(q)$ is $n - \ell(\mu)$.

7.4. Proof of Theorem 1.23: nilpotent Hessenberg varieties. Let B be the standard upper triangular Borel subgroup of $\mathrm{G} = \mathrm{GL}_n(\mathbb{F}_q)$. For $\mathsf{h} \in \mathcal{H}_n$ and X a nilpotent $n \times n$ matrix with Jordan form J_μ we define

$$\mathcal{B}_\mu^{\mathsf{h}} \stackrel{\text{def}}{=} \left\{ g\mathrm{B} \in \mathrm{G}/\mathrm{B} : g^{-1}(I + X)g \in \mathsf{U}_{\mathsf{h}} \right\}.$$

Note that U_h is a normal subgroup of B. One can easily see that

$$\#\left\{g : g^{-1}(I+X)g \in \mathsf{U}_\mathsf{h}\right\} = \mid \mathrm{B} \mid \cdot \mid \mathcal{B}_\mu^\mathsf{h} \mid .$$

Then from Eqs. (22) and (24) and Theorem 1.13 we deduce

$$\#\mathcal{H}\mathrm{ess}_{nil}(\mathsf{h}, X) = |\mathcal{B}_\mu^\mathsf{h}| = \frac{|Z_G(I+J_\mu)|}{|\mathrm{B}|}\mathrm{F}_{\mu\mathsf{h}}(q) = q^{n^2-|\mathsf{h}|-\binom{n}{2}}\langle \tilde{\mathrm{H}}_\mu(\mathbf{x};q), X_{\mathsf{G}(\mathsf{h})}(\mathbf{x};q)\rangle.$$

The result now follows from the fact that $E_\mathsf{h} = \sum_i(\mathsf{h}(i)-i)$.

§ 8. Counting solutions of $X^2 = 0$ in $\mathfrak{u}_\lambda(\mathbb{F}_q)$: proofs of Theorems 1.24, 1.27 and 1.29

Let Λ be a composition of n and let $\lambda = \mathrm{sort}(\Lambda)$. We define

$$\mathrm{C}_\Lambda(q) \overset{\text{def}}{=} \#\{X \in \mathfrak{u}_\Lambda(\mathbb{F}_q) : X^2 = 0\}. \tag{93}$$

The condition $X^2 = 0$ is equivalent to having only blocks of size at most 2 in the Jordan form. Thus, for $X \in \mathcal{C}_\mu \cap \mathfrak{u}_\Lambda(\mathbb{F}_q)$ we have $X^2 = 0$ if and only if $\ell(\mu') \le 2$. From this and noting that $\mathrm{F}_{(1^n)\Lambda}(q) = 1$ we deduce:

$$\mathrm{C}_\Lambda(q) = 1 + \sum_{\substack{\lambda \trianglelefteq \mu' \\ \ell(\mu')=2}} \mathrm{F}_{\mu\Lambda}(q). \tag{94}$$

Since $\mathrm{F}_{\mu\Lambda}(q) = \mathrm{F}_{\mu\lambda}(q)$, by (94) it follows that $\mathrm{C}_\Lambda(q) = \mathrm{C}_\lambda(q)$. Our first application of (94) is the following estimate of $\mathrm{C}_\lambda(q)$.

Proposition 8.1. *Let $\lambda = (\lambda_1, \dots, \lambda_s)$ be a partition of n and set $\rho_\alpha = (n-\alpha, \alpha)$ where $\alpha = \min\{\lfloor n/2 \rfloor, n-\lambda_1\}$. Then $\mathrm{C}_\lambda(q) = \mathrm{K}_{\rho_\alpha\lambda}\, q^{n\alpha-\alpha^2}\,(1+o_q(1))$.*

Proof. The conjugates of partitions μ of n satisfying $\lambda \trianglelefteq \mu'$ and $\ell(\mu') = 2$ must be of the form $\mu' = (n-k, k)$, where $k \le \min\{\lfloor n/2 \rfloor, n-\lambda_1\}$. Thanks to Theorem 1.7, we know that $\mathrm{F}_{\mu\lambda}(q)$ is a polynomial of degree

$$\binom{n}{2} - \mathsf{n}(\mu) = \binom{n}{2} - \binom{n-k}{2} - \binom{k}{2} = nk - k^2, \tag{95}$$

with leading coefficient $\mathrm{K}_{\mu'\lambda}$. Note that $nk-k^2$ is increasing on $k \le n/2$ and so $\mathrm{K}_{\rho_\alpha\lambda}\, q^{n\alpha-\alpha^2}$ dominates the terms in (94). This proves the claim. □

Remark 8.2. For $\lambda = (1^n)$ we have

$$\rho_\alpha = \begin{cases} (n/2, n/2) & n \text{ is even;} \\ ((n+1)/2, (n-1)/2) & n \text{ is odd.} \end{cases} \tag{96}$$

In this case $\mathrm{K}_{\rho_\alpha\lambda}$ is the number of standard Young tableaux of shape ρ_α. It is well-known that the number of standard Young tableaux of shape (k,k) (or of shape $(k, k-1)$) is equal to the Catalan number $\frac{1}{k+1}\binom{2k}{k}$.

Notation 8.3. In the rest of this section we work with symmetric polynomials rather than with symmetric functions. Thus, for example we assume $\mathbf{x} = (x_1, \dots, x_s)$ for suitable $s \ge 1$ and then $\mathrm{P}_\mu(\mathbf{x}; q, t)$ means the Macdonald polynomial in s variables $x_1, \dots, x_s$.

We now turn our attention to $a_{nk}(\lambda)$ defined in (31). As usual, set $s = \ell(\lambda)$. From (12) and (95), we obtain

$$a_{nk}(\lambda) = (q-1)^n q^{nk-k^2-\mathsf{n}(\lambda')-n}\left(\prod_{1\le i\le s}[\lambda_i]_q!\right)[\mathbf{x}^\lambda]\mathrm{Q}_{\mu'}(\mathbf{x}; 1/q, 0), \tag{97}$$

where $\mathbf{x} = (x_1, \dots, x_s)$. Evidently $a_{nk}(\lambda) = 0$ when $k > n/2$.

8.1. The Cauchy identity and the proof of Theorem 1.24. Our next goal is to demonstrate that a generalization of Kirillov's mysterious relation (25) follows from the Cauchy identity for Macdonald polynomials, which we now recall. Let $\mathbf{x} = (x_1, \dots, x_s)$ and $\mathbf{y} = (y_1, \dots, y_r)$ be two sequences of independent indeterminates, and define

$$\Pi(\mathbf{x}, \mathbf{y}; q, t) = \prod_{1 \le i \le s} \prod_{1 \le j \le r} \frac{(t x_i y_j \,;\, q)_\infty}{(x_i y_j \,;\, q)_\infty}, \qquad (a \,;\, q)_\infty := \prod_{l \ge 0} \left(1 - a q^l\right). \tag{98}$$

The Cauchy identity for Macdonald polynomials [Mac15, p. 324, Eq. (4.13)] states that

$$\sum_{\mu} \mathrm{P}_\mu(\mathbf{y}; q, t) \mathrm{Q}_\mu(\mathbf{x}; q, t) = \Pi(\mathbf{x}, \mathbf{y}; q, t).$$

One can express $\Pi(\mathbf{x}, \mathbf{y}; q, t)$ in terms of the monomial symmetric polynomials. Let $g_l(\mathbf{x}; q, t)$ denote the coefficient of y^l in the power series expansion of the infinite product

$$\prod_{1 \le i \le s} \frac{(t x_i y \,;\, q)_\infty}{(x_i y \,;\, q)_\infty} = \sum_{l \ge 0} g_l(\mathbf{x}; q, t) y^l. \tag{99}$$

This coefficient can be calculated explicitly. From [Mac15, p. 314, Example 1] we have

$$g_l(\mathbf{x}; q, t) = \sum_{\nu \vdash l} \frac{(t \,;\, q)_\nu}{(q \,;\, q)_\nu} m_\nu(\mathbf{x}), \qquad (a \,;\, q)_\nu = \prod_{i \ge 1} (a \,;\, q)_{\nu_i}, \tag{100}$$

where $m_\nu(\mathbf{x})$ is the monomial symmetric polynomial. For any partition $\rho = (\rho_1, \rho_2, \dots)$ define

$$g_\rho(\mathbf{x}; q, t) := \prod_{i \ge 1} g_{\rho_i}(\mathbf{x}; q, t).$$

Moreover we have (see [Mac15, p. 311, Eq. (2.10)] for a proof) that

$$\Pi(\mathbf{x}, \mathbf{y}; q, t) = \Pi(\mathbf{y}, \mathbf{x}; q, t) = \sum_{\rho} g_\rho(\mathbf{y}; q, t) m_\rho(\mathbf{x}). \tag{101}$$

Since $\mathrm{P}_\rho(\mathbf{y}; q, t) = 0$ when $\ell(\rho) > r$, if we set $\mathbf{y} = (y_1, y_2)$ in the Cauchy identity, we obtain

$$\sum_{\ell(\mu') \le 2} \mathrm{P}_{\mu'}(y_1, y_2; q, 0) \mathrm{Q}_{\mu'}(\mathbf{x}; q, 0) = \sum_{\rho} g_\rho(y_1, y_2; q, 0)\, m_\rho(\mathbf{x}). \tag{102}$$

This in particular implies:

$$\sum_{\substack{\mu' = (n-k,k) \\ 0 \le k \le n/2}} \mathrm{P}_{\mu'}(y_1, y_2; 1/q, 0)\, [\mathbf{x}^\lambda] \mathrm{Q}_{\mu'}(\mathbf{x}; 1/q, 0) = g_\lambda(y_1, y_2; 1/q, 0). \tag{103}$$

From (103) and (97) we deduce the following result.

Theorem 8.4. *Let $\lambda = (\lambda_1, \dots, \lambda_s)$ be a partition of n, and let $a_{nk}(\lambda)$ denote the number of matrices X in $\mathfrak{u}_\lambda(\mathbb{F}_q)$ of rank k such that $X^2 = 0$. Then*

$$\sum_{0 \le k \le n/2} a_{nk}(\lambda)\, q^{k(k-n)} \mathrm{P}_{(n-k,k)}(y_1, y_2; 1/q, 0) = [\lambda]_q!\, (q-1)^n q^{-n-\mathfrak{n}(\lambda)} g_\lambda(y_1, y_2; 1/q, 0), \tag{104}$$

where $[\lambda]_q! := [\lambda_1]_q! \dots [\lambda_s]_q!$.

We will now proceed with finding explicit formulas for $\mathrm{P}_{(n-k,k)}(y_1, y_2; 1/q, 0)$ and $g_\lambda(y_1, y_2; q, 0)$. From [Mac15, p. 323, Eq. (4.9)] we have

$$\mathrm{P}_{(l)}(y_1, y_2; q, 0) = (q \,;\, q)_l\, g_l(y_1, y_2; q, 0).$$

Note that $m_\nu(y_1, y_2) = 0$ when $\ell(\nu) \ge 3$. Thus in (100) we only need to consider partitions $\nu = (l - j, j)$ with $j \le l/2$. Moreover we have

$$m_{(l-j,j)}(y_1, y_2) = \begin{cases} y_1^{l-j} y_2^j + y_1^j y_2^{l-j} & j < l - j; \\ y_1^{l/2} y_2^{l/2} & j = l - j. \end{cases}$$

As a consequence of these we obtain

$$\mathrm{P}_{(l)}(y_1,y_2;q,0)=\sum_{l-j\geq j\geq 0}\frac{(q\,;q)_l}{(q\,;q)_{l-j}\,(q\,;q)_j}m_{(l-j,j)}(y_1,y_2)=\sum_{j=0}^{l}\begin{bmatrix}l\\ j\end{bmatrix}_q y_1^{l-j}y_2^j.$$

But $\mathrm{P}_{(n-k,k)}(y_1,y_2;q,0)=(y_1y_2)^k\,\mathrm{P}_{(n-2k)}(y_1,y_2;q,0)$ by [Mac15, p. 325, Eq. (4.17)]), hence for $0\leq k\leq n/2$ we have

$$\mathrm{P}_{(n-k,k)}(y_1,y_2;1/q,0)=(y_1y_2)^k\sum_{j=0}^{n-2k}\begin{bmatrix}n-2k\\ j\end{bmatrix}_{1/q}y_1^{n-2k-j}y_2^j. \tag{105}$$

Remark 8.5. The formula for $\mathrm{P}_{(n-k,k)}(y_1,y_2;1/q,0)$ in (105) is probably well-known, but we have included a proof for the reader's convenience.

Remark 8.6. By setting $y_1=e^{i\theta}$ and $y_2=e^{-i\theta}$ in (105) we obtain

$$\mathrm{P}_{(n-k,k)}(e^{i\theta},e^{-i\theta};1/q,0)=\sum_{l=0}^{n-2k}\begin{bmatrix}n-2k\\ l\end{bmatrix}_{1/q}e^{(n-2k-2l)i\theta}=H_{n-2k}(\cos(\theta)\,|\,1/q). \tag{106}$$

where $H_m(w\,|\,q)$ is the q-Hermite polynomial.

We turn our attention to $g_\lambda(y_1,y_2;q,0)$ by expressing it in terms of power sum symmetric polynomials. It is known [Mac15, p. 311, Eq. (2.9)] that

$$g_n(y_1,y_2;q,0)=\sum_{\rho\vdash n}z_\rho(q,0)^{-1}\,p_\rho(y_1,y_2),\qquad\text{where}\qquad z_\rho(q,t)\overset{\text{def}}{=}z_\rho\prod_{i\geq 1}\frac{1-q^{\rho_i}}{1-t^{\rho_i}}, \tag{107}$$

with z_ρ as in (30) and $p_\rho(y_1,y_2)$ denoting the power sum symmetric polynomial defined as

$$p_\rho(y_1,y_2)\overset{\text{def}}{=}\prod_{i\geq 1}p_{\rho_i}(y_1,y_2),\qquad p_k(y_1,y_2)=y_1^k+y_2^k.$$

It is straightforward to show that

$$z_\rho(1/q,0)^{-1}=q^{|\rho|}z_\rho^{-1}\prod_{i\geq 1}\frac{1}{q^{\rho_i}-1},$$

which implies

$$g_\lambda(y_1,y_2;1/q,0)=q^n\prod_{i=1}^{s}\sum_{\rho\vdash\lambda_i}\frac{1}{z_\rho\prod_{i\geq 1}(q^{\rho_i}-1)}p_\rho(y_1,y_2). \tag{108}$$

Recall the definition of the Chebyshev polynomials $T_\rho(w)$ from (29). With this preparation, we are ready to prove Theorem 1.24.

Proof of Theorem 1.24. We prove this by specializing $y_1=e^{i\theta}$ and $y_2=e^{-i\theta}$ in (104). We start by simplifying the right side of the equation. From the definition of $p_k(y_1,y_2)$ we get

$$p_k\left(e^{i\theta},e^{-i\theta}\right)=2T_k(\cos(\theta)).$$

Hence from (108) we infer that

$$g_\lambda\left(e^{i\theta},e^{-i\theta};1/q,0\right)=q^n\prod_{i=1}^{s}\sum_{\rho\vdash\lambda_i}\frac{2^{\ell(\rho)}T_\rho(\cos(\theta))}{z_\rho\prod_{i=1}^{\ell(\rho)}(q^{\rho_i}-1)}.$$

Thus the right hand side of (104) becomes

$$[\lambda]_q!(q-1)^nq^{-\mathsf{n}(\lambda')}\prod_{i=1}^{s}\sum_{\rho\vdash\lambda_i}\frac{2^{\ell(\rho)}T_\rho(\cos(\theta))}{z_\rho\prod_{i=1}^{\ell(\rho)}(q^{\rho_i}-1)}.$$

On the other hand, from (106), the left hand side of (104) becomes

$$\sum_{k=0}^{n} a_{nk}(\lambda) q^{k(k-n)} H_{n-2k}(\cos(\theta)\,|\,1/q).$$

This finishes the proof. □

Remark 8.7. Specializing $y_1 = 1$ and $y_2 = -1$ in (104) gives us another interesting identity. Let λ be a partition of $n = 2m$ with at least one odd part. Obviously $p_k(y_1, y_2) = 0$ when k is odd. Thus, from (108), we infer that the right hand side of (104) is zero. One can also show

$$\sum_{j=0}^{n} (-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_q = \begin{cases} 0 & \text{if } n \text{ odd;} \\ \prod_{i=1}^{n/2} (1 - q^{2j-1}) & \text{if } n \text{ is even.} \end{cases}$$

This in particular implies

$$\mathrm{P}_{(n-k,k)}(y_1, y_2; 1/q, 0) = (-1)^k \prod_{j=1}^{m-k} \left(1 - q^{1-2j}\right) = (-1)^m \prod_{j=1}^{m-k} \left(q^{1-2j} - 1\right).$$

Therefore we deduce

$$\sum_{k=0}^{m} a_{nk}(\lambda) q^{k(k-n)} \prod_{j=1}^{m-k} \left(q^{1-2j} - 1\right) = 0.$$

8.2. Proofs of Theorem 1.27, Corollary 1.28, and Theorem 1.29. Recall that $\mathrm{Q}_\lambda(\mathbf{x}, q, t)$ denotes the dual Macdonald polynomial indexed by λ. For integers $r \geq s \geq 0$ set

$$\mathrm{Q}_{(r,s)}(\mathbf{x}, q) := \mathrm{Q}_{(r,s)}(\mathbf{x}\,; q, 0).$$

Here (r, s) denotes a partition of length at most two. We use the main theorem of Jing and Józefiak [JJ92, Eq. (1)], which gives a formula for the $Q_{(r,s)}$ in terms of the $\mathrm{Q}_{(r+i)}$ and $\mathrm{Q}_{(s-i)}$. Setting $t = 0$, $r = n - k$, and $s = k$ in the formula [JJ92, Eq. (1)], we obtain

$$\mathrm{Q}_{(n-k,k)}(\mathbf{x}, q) = \sum_{j=0}^{k} (-1)^j q^{\binom{j}{2}} \begin{bmatrix} n-2k+j \\ j \end{bmatrix}_q \frac{1 - q^{n-2k+2j}}{1 - q^{n-2k+j}} \mathrm{Q}_{(n-k+j)}(\mathbf{x}, q) \mathrm{Q}_{(k-j)}(\mathbf{x}, q), \tag{109}$$

where we use the conventions $\mathrm{Q}_{(0)}(\mathbf{x}\,; q, t) = 1$ and $(1 - q^{n-2k+2j})/(1 - q^{n-2k+j}) = 1$ when $n - 2k = j = 0$. Moreover by [Mac15, p. 329, Eq. (2.9)] we have

$$\mathrm{Q}_{(l)}(\mathbf{x}\,; q, t) = g_l(\mathbf{x}\,; q, t),$$

where $g_l(\mathbf{x}\,; q, t)$ was defined in (99). Henceforth we set

$$g_l(\mathbf{x}, q) := g_l(\mathbf{x}\,; q, 0).$$

Thus from (100) we obtain

$$\begin{aligned} [\mathbf{x}^\lambda] \mathrm{Q}_{(n-k,n)}(\mathbf{x}, q) &= \sum_{j=0}^{k} (-1)^j q^{\binom{j}{2}} \begin{bmatrix} n-2k+j \\ j \end{bmatrix}_q \frac{1 - q^{n-2k+2j}}{1 - q^{n-2k+j}} [\mathbf{x}^\lambda] g_{n-k+j}(\mathbf{x}, q) g_{k-j}(\mathbf{x}, q) \\ &= \sum_{j=0}^{k} (-1)^j q^{\binom{j}{2}} \begin{bmatrix} n-2k+j \\ j \end{bmatrix}_q \frac{1 - q^{n-2k+2j}}{1 - q^{n-2k+j}} [\mathbf{x}^\lambda] \sum_{\substack{\nu \vdash n-k+j \\ \rho \vdash k-j}} \frac{m_\nu(\mathbf{x}) m_\rho(\mathbf{x})}{(q\,;q)_\nu\,(q\,;q)_\rho}. \end{aligned}$$

The monomial symmetric polynomial is a specialization of the Hall-Littlewood polynomial at $t = 1$. Thus from [Mac15, Sec. III.3, p. 215] we have

$$[\mathbf{x}^\lambda] \sum_{\substack{\nu \vdash n-k+j \\ \rho \vdash k-j}} \frac{m_\nu(\mathbf{x}) m_\rho(\mathbf{x})}{(q\,;q)_\nu\,(q\,;q)_\rho} = \sum_{\substack{\nu \vdash n-k+j \\ \rho \vdash k-j}} \frac{f^\lambda_{\nu\rho}(1)}{(q\,;q)_\nu\,(q\,;q)_\rho},$$

where, as before, $f^{\lambda}_{\nu\rho}(t)$ is a structure constant associated with products of Hall-Littlewood polynomials. But

$$(1/q\,;1/q)_n = \prod_{j=1}^{n}\left(1-\frac{1}{q^j}\right) = \frac{(q-1)^n}{q^{n+\binom{n}{2}}}[n]_q! \quad \text{and} \quad \begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)}\begin{bmatrix} n \\ k \end{bmatrix}_{1/q}.$$

Hence $[\mathbf{x}^{\lambda}]\mathrm{Q}_{(n-k,k)}(\mathbf{x}\,;1/q)$ is equal to

$$\frac{q^n}{(q-1)^n}\sum_{j=0}^{k}(-1)^j q^{-\binom{j+1}{2}-(n-2k)j}\begin{bmatrix} n-2k+j \\ j \end{bmatrix}_q \frac{q^{n-2k+2j}-1}{q^{n-2k+j}-1}\sum_{\substack{\nu\vdash n-k+j \\ \rho\vdash k-j}}\frac{q^{\mathsf{n}(\nu')+\mathsf{n}(\rho')}f^{\lambda}_{\nu\rho}(1)}{[\nu]![\rho]!}.$$

This combined with (97) proves Theorem 1.27.

Remark 8.8. The formula (109) of Jing and Józefiak has been generalized by Lassalle and Schlosser [LS06, Theorem 4.1]. It might be possible to use their result to count the number of matrices $X \in \mathfrak{u}_{\lambda}(\mathbb{F}_q)$ satisfying $X^k = 0$.

We now proceed towards the proof of Theorem 1.29. Since $a_{nk} = 0$ for $k > \lfloor n/2 \rfloor$, from (35) it follows immediately that

$$\mathrm{C}_n = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor}\sum_{j=0}^{k}(-1)^j q^{nk-k^2-\binom{j+1}{2}-(n-2k)j}\begin{bmatrix} n-2k+j \\ j \end{bmatrix}_q \binom{n}{k-j}\frac{q^{n-2k+2j}-1}{q^{n-2k+j}-1}.$$

For $k_1 = k-j$ and $k_2 = j$ we have $0 \le k_1+k_2 \le \lfloor \frac{n}{2} \rfloor$ and we can rearrange the right hand summation as

$$\mathrm{C}_n = \sum_{0\le k_1+k_2\le \lfloor \frac{n}{2} \rfloor} q^{nk_1-k_1^2}\binom{n}{k_1}(-1)^{k_2}q^{\binom{k_2}{2}}\begin{bmatrix} n-2k_1-k_2 \\ k_2 \end{bmatrix}_q \frac{[n-2k_1]_q}{[n-2k_1-k_2]_q}, \tag{110}$$

where on the right hand side we use the convention $(q^0-1)/(q^0-1) = 1$ when $n-2k_1 = 0$. This proves Corollary 1.28. Next we define

$$\mathrm{B}_n := \begin{cases} 1 & n = 0; \\ \displaystyle\sum_{j=0}^{\lfloor n/2 \rfloor}(-1)^j q^{\binom{j}{2}}\begin{bmatrix} n-j \\ j \end{bmatrix}_q \frac{[n]_q}{[n-j]_q} & n \ge 1. \end{cases}$$

Thus (110) can be rewritten as

$$\mathrm{C}_n = \sum_{k_1=0}^{\lfloor n/2 \rfloor} q^{nk_1-k_1^2}\binom{n}{k_1}\mathrm{B}_{n-2k_1}. \tag{111}$$

Our next task is to simplify B_n. We need the following lemma, which was first proved in [EZ96, Lemma 2] by the "A=B" method, using a computer. For a q-hypergeometric proof see [War04].

Lemma 8.9. *Let* $\mathrm{A}_n := \sum_{j=0}^{\lfloor n/2 \rfloor}(-1)^j q^{\binom{j}{2}}\begin{bmatrix} n-j \\ j \end{bmatrix}_q$ *for* $n \ge 0$. *Then* $\mathrm{A}_n = (-1)^n\chi_3(n+1)q^{\frac{1}{3}\binom{n}{2}}$, *where* χ_3 *is the Dirichlet character defined in (36).*

We let $\mathrm{A}_n = 0$ when $n < 0$. Note that $[n]_q = [n-j]_q + q^{n-j}[j]_q$ and so for each $n \ge 0$ we have $\mathrm{B}_n = \mathrm{A}_n - q^{n-1}\mathrm{A}_{n-2}$. Thus by Lemma 8.9 we have

$$\mathrm{B}_n = (-1)^n\left[\chi_3(n+1)q^{\frac{1}{3}\binom{n}{2}} - \chi_3(n-1)q^{\frac{1}{3}\binom{n+1}{2}}\right] \qquad \text{for } n \ge 1. \tag{112}$$

We only prove Theorem 1.29 for C_{2n+1}, as the proof for C_{2n} is analogous. By (111) and (112),

$$\begin{aligned}
\mathrm{C}_{2n+1} &= \sum_{j=0}^{n} q^{(2n+1)j-j^2} \binom{2n+1}{j} \mathrm{B}_{2n+1-2j} \\
&= \sum_{j=0}^{n} q^{(2n+1)j-j^2} \binom{2n+1}{j} \left[\chi_3(2(n-j)) q^{\frac{1}{3}\binom{2(n-j)+2}{2}} - \chi_3(2(n-j+1)) q^{\frac{1}{3}\binom{2(n-j)+1}{2}}\right] \\
&= \sum_{j=0}^{n} q^{(2n+1)j-j^2} \binom{2n+1}{j} \left[\chi_3(n-j+1) q^{\frac{1}{3}\binom{2(n-j)+1}{2}} - \chi_3(n-j) q^{\frac{1}{3}\binom{2(n-j)+2}{2}}\right] \\
&= \sum_{j=0}^{n} q^{n^2+n-j^2-j} \binom{2n+1}{n-j} \left[\chi_3(j+1) q^{\frac{1}{3}\binom{2j+1}{2}} - \chi_3(j) q^{\frac{1}{3}\binom{2j+2}{2}}\right].
\end{aligned}$$

This proves the first formula in Theorem 1.29 for C_{2n+1}. To prove the second formula for C_{2n+1}, define $i \mapsto \mathbb{1}_i$ to be the characteristic function of $\{0, 1, \ldots, n\}$, i.e., $\mathbb{1}_i = 1$ for $0 \le i \le n$ and $\mathbb{1}_i = 0$ otherwise. Then from the last formula for C_{2n+1} we obtain

$$\begin{aligned}
\frac{\mathrm{C}_{2n+1}}{q^{n^2+n}} &= \sum_{i=0}^{2} \sum_{j\in\mathbb{Z}} q^{-(3j+i)^2-(3j+i)} \binom{2n+1}{n-3j-i} \mathbb{1}_{3j+i} \left[\chi_3(i+1) q^{\frac{1}{3}\binom{2(3j+i)+1}{2}} - \chi_3(i) q^{\frac{1}{3}\binom{2(3j+i)+2}{2}}\right] \\
&= I + II,
\end{aligned}$$

where

$$I = \sum_{j\in\mathbb{Z}} q^{-3j^2-2j} \left[\mathbb{1}_{3j} \binom{2n+1}{n-3j} - \mathbb{1}_{3j+1} \binom{2n+1}{n-3j-1}\right],$$

and

$$II = \sum_{j\in\mathbb{Z}} q^{-3j^2-4j-1} \left[\mathbb{1}_{3j+2} \binom{2n+1}{n-3j-2} - \mathbb{1}_{3j+1} \binom{2n+1}{n-3j-1}\right].$$

Using $\binom{2n+1}{a} = \binom{2n+1}{2n+1-a}$ and then substituting j by $-j-1$ in II, we obtain

$$\begin{aligned}
II &= \sum_{j\in\mathbb{Z}} q^{-3j^2-4j-1} \left[\mathbb{1}_{3j+2} \binom{2n+1}{n+3j+3} - \mathbb{1}_{3j+1} \binom{2n+1}{n+3j+2}\right] \\
&= \sum_{j\in\mathbb{Z}} q^{-3j^2-2j} \left[\mathbb{1}_{-3j-1} \binom{2n+1}{n-3j} - \mathbb{1}_{-3j-2} \binom{2n+1}{n-3j-1}\right].
\end{aligned}$$

Finally, note that

$$\binom{2n+1}{n-3j} = \mathbb{1}_{3j} \binom{2n+1}{n-3j} + \mathbb{1}_{-3j-1} \binom{2n+1}{n-3j},$$

and

$$\binom{2n+1}{n-3j-1} = \mathbb{1}_{3j+1} \binom{2n+1}{n-3j-1} + \mathbb{1}_{-3j-2} \binom{2n+1}{n-3j-1}.$$

Using the latter relations in $I + II$ proves the second formula for C_{2n+1} in Theorem 1.29.

§ 9. Hook partitions and the proof of Theorem 1.36

The goal of this section is to prove Theorem 1.36 using Theorem 1.7 combined with (39). The conjugate of μ is $\mu' = (n-k, 1^k)$. Then the exponent of $q-1$ in Theorem 1.36, using Theorem 1.7, is $n - (n-k) = k$ and the exponent of q is

$$\binom{n}{2} + \ell(\mu) - \mathsf{n}(\mu) - n = \binom{n}{2} + n - k - \binom{n-k}{2} - n = \binom{n}{2} - \binom{n-k}{2} - k.$$

Next we prove this simple lemma.

Lemma 9.1. *Let $\lambda = (\lambda_1, \dots, \lambda_s)$ be a partition of n and let $\mu' = (n-k, 1^k)$ be a hook partition. Then $\lambda \trianglelefteq \mu'$ if and only if $k \leq s-1$.*

Proof. Obviously $\lambda \trianglelefteq \mu'$ implies $\ell(\mu') \leq \ell(\lambda)$ and so $k \leq s-1$. Now assume $k \leq s-1$. Then

$$\lambda_1 + (s-1) \leq \lambda_1 + \cdots + \lambda_s = n - k + k \Rightarrow \lambda_1 \leq n - k + (k - (s-1)) \Rightarrow \lambda_1 \leq n - k.$$

Similarly $\lambda_1 + \lambda_2 + (s-2) \leq n-k+1+(k-1)$, which gives $\lambda_1 + \lambda_2 \leq n-k+1+(k-1-(s-2))$ and so $\lambda_1 + \lambda_2 \leq n-k+1$ since $k \leq s-1$. The inequality for $\lambda_1 + \cdots + \lambda_i$ is deduced from the same pattern. □

To apply Theorem 1.7, we need to calculate $s_q(T) r_q(T)$ explicitly. For this purpose, we must describe all semistandard Young tableaux of shape μ' and content λ. We claim that any such semistandard Young tableau arises from a subset of $\{2, \dots, s\}$ of size k. The existence of such a subset is guaranteed by Lemma 9.1.

Note that the top-left box of any tableau of shape μ' and content λ must be 1. Given a semistandard tableau of shape μ' and content λ, the set of entries in rows 2 through $k+1$ of μ' determine a unique size subset of $2, \cdots, s$ of size k. this proves the claim and show that

$$\mathrm{K}_{\mu'\lambda} = \binom{s-1}{k}.$$

Now fix $2 \leq j \leq s-k+1$ and pick A, a subset of $\{2, \dots, s\}$, with $\min A = j$. This subset produces a unique tableau T. Note that numbers strictly less than j must only appear in the first row of T. Because of this, the number of entries in the first row of T that are strictly less than j is $\lambda_1 + \cdots + \lambda_{j-1}$. Thus

$$r_q(T) = [\lambda_1 + \cdots + \lambda_{j-1}]_q \,.$$

For any $i \notin A$ we have $\beta^i = (\lambda_i, 0, \dots)$, on the other hand $\beta^i = (\lambda_i - 1, 1, 0, \cdots)$ when $i \in A$. Therefore

$$s_q(T) = \prod_{i \in A} [\lambda_i]_q.$$

By combining these facts with Theorem 1.7, we deduce Theorem 1.36.

§ 10. Double cosets and the proof of Theorem 1.38

Let $\mathsf{h}_1 \in \mathcal{H}_n$ be a Hessenberg function. By the bijective correspondence $X \mapsto I + X$ between $\mathfrak{u}_{\mathsf{h}_1}(\mathbb{F}_q)$ and $\mathsf{U}_{\mathsf{h}_1} = \mathsf{U}_{\mathsf{h}_1}(\mathbb{F}_q)$, the number of matrices in $\mathsf{U}_{\mathsf{h}_1}$ of Jordan type μ is equal to $\mathrm{F}_{\mu \mathsf{h}_1}(q)$. Let θ_{h_1} denote the character of $\mathrm{Ind}_{\mathsf{U}_{\mathsf{h}_1}}^{\mathrm{G}} 1$, where $\mathrm{G} = \mathrm{GL}_n(\mathbb{F}_q)$. Note that $|\mathsf{U}_{\mathsf{h}_1}| = q^{n^2 - |\mathsf{h}_1|}$ and so from (22) we obtain

$$\theta_{\mathsf{h}_1}(u) = \begin{cases} 0 & u \text{ is not a unipotent matrix;} \\ \frac{|\mathrm{Z}_{\mathrm{G}}(I+J_\mu)|}{q^{n^2-|\mathsf{h}_1|}} \mathrm{F}_{\mu \mathsf{h}_1}(q) & u \in \mathrm{Cl}_{\mathrm{G}}(I + J_\mu). \end{cases} \tag{113}$$

Now let $\mathsf{h}_2 \in \mathcal{H}_n$ be another Hessenberg function. We have the well-known equality

$$\langle \theta_{\mathsf{h}_1}, \theta_{\mathsf{h}_2} \rangle = |\mathsf{U}_{\mathsf{h}_1} \backslash \mathrm{G} / \mathsf{U}_{\mathsf{h}_2}| \,, \tag{114}$$

which follows from Frobenius reciprocity and Mackey decomposition. Let $\mathbf{U} = \bigsqcup_{\mu\vdash n} \mathrm{Cl}_{\mathrm{G}}(I+J_\mu)$ denote the set of all unipotent matrices. Then

$$\begin{aligned}\langle\theta_{\mathsf{h}_1},\theta_{\mathsf{h}_2}\rangle &= \frac{1}{|\mathrm{G}|}\sum_{u\in\mathrm{G}}\theta_{\mathsf{h}_1}(u)\overline{\theta_{\mathsf{h}_2}(u)} = \frac{1}{|\mathrm{G}|}\sum_{u\in\mathbf{U}}\theta_{\mathsf{h}_1}(u)\overline{\theta_{\mathsf{h}_2}(u)}\\ &= \frac{1}{|\mathrm{G}|}\sum_{\mu\vdash n}\sum_{u\in\mathrm{Cl}_{\mathrm{G}}(I+J_\mu)}\frac{|\mathrm{Z}_{\mathrm{G}}(I+J_\mu)|^2}{q^{2n^2-|\mathsf{h}_1|-|\mathsf{h}_2|}}\mathrm{F}_{\mu\mathsf{h}_1}(q)\,\mathrm{F}_{\mu\mathsf{h}_2}(q)\\ &= \frac{1}{q^{2n^2-|\mathsf{h}_1|-|\mathsf{h}_2|}}\sum_{\mu\vdash n}|\mathrm{Z}_{\mathrm{G}}(I+J_\mu)|\mathrm{F}_{\mu\mathsf{h}_1}(q)\,\mathrm{F}_{\mu\mathsf{h}_2}(q).\end{aligned} \tag{115}$$

This calculation combined with (114) proves Theorem 1.38.

§ 11. Appendix: Counting matrices with a given rank sequence

Recall that C_l^r is defined in (61). The following lemma is probably well-known.

Lemma 11.1. *Suppose $l \geq 1$ is an arbitrary positive integer and let $0 \leq r \leq l$. Let W be an r-dimensional vector space over $\mathbb{F}_q$. Then the number of l-tuples $(w_1,\dots,w_l) \in W^{\oplus l}$ with the property that $W = \mathrm{span}(v_1,\dots,v_l)$ is equal to C_l^r.*

Proof. By expressing elements of W in a chosen basis, one observes that the assertion is equivalent to counting the number of $l\times r$ matrices over $\mathbb{F}_q$ of rank r. A classical formula due to Landsberg gives the number of $l\times n$ matrices of rank r as

$$\begin{bmatrix}n\\r\end{bmatrix}_q \left(q^l-1\right)\left(q^l-q\right)\cdots\left(q^l-q^{r-1}\right) = \begin{bmatrix}n\\r\end{bmatrix}_q q^{\binom{r}{2}}(q-1)^r\frac{[l]_q!}{[l-r]_q!}.$$

Setting $n=r$ yields the lemma. □

Lemma 11.2. *Let $n=n_1\geq n_2\geq\cdots\geq n_\ell > n_{\ell+1}=0$ be a sequence of positive integers, and let $m=r_0\geq r_1\geq\cdots\geq r_\ell\geq r_{\ell+1}=0$ be a sequence of non-negative integers, with the property that $r_i-r_{i+1}\leq n_i-n_{i+1}$ for $1\leq i\leq\ell$. Then, the number of $n\times m$ matrices over $\mathbb{F}_q$ with $\mathrm{rank}(A_i)=r_i$ for $1\leq i\leq\ell$, where A_i is the submatrix of A formed by the first n_i rows of A, is*

$$\prod_{j=1}^{\ell} q^{r_{j+1}(n_j-n_{j+1})}\begin{bmatrix}m-r_{j+1}\\r_j-r_{j+1}\end{bmatrix}_q \mathrm{C}_{n_j-n_{j+1}}^{r_j-r_{j+1}}. \tag{116}$$

Proof. The proof is by induction on ℓ. For $\ell=1$ the assertion follows from Lemma 11.1 and the fact that there are $\begin{bmatrix}m\\r_1\end{bmatrix}_q$ subspaces of dimension r_1 in an $\mathbb{F}_q$-vector space of dimension m. Next we assume the assertion holds for some ℓ, and we prove it for $\ell+1$. Given vectors $w_1,\dots,w_{n_2}\in\mathbb{F}_q^m$ such that their span, denoted by W_2, has dimension r_2, we aim to count the number of (n_1-n_2)-tuples $(w_{n_2+1},\dots,w_{n_1})$ in $\mathbb{F}_q^m$ such that $\mathrm{span}\{w_1,\dots,w_{n_1}\}$ is r_1-dimensional. To this end, we first select a subspace W_1 of $\mathbb{F}_q^m$ of dimension r_1 that contains W_2. There are $\begin{bmatrix}m-r_2\\r_1-r_2\end{bmatrix}_q$ such subspaces. Now let $W_1\supseteq W_2$ be fixed. Next, we count the number of (n_1-n_2)-tuples $(w_{n_2+1},\dots,w_{n_1})$ of vectors in W_1 such that $\mathrm{span}\{w_1,\dots,w_{n_1}\}$ is r_1-dimensional. Since the first n_2 vectors already generate W_2, these n_1-n_2 vectors must be chosen in such a way that their images in W_1/W_2 under the natural projection $\mathrm{span}\ W_1/W_2$. To do this, we first choose a spanning set for W_1/W_2 and then lift the vectors to W_1. By Lemma 11.1, there are $C_{n_1-n_2}^{r_1-r_2}$ spanning sets for W_1/W_2. Since each vector in W_1/W_2 has q^{r_2} lifts to W_1, the total number of ways to choose $w_{n_2+1},\cdots,w_{n-1}$ is $q^{r_2(n_1-n_2)}\mathrm{C}_{n_1-n_2}^{r_1-r_2}$. Therefore, the number of (n_1-n_2)-tuples $(w_{n_2+1},\dots,w_{n_1})$ with the desired property is

$$q^{r_2(n_1-n_2)}\begin{bmatrix}m-r_2\\r_1-r_2\end{bmatrix}_q \mathrm{C}_{n_1-n_2}^{r_1-r_2}.$$

Combining the latter formula with the hypothesis of induction readily implies the claim of the lemma. □

Acknowledgements

We would like to thank Ferenc Balogh, Steven Karp, Siddhartha Sahi, Shahab Shaabani, Majid Shahabi, Jasem Tati and Ole Warnaar for many fruitful discussions. K.M.K. was supported by the German Research Foundation (DFG) Grant DI 506/19-1. S.R. gratefully acknowledges support from an Indo-Russian project DST/INT/RUS/RSF/P41/2021. The research of H.S. was partially supported by NSERC Discovery Grant RGPIN-2024-04030. H.S. would also like to thank WestJet Airlines for a significantly delayed flight on 03/06/2024. The ensuing sleepless night at the airport played a crucial role in the formulation of Lemma 2.8.

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